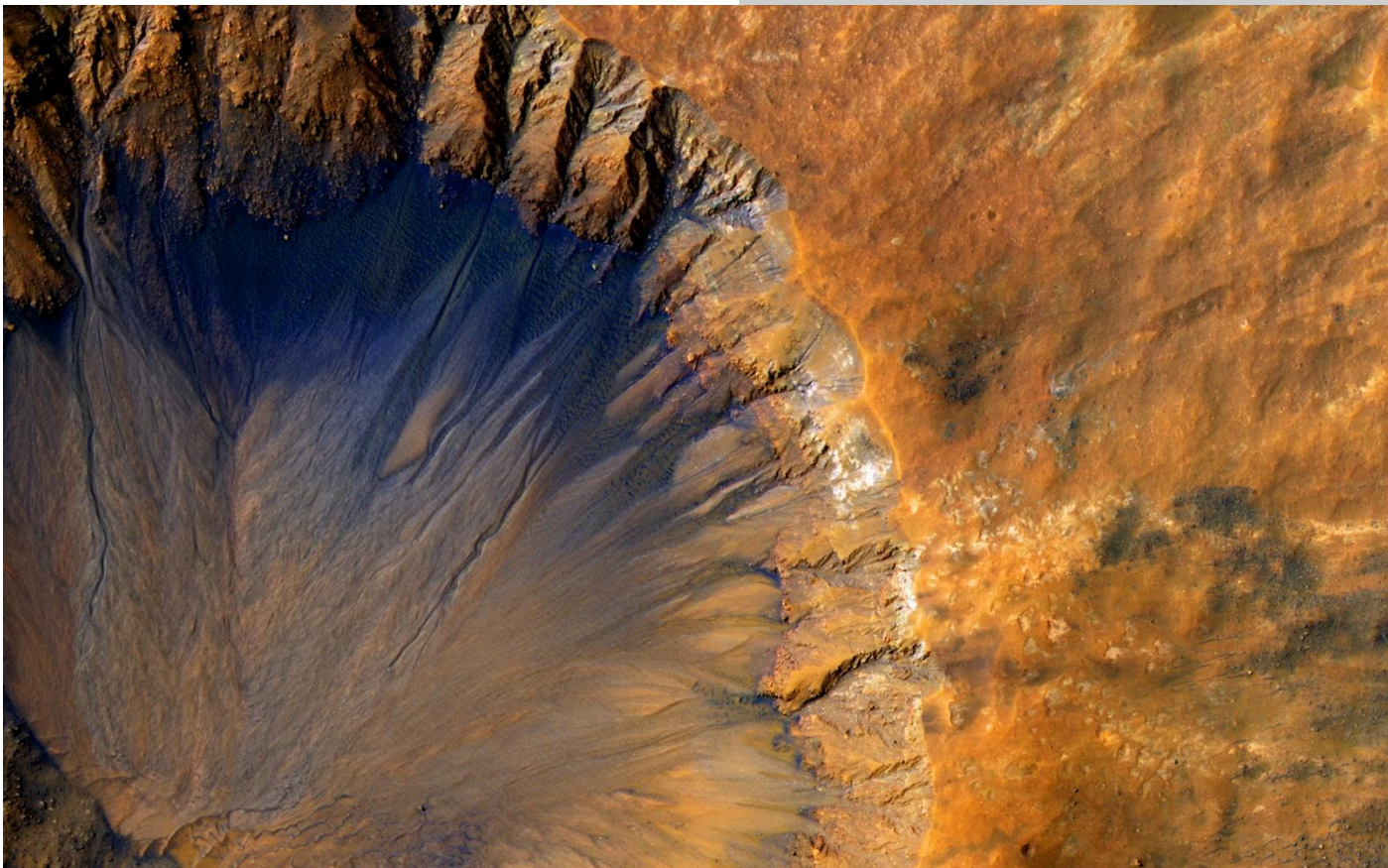


Earth, Space, and Environmental Science Explorations with ArcGIS Pro

Written for ArcGIS Pro 3.0
Second Edition



Sirenum Fossae, Mars
Photo by NASA on Unsplash

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Edited by Diana Valdez

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Preface

When I first started teaching Geographic Information Systems (GIS) over 15 years ago, there were only a handful of GIS tutorial books on the market. While a few of them focused on a particular theme, many of them were mainly technical publications with no unifying theme. These publications are perfect for teaching new users about basic GIS operations and continue to be invaluable to many users today. However, these publications consisted of tutorials focusing on vastly different real-world scenarios, and felt a little disjointed to integrate into a class that focuses on a specific field (e.g. Environmental Science, Geography, Urban Planning). Since then, I am happy to see that there has been a proliferation of publications that are not only focused on introducing the technology, but also provide an introduction to a specific field as well as GIS.

It is with this consideration in mind that I wrote this book to not only introduce students to the tools in ArcGIS Pro, but also to provide a glimpse into some of the topics that may be covered in an introductory course in Geography, Environmental Science, or related disciplines. I aimed to achieve a balance of technical instruction and information about Earth, Space, and Environmental Sciences, to make this publication helpful for new GIS learners, as well as students who may be interested in learning about the applications of GIS in the Earth, Space, and Environmental Sciences.

Before you begin with the book, please download and unzip the data for the book at <https://tinyurl.com/ESESGIS>. Each chapter will begin with a list of the technical objectives for the lab exercise, which is particularly useful to refer to should you need a refresher on how to perform a particular task at a later date. The list of technical objectives is followed by a short introduction that provides the context for the exercise. Subsequently, you will find the step-by-step directions for the exercises with questions embedded throughout (in yellow highlight) to ensure that you are on the right track. To obtain a copy of the answer key for the questions, please email me at wcheung@palomar.edu with a copy of your current faculty identification or other forms of professional identification.

To complete the exercises in this book, you will need access to ArcGIS Pro 3.0 (and ArcGIS Online with credits for some chapters). Please remember to first download the exercise data for the book at <https://tinyurl.com/ESESGIS>.

I wish you a wonderful time exploring various topics in GIS and Earth, Space, and Environmental Sciences in this book.

Wing Cheung PhD, GISP

Acknowledgments

I will like to thank my former student and editor, Diana Valdez, for offering feedback on the directions and suggesting clarifications in the text. I also want to thank students in my GIS classes, who also offer valuable feedback on the exercises and tested the directions. I will also like to express my sincerest thanks to prior authors whose works inspired some of the chapters in this book.

I also want to thank my colleagues, friends, and family for offering their technical and personal support to help me complete this project.

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Chapter 1: Where are our flood hazard areas? And who is living there?

In this exercise, you will:

- Create a new project and map
- Add layers to a new map
- Share and publish a web layer
- Analyze spatial relationships using the “enrich layer” function in ArcGIS Online
- Explore summary statistics
- Filter out missing data values and features

Introduction

A floodplain is defined as “the flat low-lying area flanking a stream channel that is subjected to recurrent flooding” (Christopherson, 2002). Floodplains are frequently formed when a stream meanders (i.e. shifts) across a flat and wide low-lying area. (McKnight and Hess, 2009) Floodplains are occasionally inundated during periods of heavy precipitation when water spills over the normal stream channel.

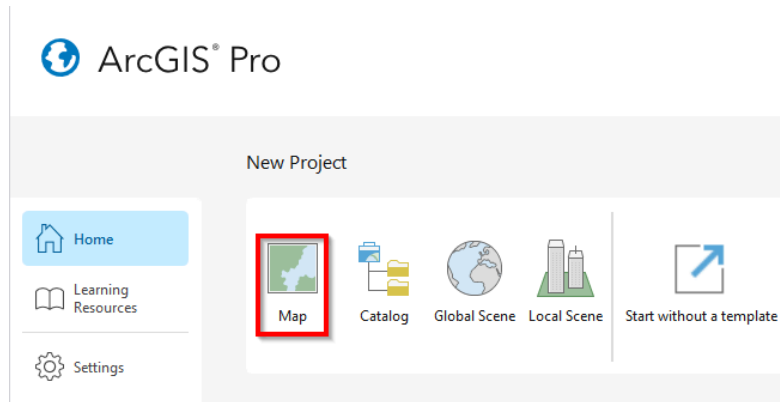


THE PARANÁ RIVER APPEARS AS A WIDE, BLUE STRIP WITH THE MUDDY BROWN WATER OF THE SMALLER VERDE RIVER ENTERING FROM THE NORTHWEST (TOP LEFT). (SOURCE: NASA ASTRONAUT PHOTOGRAPH ISS030-E-90012, PUBLIC DOMAIN, VIA WIKIMEDIA COMMONS)

While a floodplain is usually associated with riverine flooding, flooding may also occur in coastal regions as a result of high tides, storm surges, and precipitation. Hence, a more inclusive term used to describe areas subjected to occasional flooding is flood hazard areas. In particular, the Federal Emergency Management Agency (FEMA) calls areas that have a 1 percent chance of being flooded in a given year the 100-year floodplain, and identifies these Special Flood Hazard Areas (SFHA) in its flood insurance rate maps (FEMA, 2019). Within a flood insurance rate map, 100-year floodplains may include Zone AE (shown in neon blue in the graphic below) as well as other zones. Homes and businesses within these areas “with mortgages from federally regulated or insured lenders are required to have flood insurance” (FEMA, 2018). You can explore flood hazard areas in your region by visiting [FEMA’s Flood Map Changes Viewer](#).

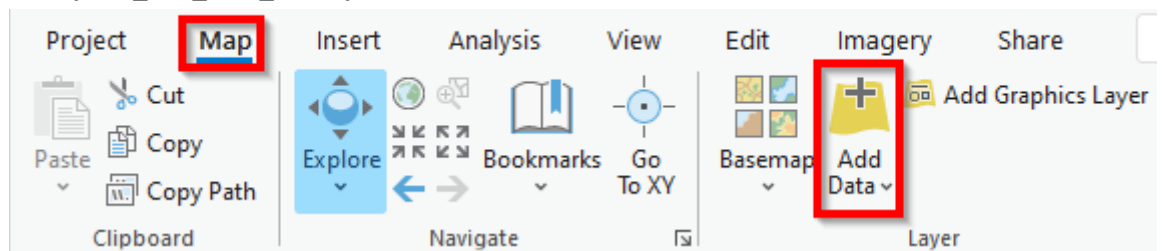
In this exercise, you will explore the location of the 100-year floodplains in San Diego County, and analyze the socioeconomic characteristics of the residents in these areas. Specifically, you will use ArcGIS Online as your primary platform for visualization and analysis in this exercise.

- Start **ArcGIS Pro**. 
- In the Welcome screen, click **Map** to start a new project, name it something descriptive (e.g. Floodplain), and save it in a safe location.



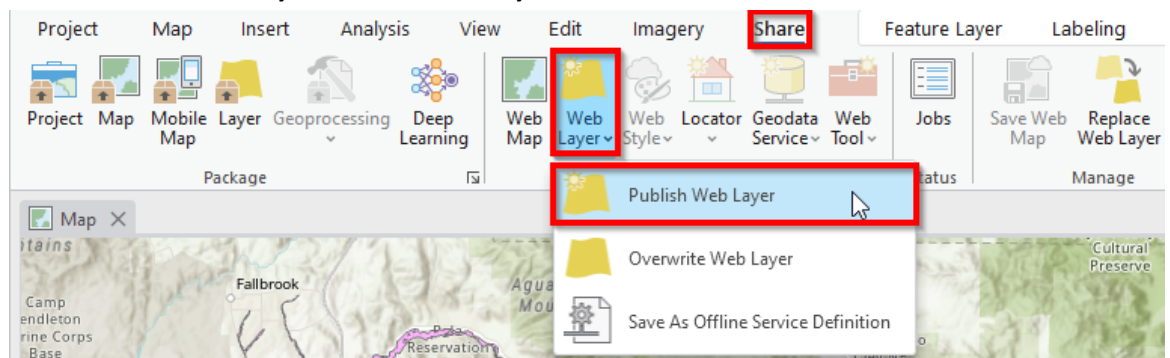
- c. Add a layer that shows the areas within San Diego County that has a 1% chance of being flooded in any given year according to the Federal Emergency Management Agency (FEMA), these areas are also called 100-year floodplains.

Go to **Add Data** (found under the **Map** tab in the main menu), and add in the **Floodplain_100_Year_CN.shp**



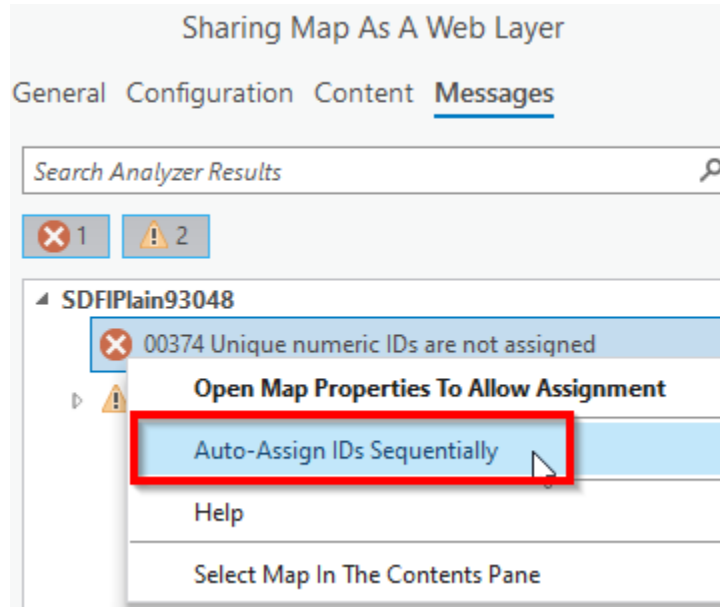
2. Share your layer online

- a. To share the layer as a web layer on ArcGIS Online, go to the **Share** tab in the main menu. Click on **Web Layer**>**Publish Web Layer**.



- b. In the Sharing Map As A Web Layer pane, enter the following:
 - i. Name: **SDFloodplain_(first initial and last name)**
Tip: Each item published to ArcGIS Online must have a unique name within the organization
 - ii. Summary: **FEMA floodplain**
 - iii. Tags: **Flood Hazard**

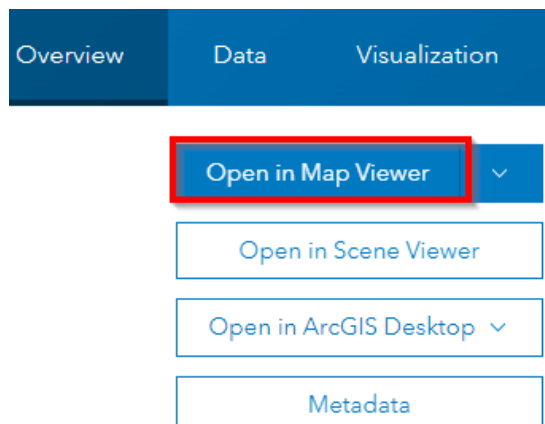
- iv. Layer Type: **Feature**
- v. Share with: **Everyone**
- vi. Click **Analyze** (Right-click the error , select **Auto-Assign IDs Sequentially**, and ignore the warnings .



- vii. Click **Publish**.
- c. Once the publication process is complete, you should see a green box at the bottom of the window telling you that ArcGIS has successfully published the web layer.



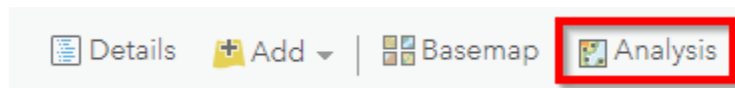
- d. Click the **Manage the web layer** link at the bottom of the green box to view your web layer on ArcGIS Online.
- e. Click **Open in Map Viewer** within the item details page to view your layer in your internet browser.



Q1. Hypothesize the relationship between income and 100-year floodplain locations. In other words, do you think the median household income of families living within the 100-year floodplain is higher or lower than the county average (the median household income for a family in San Diego County is \$82,426 in 2020)? Why? Write down your hypothesis in 2-3 sentences.

3. Analyze spatial relationships in ArcGIS Online

- In your internet browser that is displaying the 100-year floodplain layer, click **Sign In** and enter your ArcGIS Online account information. The following steps and tools that we will use are only available to paid ArcGIS Online users.
- Click **Open in Map Viewer Classic** at the top of the window. (Analysis tools used in this exercise were only available in Map Viewer Classic at the time of writing)
- To test your hypothesis, click **Analysis**. (If the Analysis icon is not available, check to make sure you are signed in and not using a free, public ArcGIS Online account)



- Select **Data Enrichment>Enrich Layer**.



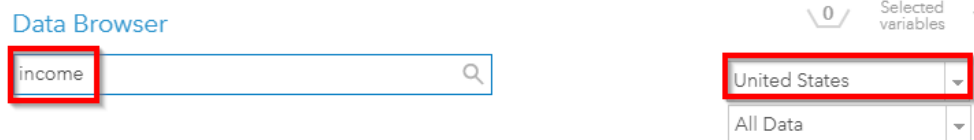
****Tip: This capability costs credit/money, so use it sparingly****

- Within the Enrich Layer dialog, enter the following:
 - Choose layer to enrich with new data: SDFloodplain_(first initial and last name)

- Click the **Select Variables** box.

SELECT VARIABLES

- Select **United States** from the dropdown. Type in **Income** and hit Enter on your keyboard.



- On the next window, expand **2022 Income (Esri)**, and check the box next to **2022 Median Household Income (Esri)**. Click **Apply**. ****Tip: Select the income data for the current year, 2022 data is used as an example in this exercise. Check with the instructor to verify the data to use.****

Data Browser

 ALL VARIABLES

'income' variables (874)

- ☐ 2016-2020 Gross Rent (ACS)
- ☐ 2016-2020 Household Income (ACS)
- ☐ 2016-2020 Income (ACS)
- ☐ 2016-2020 Income by Age (ACS)
- ☐ 2016-2020 Mortgage Status (ACS)
- ☐ 2016-2020 Poverty Status and Household Type (ACS)
- ☐ 2016-2020 Ratio of Income to Poverty (ACS)
- ☐ 2022 Disposable Income (Esri)
- ☐ 2022 Disposable Income by Age (Esri)
- ☐ 2022 Financial (Consumer Spending)
- ☒ 2022 Income (Esri)
- ☐ 2022 Average Household Income (Esri)
- ☒ 2022 Median Household Income (Esri)

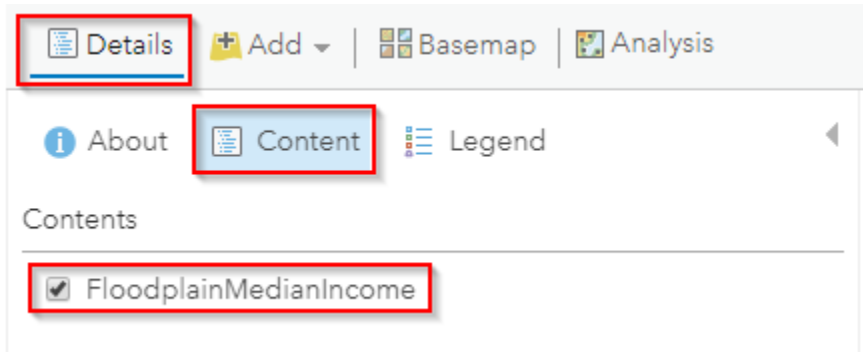
- v. Give the Result layer a descriptive name (e.g. FloodplainMedianIncome)
- vi. **Uncheck** the Use current map extent box
- vii. Click **Run Analysis** (this may take a few minutes)

4 Result layer name

Save result in

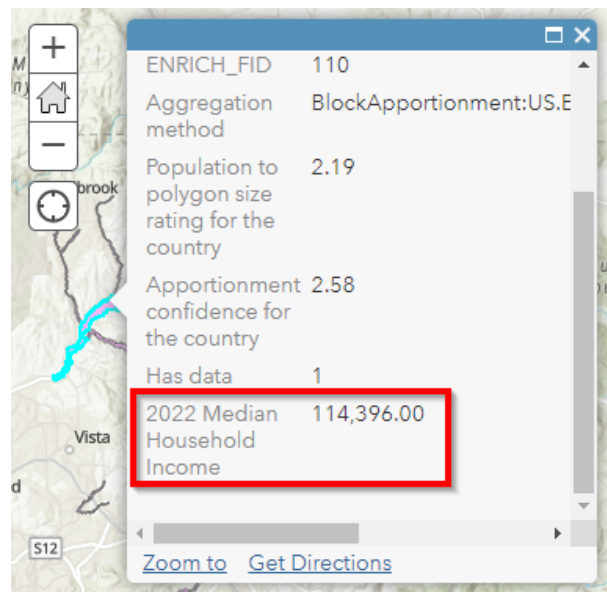
☐ Use current map extent [Show credits](#)

- f. Once the analysis is complete, click **Details>Content**, and you should see your new layer (i.e. FloodplainMedianIncome).



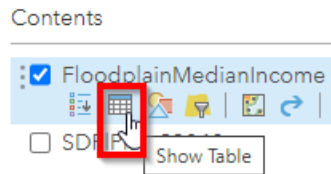
4. Explore and filter your data

- In the Contents pane, turn off the SDFloodplain_(first initial and last name) layer by unchecking it. The only layer that should be checked on now is the FloodplainMedianIncome layer.
- Explore the FloodplainMedianIncome layer by clicking on some of the polygons and reading the information in the pop-up window. The information of most interest to us is the value for 2022 Median Household Income within the pop-up window. This is the median (i.e. middle or 50th percentile) income of all the households within the specific floodplain polygon that you clicked on. You may have to scroll to the lower part of the window to see it.



****Tip: Depending on where you clicked, many of the polygons' Median Household Income will report 0.00 because they are in the middle of a lake or river, and no one is living within those areas****

- c. To see if our hypothesis is correct, hover your mouse over the FloodplainMedianIncome layer's name in the Contents pane and click on the **Show Table** icon .



- d. You should now see the FloodplainMedianIncome layer's attribute table, and scroll until you see the 2022 Median Household Income fields/columns in the table. Click on the column name, and select **Statistics**.

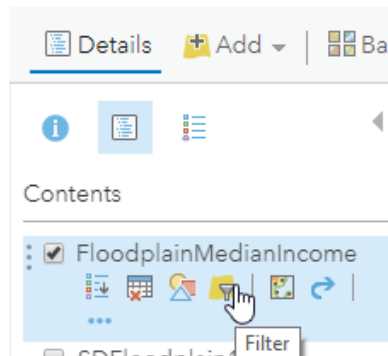
Population to polygon size rating for the country	Apportionment confidence for the country	Has data	2022 Median Household Income
2.19	2.58	0	Sort Ascending Sort Descending Statistics Calculate Delete
2.19	2.58	0	

Q2. In the pop-up window, what is reported as the **Average 2022 Median Household Income** among families living within the 100-year floodplain in San Diego County?

Given that many of the polygons are in the middle of a lake or river with no data to report, there are many 0.00 values within the Median Household Income field that are skewing/inaccurately depressing the Average Median Household Income. Thus, we must first filter out those 0.00 values before calculating an average again.

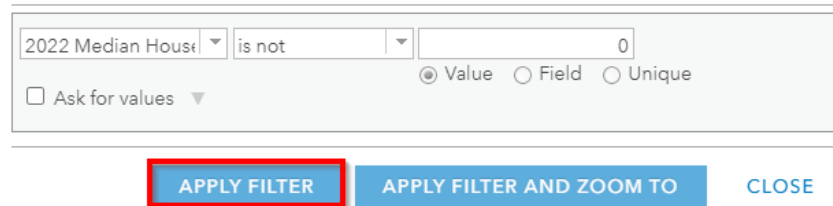
- e. Hover your mouse over the FloodplainMedianIncome layer's name in the Contents pane

and click on the **Filter** icon .



- f. In the Filter: FloodplainMedianIncome window, enter the following:
- First dropdown: **2022 Median Household Income**
 - Second dropdown: **is not**
 - Third box: type in **0**
 - Click **Apply Filter**.

This filter operation ensures that we are only retaining areas with income-generating households for our subsequent operations.



- g. Repeat step d in this section, and find the Average 2022 Median Household Income among families living within the 100-year floodplain in San Diego County.

Q3. What is the updated **Average 2022 Median Household Income** among families living within the 100-year floodplain, now that we have excluded the polygons without income-generating households?

Q4. Was your hypothesis correct or incorrect? In two to three sentences, what may this result suggest about the actual relationship between income and natural hazards?

Works Cited

Christopherson, Robert W. *Geosystems : An Introduction to Physical Geography*. Upper Saddle River (N.J.), Prentice Hall, 2002.

FEMA. "Flood Zones." *Fema.Gov*, Federal Emergency Management Agency, 2019, www.fema.gov/flood-zones. Accessed 26 June 2019.

FEMA. "How Do I Buy Flood Insurance?" *Fema.Gov*, Federal Emergency Management Agency, 2018, www.fema.gov/national-flood-insurance-program/How-Buy-Flood-Insurance. Accessed 26 June 2019.

Mcknight, Tom L, and Darrel Hess. *Physical Geography*. Boston, Ma, Pearson Education, 2009.

Chapter 2: Exploring the characteristics and impacts of historical hurricanes

In this exercise, you will:

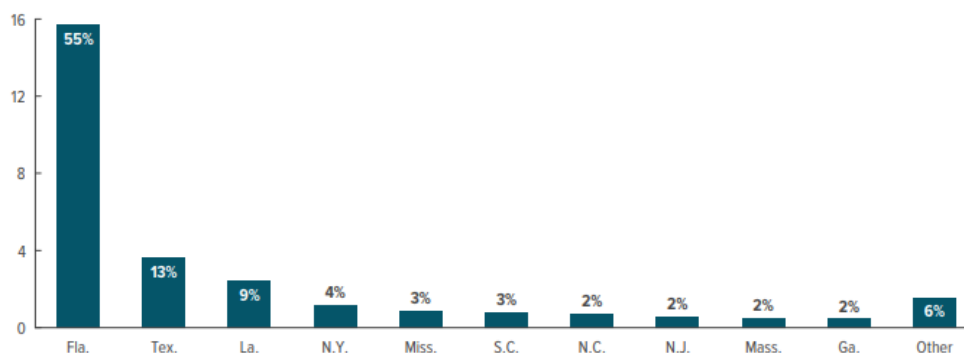
- *Create a new project and map*
- *Add layers to a new map*
- *Rename layers*
- *Examine a layer's attributes*
- *Perform definition queries*
- *Explore summary statistics*
- *Create scatter plots*
- *Select features by their attributes*
- *Select features by their locations*

Introduction

The Congressional Budget Office (CBO) estimated that hurricanes caused approximately \$28 billion in annual damage nationwide in 2016. Partly as a result of coastal development and climate change, the CBO projected that the cost of damages will increase by nearly 40 percent by 2075 (Plumer, 2017). Meanwhile, federal spending in response to these disasters is expected to increase from \$18 billion in 2016 to \$24 billion in 2075 (CBO, 2016). Moreover, the distribution of hurricane-related damage is not evenly distributed among coastal states, as states such as Florida alone account for more than half of the damage (CBO, 2016).

Expected Annual Hurricane Damage in CBO's Reference Case, by State

Billions of Dollars



Source: Congressional Budget Office, using data from Risk Management Solutions.

CBO's estimate of expected annual hurricane damage at present—the "reference case"—is based on estimates of the current frequency of hurricanes, current state-specific sea levels, and the current valuation of property exposed to hurricanes.

Percentages indicate each state's share of the total damage in CBO's reference case.

Dollar amounts are expressed in 2015 dollars.

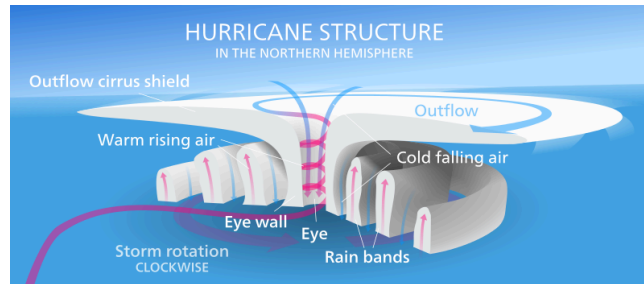
"Other" includes the following: Alabama, Connecticut, Delaware, Maine, Maryland, New Hampshire, Pennsylvania, Rhode Island, Vermont, Virginia, Washington, D.C., and West Virginia.

THE DISTRIBUTION OF ANNUAL HURRICANE-RELATED DAMAGE BY STATE BASED ON 2015 HURRICANE FREQUENCY, SEA LEVEL, AND PROPERTY VALUE INFORMATION. (SOURCE: US CONGRESSIONAL BUDGET OFFICE)

There are five major ways in which hurricanes inflict damage (Amadeo, 2019), they are:

(1) **high winds**, which are used to classify whether a hurricane is a category 1,2,3,4, or 5 event according to the Saffir-Simpson Hurricane Wind Scale (NOAA, 2012).

Category	Sustained Wind Speed
1	74-95 mph
2	96-110 mph
3	111-129 mph
4	130-156 mph
5	157 mph or higher



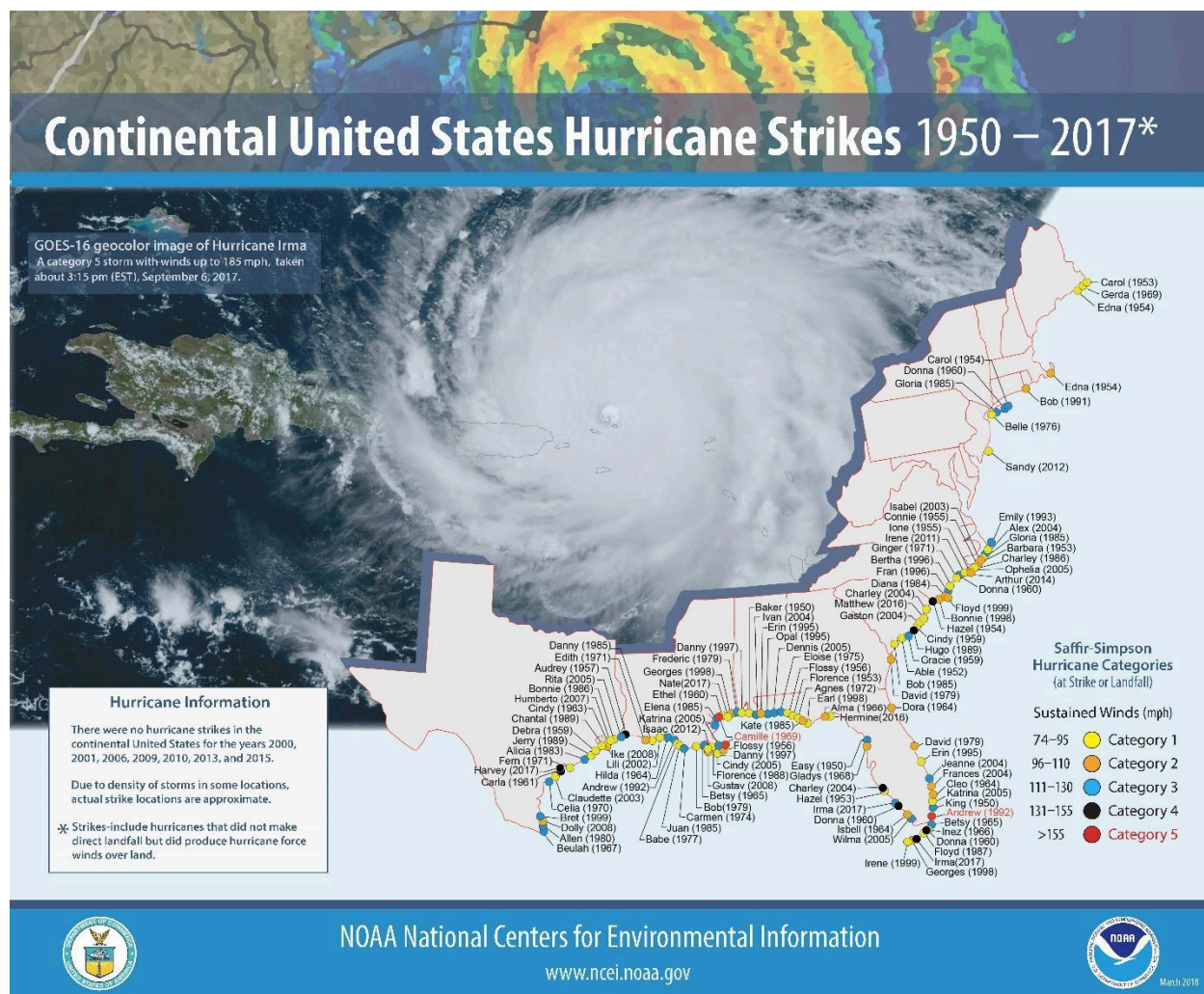
THE SAFFIR-SIMPSON HURRICANE WIND SCALE (LEFT). DIAGRAM OF A NORTHERN HEMISPHERE HURRICANE (RIGHT). (SOURCE: KELVINSONG, CC BY 3.0, VIA WIKIMEDIA COMMONS)

(2) **storm surge**, which is the rise in water above normal high tide.

(3) **extreme rainfall**, which may cause coastal floods, flash floods, river overtopping, and dam and levee failures, among other disasters that can lead to the loss of lives and properties.

(4) **location**. Given that hurricanes form over warm ocean waters, and the unprecedented rate of development in many coastal communities, the number of vulnerable properties and people has dramatically increased over time.

(5) **preparedness**. Timely and orderly evacuation can reduce the loss of lives associated with hurricanes. However, time and again, evacuees have decided to “ride out the storm” due to various reasons (e.g. care for pets, lack of shelter space, mobility issues, etc.), which may add to the hurricane’s death toll. Furthermore, even for those who evacuate, property damage caused by floodwater and wind contributes to the hurricane’s economic toll.



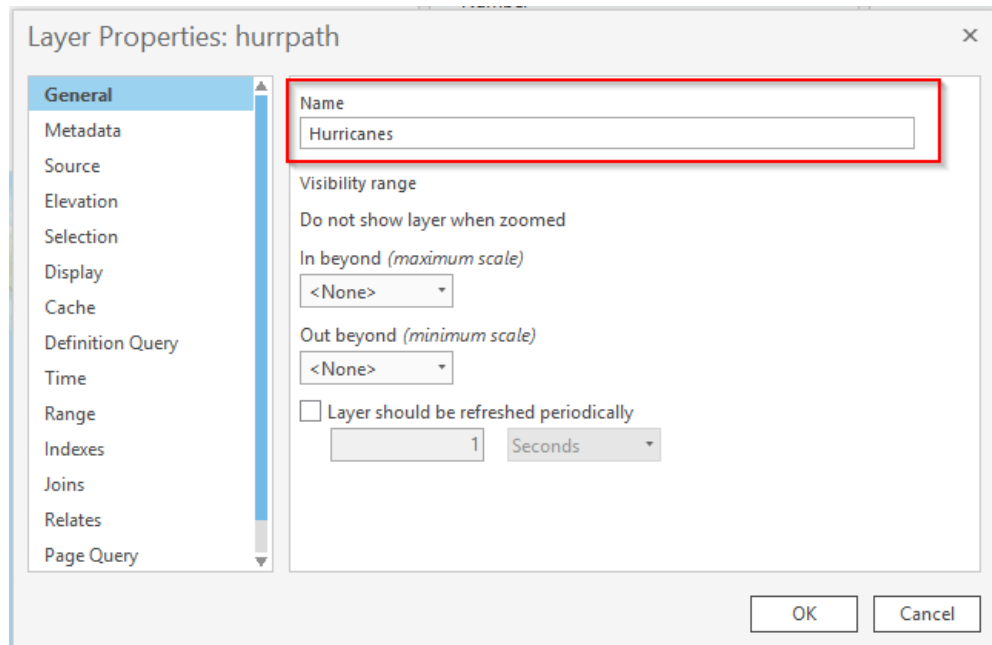
THE LOCATIONS WHERE HURRICANES MADE LANDFALL IN THE CONTINENTAL UNITED STATES BETWEEN 1950 AND 2017 CATEGORIZED BY SUSTAINED WIND SPEEDS. (SOURCE: NOAA)

In this exercise, you will analyze some of the most destructive hurricanes to hit the Atlantic coast of the United States between 1950 and 2000. In particular, you will examine the economic and human tolls associated with these past extreme weather events, and assess areas and populations that may be vulnerable to future hurricanes.

1. Start a project, rename layers, add and explore the data

- Start **ArcGIS Pro**.
- In the home screen, click **Map**  to start a new project. Give it a name (e.g. hurricanes) and save it in a safe location.
- Go to **Add data** (found under the **Map** tab), and add the **hurrrpath.shp** to your map. This layer shows the paths of hurricanes that have impacted the continental US between 1950 and 2000.

- d. In the Contents pane, right-click **hurrrpath**, and select **Properties**. In the General section of the Layer Properties window, change the Name from **hurrrpath** to **Hurricanes**. Click OK.



- e. Return to the Contents pane, right-click **Hurricanes** (formerly **hurrrpath**), and select **Attribute Table**  .

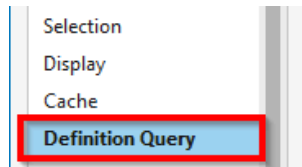
Q1. How many hurricane tracks are in the layer?

- f. Within the attribute table, right-click on the column name of **MAX_WINDSP** (which contains information on the maximum windspeed for each hurricane), and select **Sort Descending** to list the hurricane events sorted from highest windspeed to lowest windspeed.

Q2. What are the names of the two hurricane events with the strongest maximum windspeed?

2. Filter data with definition queries and compute summary statistics

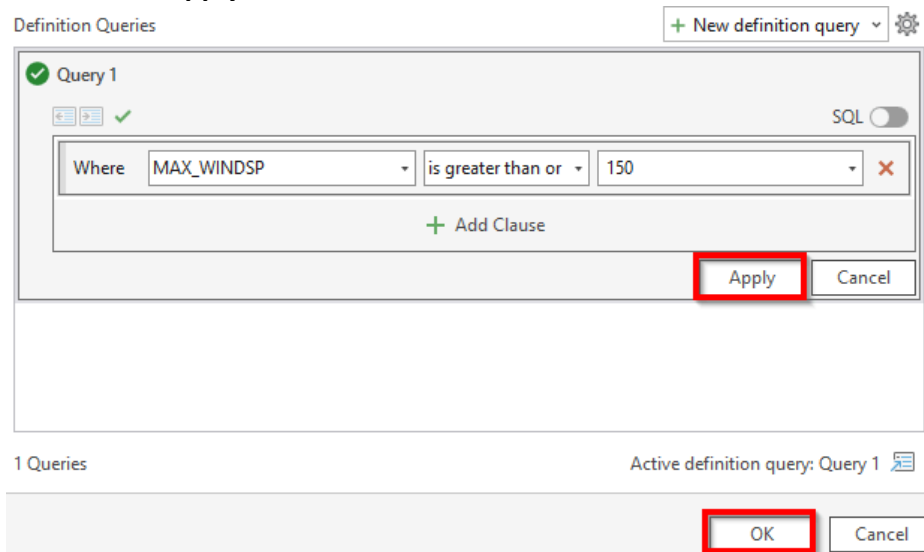
- a. Return to the Contents pane, right-click **Hurricanes**, and select **Properties**. Select **Definition Query** in the left margin.



There are no definition queries defined.

+ New definition query ▾

- b. In the Definition Query section of the Layer Properties window, click **New definition query**, and enter the following:
 - i. Select **MAX_WINDSP** for the first dropdown.
 - ii. Select **is greater than or equal to** for the second dropdown.
 - iii. Enter **150** in the third box.
 - iv. Click **Apply**. Click **OK**.



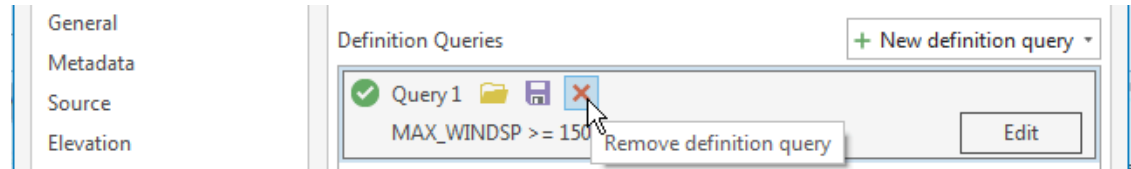
- c. Return to the Contents pane, right-click **Hurricanes**, and select **Attribute Table**.

Q3. How many hurricane tracks are left in the layer after the definition query?

- d. Within the attribute table, right-click on the column name **PATH_LEN** (which contains information on the length of the track for each hurricane), and select **Statistics**.

Q4. What is the mean (average) track length (in kilometers) of the hurricanes that are currently in the attribute table (round answer to two decimal places)?

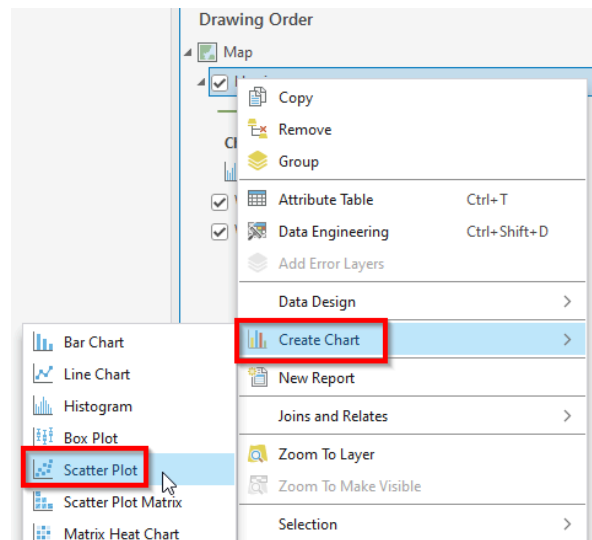
- e. To clear the definition query filter, return to the Contents pane, right-click **Hurricanes**, and select **Properties**. In the Definition Query section of the Layer Properties window, hover your mouse over the query we previously created, and select the **red X**, click **Yes**. Click **OK**.



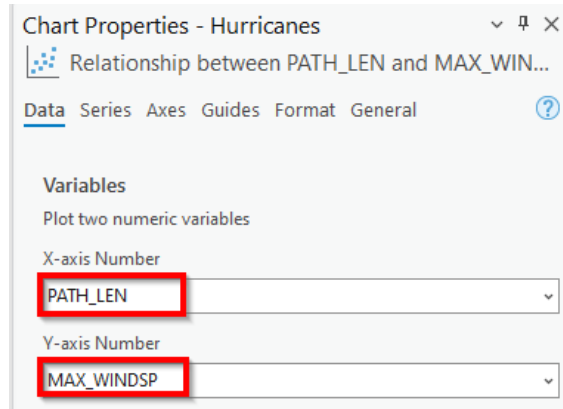
- f. Return to the Contents pane, right-click **Hurricanes**, and select **Attribute Table**. You should now see a lot more hurricane tracks in the table. This is because the Definition Query filters out a subset of features that meet a specific criterion (i.e. maximum windspeed greater than 150 mph), and once we erased the query, we can see all of the layer's features again.

3. Explore relationships with scatterplots

- a. Right-click the **Hurricanes** layer in the Contents pane, and select **Create Chart>Scatter Plot**.



- b. In the Chart Properties window, explore the relationship between hurricane track length (PATH_LEN) and maximum wind speed (MAX_WINDSP) by selecting **PATH_LEN** for the X-axis Number and **MAX_WINDSP** for the Y-axis Number.



Q5. What is the R2 value (round answer to two decimal places)?

Tip: The R2 value tells us the percentage of the variation in the hurricanes' maximum wind speed that is explainable by their track lengths.

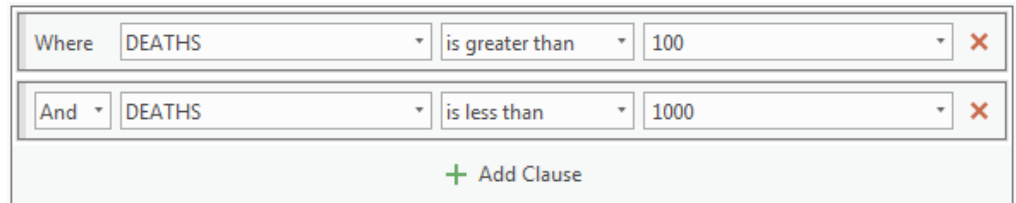
4. Select features by attributes

- a. Go to **Add data** (found under the **Map** tab), and add the **top10dam.shp** to your map. This layer shows the paths of the 10 hurricanes with the most monetary and property damage between 1950 and 2000.
- b. To see how many of the hurricanes in the top10dam layer have between 100 and

1,000 deaths, click on **Select By Attributes**  (found under the **Map** tab). In the Select Layer By Attribute window, enter the following:

- i. Input Rows: **top10dam**
- ii. Selection Type: **New selection**
- iii. Expression:
 1. Select **DEATHS** for the first dropdown, select **is greater than** for the second dropdown, and type in **100** for the third box.

2. Click **Add Clause**. Now select **And** for the first dropdown, select **DEATHS** for the second dropdown, select **is less than** for the third dropdown, and type in **1000** for the fourth box.



Where DEATHS is greater than 100

And DEATHS is less than 1000

+ Add Clause

iv. Click **OK**.

- c. Return to the Contents pane, right-click **top10dam**, and select **Attribute Table**.

Q6. How many records/features in the top10dam layer's attribute table are selected? This is the number of hurricanes in the layer that have caused between 100 and 1,000 casualties.

- d. Repeat and modify steps b and c in this section to:

- Remove the previous conditions by clicking the **Remove** icon  **Remove** in the Select By Attribute window.
- Create a new expression to find the number of hurricanes within the top10dam layer that took place between the year 1900 and the year 1950. Be sure to include the years 1900 and 1950 in your query (i.e. YEAR is greater than or equal to 1900 AND YEAR is less than or equal to 1950).

Q7. How many records/features in the top10dam layer's attribute table are now selected? This is the number of highly destructive hurricanes that took place between 1900 and 1950.

5. Select features by locations

- a. Clear any currently selected feature by clicking on **Clear**  (found under the **Map** tab). This will ensure that any selection that we make in the following steps is based on all features within the layer, rather than just a subset of previously selected features.
- b. Go to **Add data** (found under the **Map** tab), and add the **Florida.shp** to your map. This layer shows all the counties within Florida.
- c. To see how many of the top 10 hurricanes with the most monetary and property damages made landfall in Florida, click **Select By Location**  within the **Map** tab. In the Select By Location window, enter the following:

- i. Input Features: **top10dam**
- ii. Relationship: **Intersect**
- iii. Selecting Features: **Florida**
- iv. Selection type: **New selection**
- v. Click **OK**

- d. Return to the Contents pane, right-click **top10dam**, and select **Attribute Table**.

Q8. How many records/features in the **top10dam** layer's attribute table are selected? This is the number of the 10 deadliest hurricanes that made landfall in Florida.

- e. Repeat and modify steps c and d to find the number of hurricanes within the **Hurricanes** layer (**not the top10dam layer**) that made landfall in Florida.

Q9. How many records/features in the **Hurricane** layer's attribute table are selected? This is the number of hurricanes (regardless of toll or magnitude) that made landfall in Florida.

Works Cited

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Plumer, Brad. "Why Hurricane Irma Could Hurt, a Lot: Much Lies in Harm's Way." *The New York Times*, 7 Sept. 2017, www.nytimes.com/2017/09/07/climate/florida-hurricane-irma-damage.html. Accessed 25 June 2019.

Chapter 3: Want to fly a drone? Know B4UFLY!

In this exercise, you will:

- *Geocode a list of addresses*
- *Rematch problematic addresses*
- *Perform buffer analysis*
- *Select features by their locations*
- *Share information as a map package and a layer package*

Introduction

Unmanned Aircraft Systems (UAS) or drones are increasingly used for a variety of applications including agriculture, building inspection, environmental assessment, firefighting, search and rescue, and surveying and mapping. Generally, drone operators are discouraged from flying near airports because of potential interference with air traffic (Federal Aviation Administration, 2019). In fact, nefarious drone operators have caused great damage to property (Navarro and Levin, 2018) and chaos in congested airspaces (Pérez-Peña, 2018).



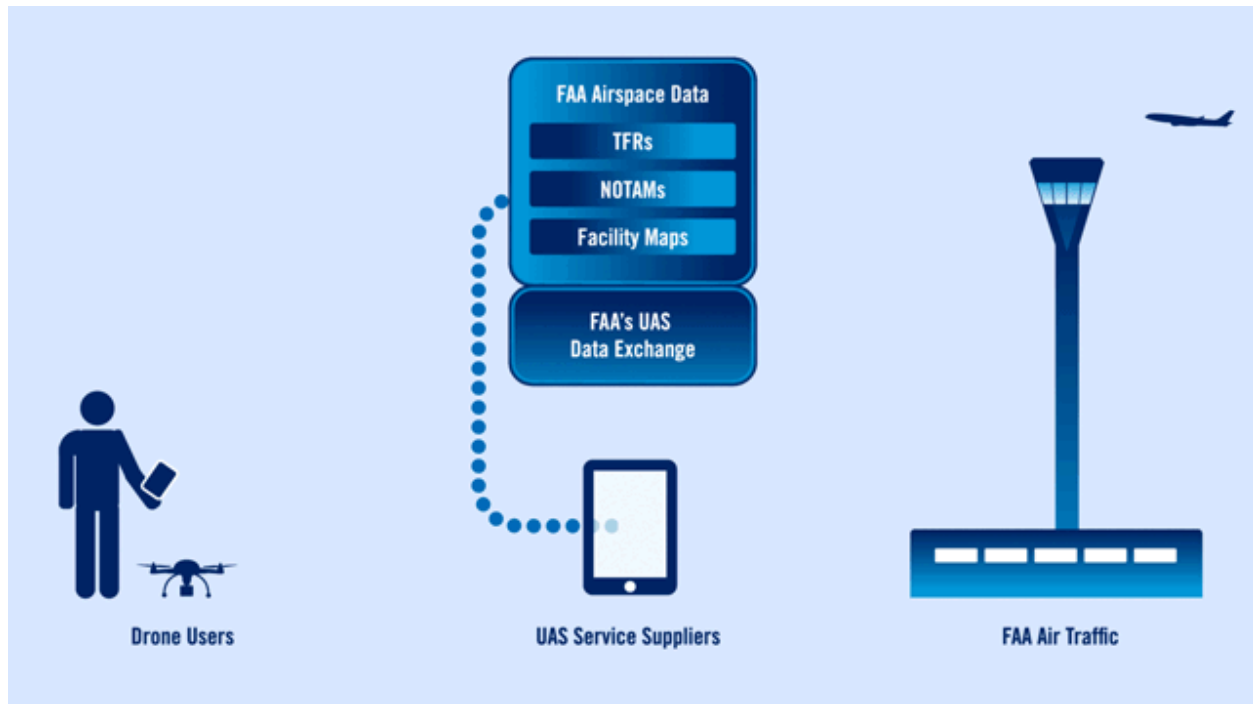
A DRONES PROHIBITED SIGN AT THE BODIE ISLAND LIGHTHOUSE, WHICH WAS BUILT IN 1871 ON THE OUTER BANKS OF NORTH CAROLINA. (SOURCE: MARTIN KRAFT, CC BY-SA 4.0, VIA WIKIMEDIA COMMONS)

While there is no uniform international regulation governing drone use near airports, governments around the world are beginning to push for the exclusion of drone operations near airports, as well as other anti-drone tactics ranging from signal jammers to eagles that are trained to take down nefarious drones (Pérez-Peña, 2018). Since the Federal Aviation Administration (FAA) Reauthorization Bill of 2018 repealed the Special Rule for Model Aircraft (i.e. Section 336 of the 2012 FAA Modernization and Reform Act), recreational drone operators in the United States are no longer prohibited from operating within five miles of an airport as long as they secure prior permission from the FAA through [LAANC](#) (Low Altitude Authorization and Notification Capability). Nonetheless, as stated before, “drone operators should avoid flying near airports because of busy air traffic in those areas (Federal Aviation Administration, 2019).”



AN ATTACK DRONE WITH AN ATTACHED NET, PART OF THE COUNTER-UNMANNED AERIAL SYSTEM DEVELOPED BY A TEAM FROM WRIGHT-PATTERSON AIR FORCE BASE, OHIO. (SOURCE: WESLEY FARNSWORTH VIA U.S. DEPARTMENT OF DEFENSE)

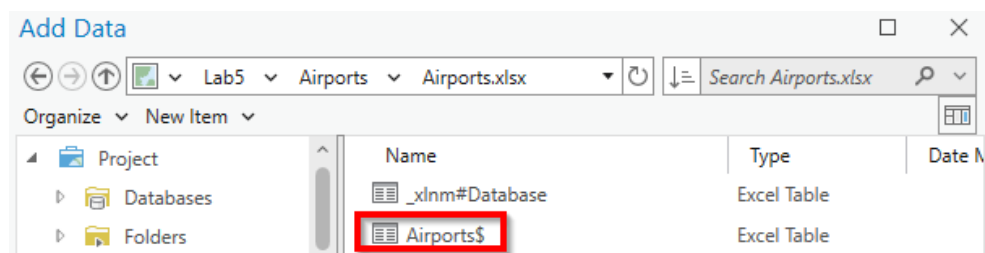
In this exercise, you will first map all the airports, heliports, glider-ports, and other airfields in San Diego County. Secondly, you will analyze how many of the colleges in San Diego County are within five miles of an airport, and may need to secure permission from the FAA before operating in the national airspace.



LAANC IN A NUTSHELL: 1) (LEFT) DRONE USERS/OPERATORS SUPPLY OPERATIONAL PARAMETERS (E.G. OPERATION DATE/TIME, FLIGHT ALTITUDE, AIRCRAFT MODEL, ETC) TO A UAS SERVICE SUPPLIER, 2) (MIDDLE) THE UAS SERVICE SUPPLIER CHECKS THE REQUEST AGAINST VARIOUS FAA AIRSPACE DATA FROM THE FAA UAS DATA EXCHANGE, 3) (RIGHT) THE DRONE USER/OPERATOR RECEIVES AUTHORIZATION OR DENIAL IN NEAR-REAL TIME. (SOURCE: U.S. FEDERAL AVIATION ADMINISTRATION)

1. Geocode airports and airfields in San Diego County

- Start **ArcGIS Pro**.
- In the home screen, click **Map** to start a new project. Give it a name (e.g. drones) and save it in a safe location.
- Go to **Add data** (found under the **Map** tab), and add the **Airport.xlsx** file to ArcGIS Pro by double clicking on the file in the Add Data window, and add in the **Airports\$** worksheet.



- Right-click the **Airports\$** worksheet in the Contents pane and select **Open** to view the contents of the worksheet.

Q1. How many records are in the worksheet?

****Tip: Remember that records refer to rows in a table and fields refer to columns in a table.****

Field: Add Calculate		Selection: Select By Attributes Zoom To		Switch Clear Delete Copy		Rows: Insert One Field	
	Name	Type	Owner	Address	City	State	Zip
1	BORREGO AIR RANCH	AIRPORT	BORREGO AIR RANCH	2580 STINSON RD.	BORREGO SPRINGS	CA	92004
2	BORREGO VALLEY	AIRPORT	COUNTY OF SAN DIEGO	1820 Palm Canyon Dr	Borrego Springs	CA	92004
3	JOHN NICHOL'S FIELD	AIRPORT	TACTICAL AIR OPERATI...	13531 Otay Lakes Rd	Jamul	CA	91935

- Geocode the data in the **Airports\$** worksheet by right-clicking on it in the Contents pane and selecting **Geocode Table**.
- In the Geocode Table pane, click **Start**. This pane which walks users through a GIS workflow (i.e. Geocoding) in a step-by-step fashion is an example of a *Task* in ArcGIS Pro.
- For step one, select the **Esri World Geocoder** for the Input Locator. Click **Next**.

Geocode Table

Step One: What locator are you using?



Input Locator

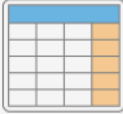
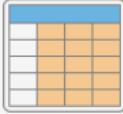
Esri World Geocoder

****Tip: If the Esri World Geocoder is not an option, be sure that you are signed into ArcGIS Pro with a paid-ArcGIS Online account with sufficient ArcGIS Online credits****

- h. For step two, make sure that the Input Table is **Airport\$**, and specify that the data we want to geocode is in **More than one field** since we will be using address, city, state, and zip code information from the input table to map our airports. Click **Next**.

Geocode Table ? v ≡ ×

Step Two: About your table

Input Table
Airports\$

How is your data structured? Is it in one field containing all components concatenated into a single field? Is it multiple fields, where different components are in separate fields? Or is there a separate country field but with other components concatenated into a single field?

More than one field v

Don't know?
Open the table.


Attribute table

- i. For step three, click the **Don't know? Open the table** icon  at the bottom of the screen to open the attribute table (if necessary) and review the table to answer the following questions:

Q2. What is the name of the field/column in the attribute table that stores the address or place data?

Q3. What is the name of the field/column in the attribute table that stores the city data?

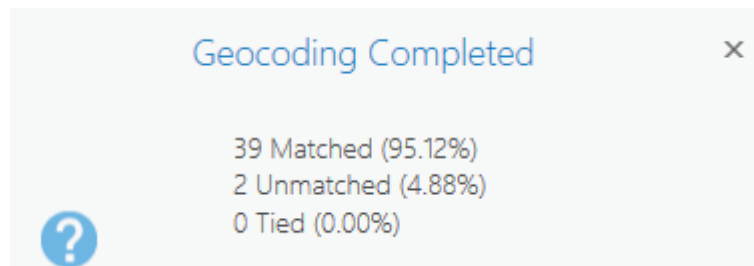
Q4. What is the name of the field/column in the attribute table that stores the state data?

Q5. What is the name of the field/column in the attribute table that stores the zip code data?

Using your answers above, ensure that the appropriate options are selected for the **Address or Place, City, State, and Zip** dropdown in the Geocode Table pane. Click **Next**.

Tip: Even though the names of the fields/columns in this example correspond nearly exactly with the type of address information that they store, this is not always the case. In most cases, you will need to manually tell ArcGIS Pro which fields contain specific address information, because ArcGIS Pro cannot identify which field in the input table contains zip code data, for example.

- j. For step four, name the output **AirportPoints** (***Tip: You must use the browse icon  to change the output name***). Make sure that the checkbox for **Add output to map** after completion is checked, select **Address Location** for the Preferred Location Type, and select **All** for Output Fields. Click **Next**.
- k. For Optional Step Five, check **United States**. Click **Next**.
- l. For Optional Step Six, check **Address**. Click **Finish**.
- m. In the Geocode Table pane, you are now presented with a summary of the options that you have chosen in the previous six steps. Click **Run**.
- n. In the Geocoding Completed result window, you should see that 39 addresses were matched with 2 addresses unmatched. However, there may be some addresses that were matched erroneously, so it is always a good practice to check where the matched addresses are. We will click **No** in the rematch process window for now but we will come back to it later.



- o. **Close** the Geocode Table pane.

2. Identify and fix problematic address matches

- a. Since we are mapping airports, glider-ports, and helipads, these features should be relatively easy to spot in an aerial photo or satellite photo.

We will check our geocoded airport points by going to the **Map** tab, and select **Basemap>Imagery**. Keep in mind that many of the points will not be directly on the airstrip or helipad since these facilities (i.e. runways) don't always have an exact address. Instead, the addresses we used may be that of a facility near the airstrip or helipad.

- b. Open the AirportPoints layer's **attribute table**.

- c. You can double-click on the **left margin** (see below) of a row in the attribute table to center the map on the point tied to that row, and then zoom in to get a closer look at each point.

	ObjectID *	Loc_name	Shape *	Status	Score	Match_type	Match_addr
1	1	World	Point	M	100	A	354 Santa Fe Dr, Encin...
2	2	World	Point	M	100	A	2580 Stinson Rd, Borre...
3	3	World	Point	M	100	A	1820 Palm Canyon Dr,...
4	4	World	Point	M	100	A	13531 Otay Lakes Rd, J...

- d. There are 4 problematic addresses in our table. Their names as listed in the USER_Name column are:
- i. FALLBROOK COMMUNITY AIRPORT*
 - ii. PAUMA VALLEY AIR PARK*
 - iii. JACUMBA
 - iv. SCE SONGS MESA

The addresses for FALLBROOK COMMUNITY AIRPORT and PAUMA VALLEY AIR PARK were entered incorrectly in the original address table, which produced the 2 unmatched addresses (*) in our results window. FALLBROOK COMMUNITY AIRPORT and PAUMA VALLEY AIR PARK should have a value of “U” in the Status column, which stands for Unmatched. We will use the rematch address tool in the following steps to correct their addresses.

While JACUMBA and SCE SONGS MESA were matched (as indicated by the value of “M” under the Status column for their rows/records), their locations are not in the correct locations. We will use the rematch address tool in the following steps to manually fix their placement.

Clear any selection in your attribute table by going to the **Map** tab, and clicking on the **Clear** icon .

- e. To access the rematch address tool, right-click on the name of the **AirportPoints** layer (not the Airports\$ table) in the Contents pane. Go to **Data**, and select **Rematch Addresses**.
- f. In the Rematch Addresses pane, click on **Unmatched** (if needed), and you should see that there is a total of 2 unmatched address records. You will correct their addresses one at a time.

Rematch Addresses

Unmatched Matched

Address or Place	22155 S Messian Rd
Address2	<Null>
Address3	<Null>
Neighborhood	<Null>
City	<Null>
Subregion	<Null>
State	<Null>
ZIP	<Null>
ZIP4	<Null>
Country	<Null>

☒ Auto Apply

1 of 2

- g. One of the unmatched addresses was entered as “22155 S Messian Rd” in the original Airports\$ table. Change the address to **2155 Mission Rd** in the Address or Place field, enter **Fallbrook** in the City field, and enter **92028** as the Zip. Make sure the **Auto Apply** box is checked.
- h. As you type in the corrected information, you should see the list of candidates is automatically updated in the lower half of the pane. If the list is not automatically updated, make sure that the Auto Apply box is checked, or try closing the rematch addresses window and reopening it (see step e above).

☒ Display cached match candidates

Address	Type	Score
A 2155 S Mission Rd, Fallbrook, California, 92028	PointAddress	99.43

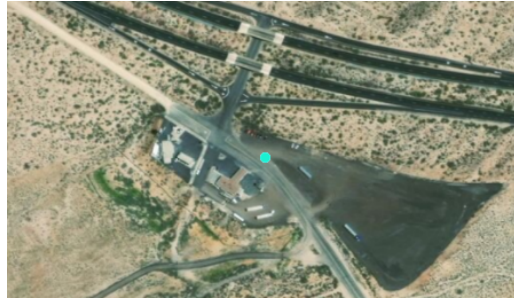
Click the **first candidate** in the list, click the **green checkmark**  to rematch the point

based on the corrected address information, and then click the **disk icon**  to save your edits.

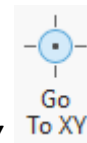
- i. Repeat step 2.f. to 2.h. to correct the remaining unmatched address of “133129 ColeG Ln”. Change its Address or Place information to **33129 Cole Grade Rd**, enter **Pauma Valley** for the City, and **92061** for the Zip. Remember to click the **green checkmark** to rematch your point, and then click the **disk icon** to save your edits. Do not close the Rematch Addresses pane.
- j. Open the AirportPoints layer’s **attribute table** if necessary, and review the Status column. “M” should be the only value found in the column (i.e. no “U”) since we just fixed the 2 unmatched addresses.

While we have fixed the 2 unmatched addresses, we still have 2 addresses that were initially placed in the wrong locations by ArcGIS Pro.

- k. Find **JACUMBA** under the USER_Name column (**Tip: Depending on where you saved your file, this column may also be called Name, between the Country and Type columns**) in the AirportPoints layer's attribute table. Double-click on the row's **left margin** to select the record and zoom into it on the map. You should see that there are no airports or helipads anywhere near its currently mapped location.



- l. Since JACUMBA Airport is in a relatively remote location without an easily identifiable address, we will use its latitude and longitude to locate it, then manually click on a location on the map to match it to the correct location.

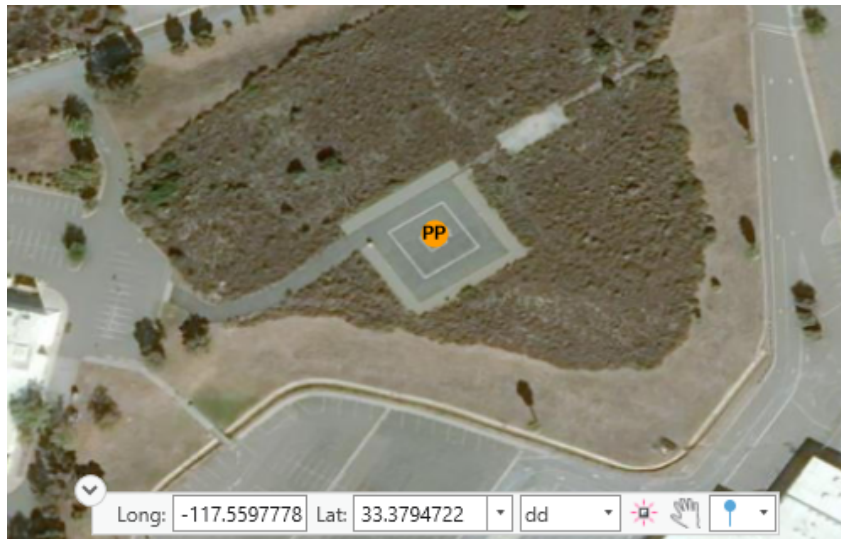


In the main menu, click on the **Map** tab, then select **Go To XY**.

- m. In the toolbar, enter **-116.1655389** for Long, **32.6158806** for Lat, and leave dd (decimal degree) selected. Click on the **Pan** tool to center your screen on the location, and click on the **Flash** tool if you wish to see the precise location on your screen again. Zoom in if necessary. This is the correct location of the JACUMBA Airport.



- n. Once you have remembered the approximate correct location of the runway for JACUMBA Airport, click the **Pick from Map** tool  in the lower half of the Rematch Addresses pane, then click the location identified by the Pan and Flash tool in the previous step. This will automatically reposition the point for JACUMBA to the clicked location. Finally, click the **green checkmark** to accept the repositioned point location, and click the **disk icon** to save your edits.
- o. Repeat steps 2.k. to 2.n. to find **SCE SONGS MESA** under the USER_Name column in the AirportPoints layer's attribute table. It is a helipad. Its current location has been incorrectly placed by ArcGIS Pro. Using the **Go To XY** tool, navigate to its correct longitude of **-117.5597778** and latitude of **33.3794722**. Double-click the **Pick from Map** tool and reposition the point to the correct location. Click the **green checkmark** to accept the repositioned point location, and the **disk icon** to save your edits. **Close** the Rematch Addresses pane.



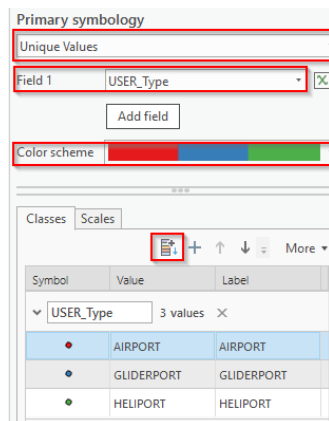
- p. Clear any selected records by using the **Clear Selection** tool  in your attribute table before continuing.

Tip: If you are still seeing a blue circle with the letter A on your map, you need to close the Rematch Address pane.

3. Symbolize facilities by category or type

- a. To symbolize our AirportPoints layer based on the type of facility being represented (e.g. airport, helipad, glider-port), select the **AirportPoints** layer in the Contents pane, and go to the **Feature Layer** tab in the main menu. Select **Symbolology** .
- b. In the symbology pane, select **Unique Values** from the dropdown list under Primary symbology. Select **USER_Type** (***Tip: This may also be called Type depending on where you saved your files, be sure to choose the Type field that is between the Name and Owner fields*) for Field 1 since that is the column/field we used to distinguish between

different types of facilities. **Change the Color scheme** to something that readily distinguishes between the three types of facilities. Click the Add All Values icon.



4. Analyze features with geoprocessing tools

- a. Since the Federal Aviation Administration generally discourages drone operators from flying near airports or other air transportation facilities (Federal Aviation Administration, 2019), we will create 5-mile buffers around all the points in the AirportPoints layer.

Go to the **Analysis** tab, click on **Tools**, and search for **Buffer** in the Geoprocessing pane. Select **Buffer (Analysis Tool)** from the results.

- b. In the Buffer tool pane, enter the following:
 - i. Input Features: **AirportPoints**
 - ii. Output Feature Class: **NoFly** (**Tip: Replace the characters after the last backslash (\) with NoFly. Do not delete everything and just type in NoFly**)
 - iii. Distance: **5 US Survey Miles**
 - iv. Method: **Planar**
 - v. Dissolve Type: **No Dissolve**
 - vi. Click **Run**
- c. To see how many colleges are within the 5-mile radius of an airport (or heliport or glider-port), **add** the **Colleges.shp** layer (from the files you download for this exercise) to your map.
- d. Look for Palomar College in your map. You can go to the **Map** tab, click on **Locate**, and enter Palomar College's Address: **1140 West Mission Road, CA 92069**.

Q6. Is Palomar College within a 5-mile radius of an airport (or heliport or glider-port)?

- e. Click on each **buffer circle** to see its attribute information in a pop-up window. Since the buffers were created by drawing circles with a 5-mile radius around each airport facility, each buffer circle's pop-up window will give you the name of the facility (i.e. Name or USER_Name) that generated the specific buffer.

****Tip: If no pop-up is showing, click the word Explore in the main menu, and ensure that Topmost Layer is checked****

Q7. Give the names of the two facilities that generated the 2 buffers that cover Palomar College in San Marcos, CA.

I:

II:

- f. To find out how many colleges in San Diego County are within a 5-mile radius of an airport facility (excluding military facilities since those were not in our Airports\$ table), go to the **Map** tab, **clear**  any selected features, and click **Select By Location** .
- g. In the Select Layer By Location pane, enter the following:
 - i. Input Feature Layer: **Colleges**
 - ii. Relationship: **Completely within**
 - iii. Selecting Features: **NoFly**
 - iv. Click **OK**
- h. Open the **Colleges** layer's attribute table to see how many features were selected at the bottom of the table.

Q8. How many colleges are within a 5-mile radius of an airport or related facilities?

Q9. What percentage of the region's colleges are within a 5-mile radius of an airport or related facilities?

- i. **Clear**  **Clear** all selected records in your attribute table.

5. Share your work with others

- a. To package and share your data with others, you must first select the name of the layer in the Contents pane. For this exercise, select the **AirportPoints** layer in the Contents pane, go to the **Share** tab and select **Layer** .
- b. In the Sharing As A Layer Package pane, enter the following:

- i. Select **Save package to file**
- ii. Name: (any name you like, but carefully note where the file is being saved)
- iii. Summary: (any summary you like)
- iv. Tags: (any keywords you like)
- v. Click **Analyze**
- vi. (If no errors or warnings are found) Click **Package**

Layer package is an excellent way to package and share individual GIS layers with others, because it basically packs up all the data, tables, and symbology for your layer into one file, so you can easily move that file between locations or email the file to others.

To open a layer package in ArcGIS Pro, go to the **Map** tab, click on **Add Data** and add in the layer package (.lpx).

- c. In order to save your entire map and all of its layers so you can share it with others, go to the **Share** tab and select **Map**  .
- d. In the Share Map As A Map Package pane, enter the following:
 - i. Select **Save package to file**
 - ii. Name: (any name you like, but carefully note where the file is being saved)
 - iii. Summary: (any summary you like)
 - iv. Tags: (any keywords you like)
 - v. Click **Analyze**
 - vi. (If no errors or warnings are found) Click **Package**

The map package packages all the layers (except standalone tables) and their properties (symbology information) into one file, so you can easily move the entire map and its data to a different location, or share your map and data with others as a single file.

To open the map package in ArcGIS Pro, go to the **Insert** tab, click on **Import Map**, and add the map package (.mpkx).

Works Cited

“FAA Reauthorization Bill Establishes New Conditions for Recreational Use of Drones.” *News and Updates*, Federal Aviation Administration, 12 Oct. 2018, www.faa.gov/news/updates/?newsId=91844.

“Flying Near Airports.” *Airspace Restrictions*, Federal Aviation Administration, 31 May. 2019, www.faa.gov/uas/recreational_fliers/where_can_i_fly/airspace_restrictions/flying_near_airports/.

Pérez-Peña, Richard. “Gatwick Airport Drone: Lots of Second-Guessing, but Not Many Answers.” *The New York Times*, 27 Dec. 2018, www.nytimes.com/2018/12/27/world/europe/gatwick-airport-drone.html.

Chapter 4: Analyzing wildfire extent and response in California

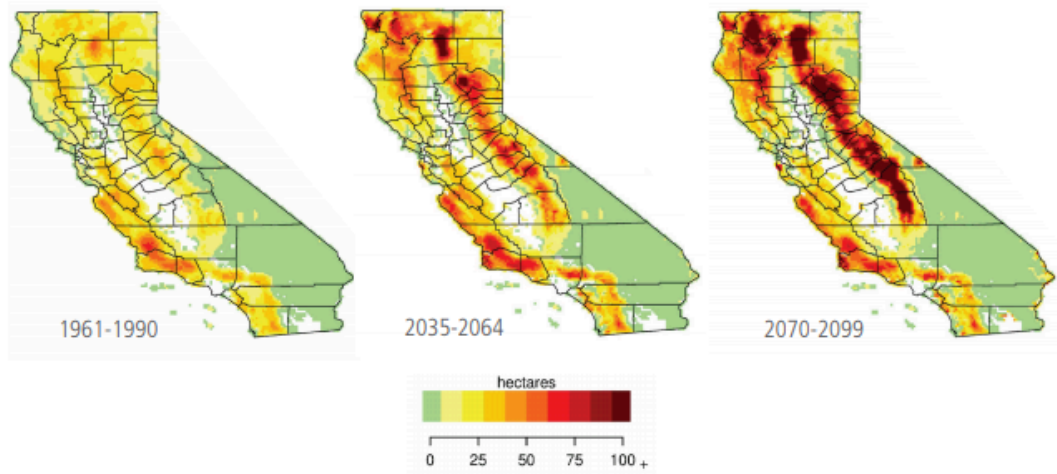
In this exercise, you will:

- *Create a new project and map*
- *Add layers to a new map*
- *Examine a layer's attributes*
- *Create a new geodatabase*
- *Import data into your geodatabase*
- *Summarize data in your layers or tables*
- *Join tabular data with vector layers*
- *Change the symbology of features*

Introduction

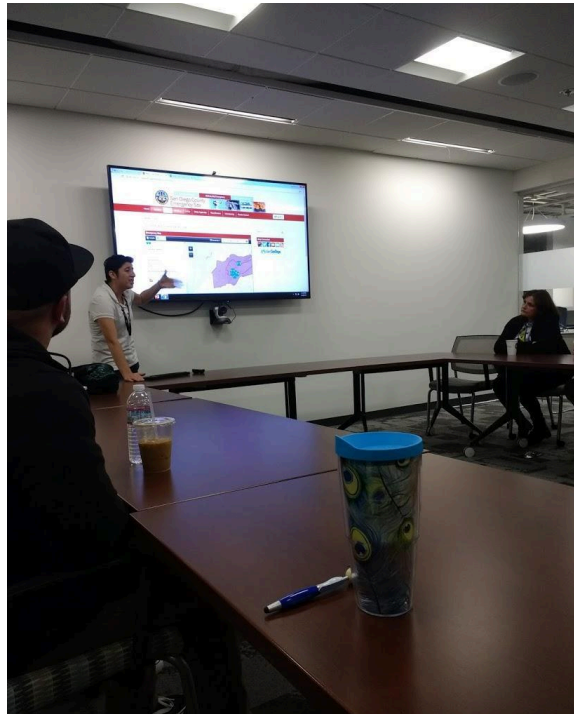
California is no stranger to wildfires. However, what is novel for the state is the intensity and length of recent fire seasons. Unlike the “typical” fire season of the past which lasts from June to October, a year-long fire season with more frequent and more destructive wildfires is becoming the new normal due to climate change (Adams, 2008; Ortiz, 2018; Thorne et al., 2018).

Specifically, a lethal mix of (1) prolonged droughts and record-breaking heat waves which left vegetation “tinder dry”, (2) an epidemic of bark beetles that caused the death of millions of trees, and (3) the usual dry and gusty Santa Ana Winds have combined to create more and more destructive and expensive wildfire seasons (Ortiz, 2018; Adolphe, 2018). As an example, the 2007 wildfire season killed 17 people, injured 140 firefighters and an unknown number of civilians, destroyed 3,069 structures, and cost property damages in excess of \$2 billion (Mitchell et al., 2007; Girnius et al., 2008). To make matters worse, as California’s population continues to grow, people and developers are increasingly building in areas of high fire risk, which increases the population’s vulnerability to wildfire hazards (Adolphe, 2018; Ortiz, 2018).



Modeled area burned by wildfires for the current time period (modeled as 1961-1990), for mid-century (2035-2064), and for late century (2070-2099). (Source: State of California's Fourth Climate Change Assessment)

Of the 6043 fires that occurred in California in 2007 alone, an estimated 1,520,362 acres were affected (CalFire, 2018). CAL FIRE responded to about 60% of the fire incidents, while Federal Firefighting Agencies and Local Governmental Agencies responded to about 32% and 8% of the fires, respectively (CalFire, 2018).



Smoke from the Lilac Fire on December 7, 2017 in Oceanside, CA (Left). Staff from 2-1-1 using ArcGIS Online maps provided by the San Diego County Office of Emergency Services to assist evacuees and residents (Right).



WILDFIRES IN NORTHERN CALIFORNIA TRIGGERED BY THUNDERSTORMS AS OF AUGUST 2008. SATELLITE IMAGE WAS CAPTURED WITH THE MODERATE RESOLUTION IMAGING SPECTRORADIOMETER (MODIS) ON NASA'S TERRA SATELLITE. (SOURCE: NASA)

In this exercise, you will explore a small subset of the wildfires that occurred during the 2007 wildfire season in California. You will not only see information such as the alarm date, and containment date, but also the fire's name and the agency that handled the primary response to the fire.

1. Start a project, add and explore the data

- a. Start **ArcGIS Pro**.
- b. In the home screen, click **Map** to start a new project. Give it a name (e.g. Fires) and save it in a safe location.
- c. We will add a layer that shows the extent of wildfires in California in 2007. Go to **Add data** (found under the **Map** tab), and add the **CAFires2007.shp** to your map.
- d. Right-click on the **CAFires2007 layer** in the Contents pane, and select **Attribute Table** to examine the attribute table of the CAFires2007.shp file.

Q1. How many fields are present in the CAFires2007 layer?

Q2. What other fields may be useful for researchers in a layer depicting the size of wildfire extents?

2. Create a geodatabase and import data

- a. Go to the Catalog pane (or go to **View>Catalog Pane** if the Catalog pane is not already open), and expand the **Databases** icon . Note that a database with the same name (e.g. Fires.gdb) as your project has already been automatically created for you when you started this new project.
- b. We will create our own geodatabase to store the data that we will be working with in this exercise.

Right-click on the **Databases** icon within the Catalog tab, and select **New File Geodatabase**. Navigate to the location where you want to save this new database. Name the new database **Wildfires**. You should now see Wildfires.gdb under the Database icon.

- c. We will now import the CAFires2007 shapefile and a standalone table into the newly created Wildfires geodatabase.

Right-click on the **Wildfires.gdb** within the Catalog pane, and select **Import>Feature Class(es)...**

- i. Select **CA Fires2007** as the Input Features
- ii. Make sure that **Wildfires.gdb** is selected as the Output Location
- iii. Click **Run**

You should now see a new feature class called CAFires2007 in the Wildfires.gdb geodatabase when you expand it in the Catalog pane.

- d. Right-click on the existing **CA Fires2007** layer in the Contents pane and select **Remove** to take it off the map.
- e. **Add** the new **CA Fires2007** layer that was just created in step c to the map, by dragging it from the Catalog pane to anywhere in the map.
- f. Next, we will import the standalone table called **2007FireTable** into the Wildfires.gdb. The process is the same as the previous step, except you will select the **Import>Table(s)...** option since we are now working with tables.
- g. Once the Table to Geodatabase dialog opens, enter the following:
 - i. Input Rows: (browse  to the lab data folder and locate) **2007FireTable.dbf**.
 - ii. Output Location: **Wildfires.gdb**
 - iii. Click **Run**

You should now see a new T2007FireTable item when you **refresh** the Wildfires.gdb geodatabase in the Catalog pane (i.e. right click Wildfires.gdb and select refresh).

****Tip:** ArcGIS Pro automatically added the T to the start of the filename, because a filename cannot start with a number.**

- h. **Add** the new **T2007FireTable** that was just created in step g to the map, by dragging it from the Catalog pane to anywhere in the map.
- i. Open the **T2007FireTable** by right-clicking on it in the Contents pane, and selecting **Open** to see the information stored in the table.

Q3. How many fields are present in the T2007FireTable?

3. Summarize attribute data

- a. To see which agency handled the most wildfires in 2007 within California, right-click on the **AGENCY** field name within the T2007FireTable, and select **Summarize**. Enter the following:
 - i. Input Table: **T2007FireTable**
 - ii. Output Table: **FireInfo2007_Statistics**
 - iii. Statistics Field(s): select **AGENCY** for Field, and **Count** for Statistic Type
 - iv. Case field: **AGENCY**
 - v. Click **OK**

Summary Statistics

Input Table: T2007FireTable

Output Table: FireInfo2007_Statistics

Statistics Fields

Field	Statistic Type
AGENCY	Count

Case Field: AGENCY

You should now see the FireInfo2007_Statistics table in the Contents pane.

Q4. Open the FireInfo2007_Statistics table, which AGENCY handled the most fires in California in 2007?

Q5. How many fires did the AGENCY from question 4 handle?

4. Join a table to a polygon feature class

- a. We will now join the CAFires2007 feature class with the T2007FireTable table in order to bring the tabular information from the table to the feature class. This will allow us to symbolize the polygons in the feature class based on the information that was initially only in the T2007FireTable table.

Right-click on the **CAFires2007** layer name in the Contents pane, select **Joins and Relates>Add Join**.

- b. In the Add Join dialog, enter the following:
 - i. Input Table: **CAFires2007** (This is the layer we are attaching the table information to)
 - ii. Input Join Field: **OBJECTID** (This is the field that will map how each feature/polygon in the CAFires2007 feature class matches up with each row in the T2007FireTable table)
 - iii. Join Table: **T2007FireTable** (This is the item from which we are getting the tabular information)
 - iv. Output Join Field: **OBJECTID** (This is the field that will map how each row in the T2007FireTable table connects to each feature/polygon in the CAFires2007 feature class)
 - v. Click **OK**

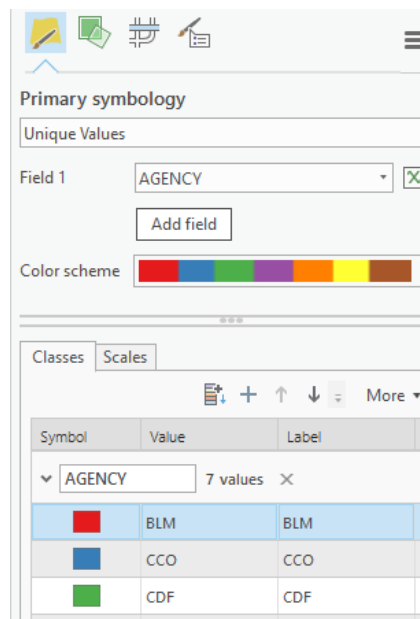
The following graphic demonstrates how the join process work, and the importance of the Input Join Field and Output Join Field. By using fields containing common data (i.e. OBJECTID, OBJECTID_1*) to match the polygons in the feature class up with the records in the table, we can temporarily bring the information from the table to the feature class for analysis and visualization purposes.

Attribute table from the CAFires2007 feature class							Attribute table from the T2007FireTable standalone table				
	OBJECTID_1 *	Shape *	OBJECTID	YEAR_	STATE	Shape		OBJECTID	UNIT_ID	FIRE_NAME	AGENCY
1	1	← Polygon	1	2007	CA	2309	1	1	LAC	OCTOBER	CCO
2	2	← Polygon	2	2007	CA	2470	2	2	LAC	MAGIC	CCO
3	3	← Polygon	3	2007	CA	20546	3	3	ANF	RANCH	USF
4	4	← Polygon	4	2007	CA	743	4	4	LAC	EMMA	CCO

- c. Open the **attribute table** for the **CAFires2007** layer, you should now see the fields from the CAFires2007 layer as well as the T2007FireTable table all in one place.

5. Adjust layer symbology and print the map

- a. To symbolize each wildfire polygon based on the agency responsible for it, select **CAFires2007** in the Contents pane, click on the **Feature Layer** tab, and select **Symbology**.
- b. In the Symbology pane, enter the following:
 - i. Select **Unique Values** under Primary symbology.
 - ii. Field 1: select **AGENCY**
 - iii. Color scheme: pick any color ramp that provides contrast between the different agency values.



- c. Go to the Contents pane, right-click on the CAFires2007 layer, and select **Zoom To Layer**.
- d. **Create a visually pleasing and informative map** with the essential map elements. Review the skills you learned in this tutorial (<https://pro.arcgis.com/en/pro-app/get-started/add-maps-to-a-layout.htm>) if necessary.

Q6. Submit a copy of your map with your answer sheet.

Works Cited

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Chapter 5: Assessing social vulnerability in areas impacted by Hurricane Florence

In this exercise, you will:

- Download data from the US Census Bureau (data.census.gov)
- Pre-process census data using Microsoft Excel
- Add and calculate new fields
- Join tabular data with vector layers
- Select features by their attributes
- Perform statistical analysis using the Ordinary Least Squares tool
- Interpret statistical output

Introduction

On September 14, 2018, Hurricane Florence made landfall as a Category 1 hurricane just south of Wrightsville Beach in North Carolina (Almansy et al., 2018). By definition, as a Category 1 hurricane on the Saffir-Simpson Hurricane Wind Scale, Florence had sustained wind speed in the range of 74 to 95 miles per hour (National Hurricane Center). While still dangerous, the wind speed of Florence is dwarfed by the wind speed of Hurricane Michael, which made landfall on October 9, 2018 near Mexico Beach, Florida as a Category 4 hurricane with a peak wind speed of 155 miles per hour (NOAA, 2019). However, unlike Hurricane Michael, Florence moved at a much slower speed of 2-3 miles per hour, which was the main cause of widespread flooding, wind damage, and the loss of property and human lives. In fact, research has shown that the speed at which hurricanes and tropical cyclones lingered over land (i.e. translation speed) has slowed globally by 10 percent between 1949 and 2016, and this is partly attributed to changes in the atmospheric circulation caused by greenhouse gas emissions and climate change (Kossin, 2018).



ASTRONAUT RICKY ARNOLD, FROM ABOARD THE INTERNATIONAL SPACE STATION, SHARED THIS IMAGE OF HURRICANE FLORENCE ON SEPT. 10, TAKEN AS THE ORBITING LABORATORY FLEW OVER THE MASSIVE STORM. (SOURCE: NASA)

In this exercise, you will look at the statistical relationship between poverty and unemployment in one of the states (i.e. South Carolina) that was hard hit by Hurricane Florence, and many of its counties were approved for federal assistance under Presidential Disaster Declarations DR-4393, DR-4394, and DR-4401. It is important to examine factors such as poverty in disaster-stricken areas, because empirical and anecdotal data from past natural disasters such as Hurricane Harvey and Hurricane Katrina have shown that residents in lower-income neighborhoods have an especially difficult time getting timely or sufficient assistance that is needed to rebuild their homes (Fernandez, 2018; Allen, 2015; Milligan, 2018).



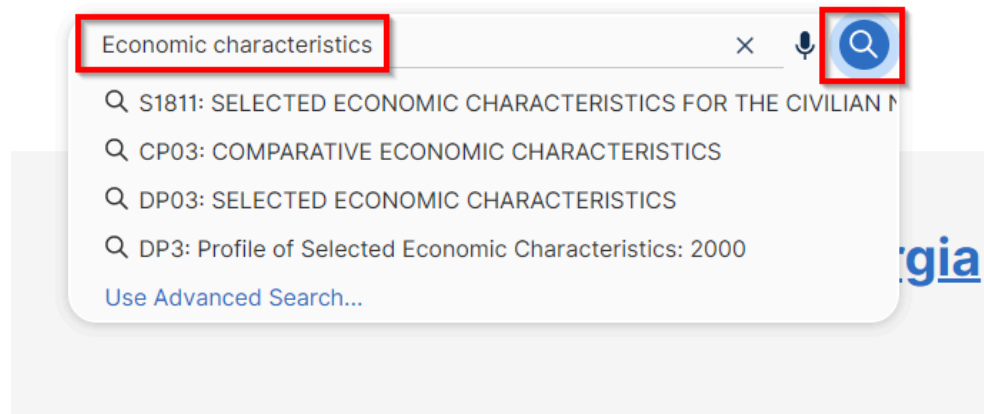
PHOTO OF WIDESPREAD FLOODING CAUSED BY HURRICANE FLORENCE NEAR LEES LANDING CIR., HORRY COUNTY, SOUTH CAROLINA.
(SOURCE: NATIONAL WEATHER SERVICE)

1. Download tabular data from the U.S. Census Bureau

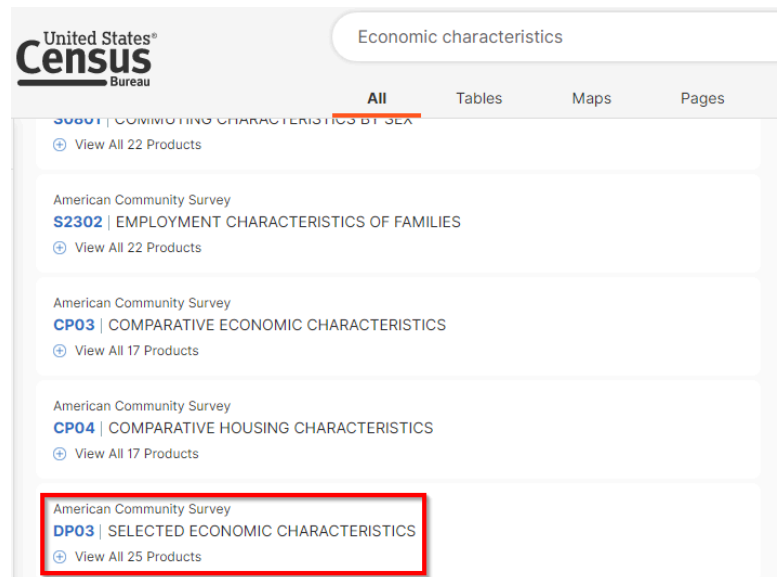
- a. In your internet browser (preferably Chrome), go to <https://data.census.gov>.
- b. In the search bar, type in **Economic characteristics**, and hit **Search**.

Explore Census Data

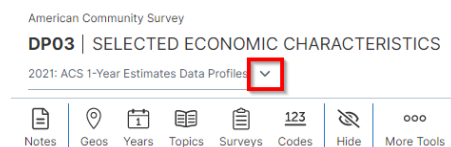
Learn about America's People, Places, and Economy

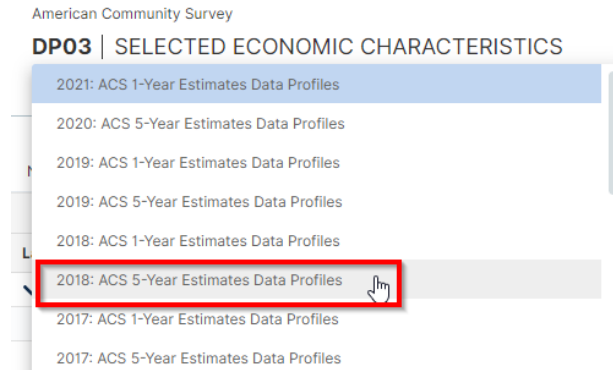


- c. In the list of results, be sure to select the one that will give you the table that was created by the **American Community Survey** program, with a Table name of **DP03**.

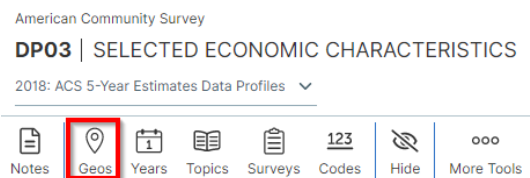


- d. Click on the **DP03** (not DP3) result.
Tip: Make sure that you are selecting the right table. Picking the wrong table will give you different results in this exercise, and you will need to redo the exercise again.
- e. Specify the exact time period you want by clicking on the Product dropdown arrow, and selecting **2018: ACS 5-Year Estimates Data Profiles**.





- f. Narrow the geographic scope of the data you are downloading by clicking the **Geos** icon.



- g. Under Most Commonly Used Geographies, select **Tract>South Carolina>All Census Tracts within South Carolina**. Click the X in the upper right to close the Select Geography window.

Most Commonly Used Geographies

Nation	State
County	Place
Zip Code Tabulation Area	Metropolitan Statistical Area
Tract	Block

Search State

- Ohio
- Oklahoma
- Oregon
- Pennsylvania
- Puerto Rico
- Rhode Island
- South Carolina
- All Census Tracts within South Carolina
- Abbeville County, South Carolina
- Aiken County, South Carolina

- h. Once the product and the geography have been properly selected, click **DOWNLOAD TABLE**.

DP03 | SELECTED ECONOMIC CHARACTERISTICS
2018: ACS 5-Year Estimates Data Profiles

Geos Years Topics Surveys Codes Margin of Error Map

census

Table is too large to display
The size of this table may exceed your browser's capabilities and may result in error or browser instability. Download the entire data or adjust your filters in the toolbar for a smaller table.

DOWNLOAD TABLE

To proceed with the table display anyway, [open the table](#).

- i. In the Download Tables window, make sure that the table being downloaded is called **DP03**, and that the **5-year estimate** published in **2018** is selected. Click **Download .CSV**.

Select Table Vintages to Download

DP03	All	2020	2019	2018	2017	2016	2015	2014	2013	2012	2011	2010
ACS 5-Year Estimates Data Profiles	☐	☐	☐	☒	☐	☐	☐	☐	☐	☐	☐	☐
ACS 5-Year Estimates Selected Population Data Profiles	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐

Compressed Size Estimate: 1.0 MB

DOWNLOAD .CSV

- j. When the progress bar of the Downloads window reaches 100%. The zip file should be automatically downloaded to your system. Find the file that you just downloaded and **unzip** it.

Tip: If nothing seems to be happening, make sure that your browser's pop-up blocker is not blocking the Census website's download window.

Downloads

Triggering Download...
ACSDP5Y2018.DP03

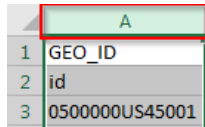
100 %

2. Prepare data in Microsoft Excel

- In the **unzipped folder**, locate the **ACSDP5Y2018.DP03-Data.csv** file, and open it with Microsoft **Excel**.
- We only need to retain and format three columns for this exercise, part c will show you how to do this.
 - The column with the label of **GEO_ID** (the description is Id2)

- ii. The column with the label of **DP03_0005PE** (the description is *Percent Estimate!!EMPLOYMENT STATUS!!Population 16 years and over!!In labor force!!Civilian labor force!!Unemployed*)
 - iii. The column with the label of **DP03_0119PE** (the description is *Percent Estimate!!PERCENTAGE OF FAMILIES AND PEOPLE WHOSE INCOME IN THE PAST 12 MONTHS IS BELOW THE POVERTY LEVEL!!All families*)
- c. We will create a new Excel spreadsheet by copying and pasting the three columns (GEO_ID, DP03_0005PE, DP03_0119PE) that we want from the original spreadsheet into the new one.

1. To start a new spreadsheet, in Excel, go to **File>New>select Blank workbook**.
2. Go back to the original spreadsheet (i.e. ACSDP5Y2018.DP03-Data.csv), and find the desired column by pressing the **Ctrl** and **F** keys at the same time on your keyboard, then enter the desired **column's name** (e.g. GEO_ID).
3. Click on the **column header** (i.e. A, B, C, D...) to select the entire column.



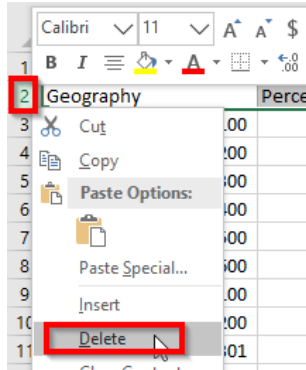
	A
1	GEO_ID
2	id
3	0500000US45001

4. With the column selected, press the **Ctrl** and **C** keys at the same time to Copy the column to your virtual clipboard.
5. Go to your new spreadsheet, click on the column header of an empty column, and press the **Ctrl** and **V** keys at the same time on your keyboard to paste the information that was stored in the virtual clipboard.
6. Go back to your original spreadsheet and **click anywhere outside** the currently selected column. **Repeat steps 2 thru step 5** in this section to find, copy, and paste the information from the **DP03_0005PE** and **DP03_0119PE** columns into the new spreadsheet.

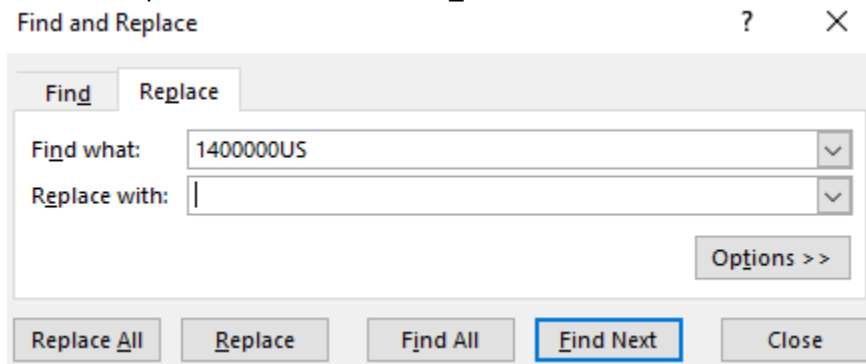
Your table should now look like the following:

	A	B	C
1	GEO_ID	DP03_0005PE	DP03_0119PE
2	Geography	Percent Estimate!!EMPLOYMEN	Percent Estimate!!PER
3	1400000US45001950100	3.8	6.6
4	1400000US45001950200	1.8	9.3
5	1400000US45001950300	2.9	12

- d. We need to clean up the names for each of our columns before opening the table in ArcGIS. Delete the second row (the row with the descriptions such as Geography, Percent Estimate!!EMPLOYMENT...; Percent Estimate!!PERCENTAGE...) by right-clicking on the **row number** and selecting **Delete**.



- e. We also need to clean up the values in the GEO_ID column in order to join it with our polygon layer later in this exercise. Click on the **column header** above the GEO_ID column's name (i.e. A) to select it.
- f. Once the GEO_ID column is selected, press the **Ctrl** and **F** keys to access the find and replace window.
- g. In the Find and Replace window, click on the **Replace** tab.
- h. Under the replace tab, enter **1400000US** in the Find what area, and leave the Replace with area blank. Click **Replace All**. This step allows us to erase the unnecessary 1400000US prefix from all of our GEO_ID values.



Your GEO_ID column should look like the following after this step.

	A	B	C
1	GEO_ID	DP03_0005PE	DP03_0119PE
2	45001950100	3.8	6.6
3	45001950200	1.8	9.3
4	45001950300	2.9	12
5	45001950400	5.8	20.2
6	45001950500	4.2	24.2
7	45001950600	4.6	21.8
8	45003020100	3.3	18.2

- i. Remember that names for columns in ArcGIS cannot have special characters or symbols such as: . > * or even spaces, we will rename the column names as follow to make them more user-friendly:
 - i. Replace the label of **GEO_ID** with **SCGeoID** (or NCGeoID when you get to working with North Carolina data later)
 - ii. Replace the label of **DP03_0005PE** with **SCNotEmployed** (or NCNotEmployed when you get to working with North Carolina data later)

- iii. Replace the label of **DP03_0119PE** with **SCUnderPoverty** (or **NCUnderPoverty** when you get to working with North Carolina data later)

Your table should now look like the following:

	A	B	C
1	SCGeoID	SCNotEmployed	SCUnderPoverty
2	45001950100	3.8	6.6
3	45001950200	1.8	9.3
4	45001950300	2.9	12
5	45001950400	5.8	20.2

- j. In Excel, go to **File>Save As** to save the file as a **CSV (Comma delimited)** file and call it **SouthCarolinatbl**, click **Save**, click **Yes** when warned by Excel about the CSV file format, and **close Excel**.

3. Download geographic layers from the U.S. Census Bureau

- a. In your internet browser (preferably Chrome), go to <https://www.census.gov/cgi-bin/geo/shapefiles/index.php>
- b. Since the tabular data you downloaded in the previous section was published in 2018, you need to download geographic layers from the same time period. Select **2018** for the year, and select **Census Tracts** for the layer type. Hit **Submit**.

TIGER/Line® Shapefiles

Select the year and layer you are interested in from the dropdown menus below and click "Submit" for a list of the available geographic areas.

Select year:

Select a layer type:

- c. On the next screen, select **South Carolina** in order to download the census tract polygons for the state. Once downloaded, **unzip** the file (tl_2018_45_tract.zip), and rename the unzipped folder **SCTracts**.

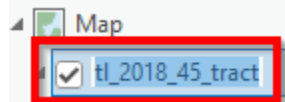
2018 TIGER/Line® Shapefiles: C

Census Tract

Select a State:

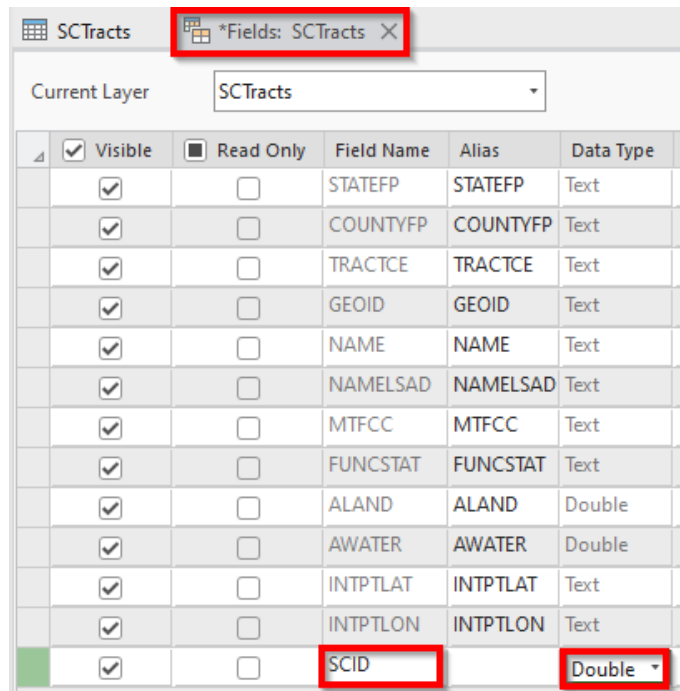
4. Add and calculate a new field

- a. Open **ArcGIS Pro**, and start a new **project** and a new **map**.
- b. Add in the **South Carolina (SCTracts)** census tract layer you downloaded in section 3, and rename it **SCTracts** in the Contents pane. ***Tip: Single click on the layer name in the Contents pane and wait until it is highlighted, then type in the new name***



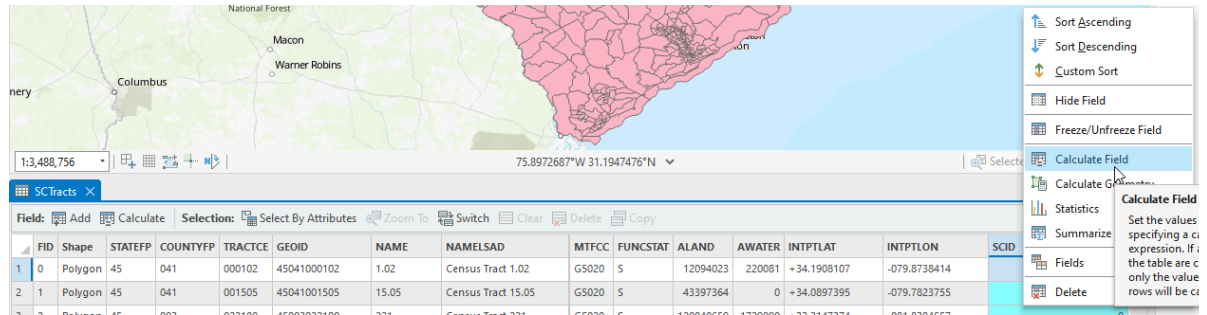
Q1. Open the attribute table of the SCTracts layer, how many fields/columns are there at this point in the exercise?

- c. Within the SCTracts layer's attribute table, click **Add**  in the upper left to create a new field.
- d. In the Fields tab within the attribute table, in the new row found at the bottom of the window, replace the word Field with **SCID** under the **Field Name** column, and replace the word Long with **Double** under the **Data Type** column. **Close** the Fields tab, and select **Yes** when prompted to save your unsaved changes.



Visible	Read Only	Field Name	Alias	Data Type
☒	☐	STATEFP	STATEFP	Text
☒	☐	COUNTYFP	COUNTYFP	Text
☒	☐	TRACTCE	TRACTCE	Text
☒	☐	GEOID	GEOID	Text
☒	☐	NAME	NAME	Text
☒	☐	NAMELSAD	NAMELSAD	Text
☒	☐	MTFCC	MTFCC	Text
☒	☐	FUNCSTAT	FUNCSTAT	Text
☒	☐	ALAND	ALAND	Double
☒	☐	AWATER	AWATER	Double
☒	☐	INTPTLAT	INTPTLAT	Text
☒	☐	INTPTLON	INTPTLON	Text
☒	☐	SCID		Double

- e. Within the SCTracts layer's attribute table (reopen it if needed), scroll to the far right to find the newly created **SCID** column.
- f. Right-click the **SCID column's name** within the attribute table, and select **Calculate Field**.



- g. Within the Calculate Field window, enter the following:
 - i. Input Table: **SCTracts**
 - ii. Field Name (Existing or New): **SCID**
 - iii. Fields: **Double click on GEOID** (to see !GEOID! populated under the SCID = field)
 - iv. Click **OK**

Tip: What you have done in steps c thru g of this section is basically copying the data from the GEOID field to the SCID field. Although the data in both columns are exactly the same, the format in which the data is stored is different in each field. Specifically, the GEOID field stores the data as Text, whereas the SCID field stores the same data as Double (a numeric data type). We need the data in a numeric format in order to perform the join operation in the next section.

- h. Repeat steps c thru g in this section to [1] create a new field with the **Field Name** of **TID** and the **Data Type** of **Short**. [2] Then use the **Calculated Field** tool to populate the **TID** field by copying the data from the **FID** field.

Your finished attribute table should look similar to the following:

	FID	Shape	STATEFP	COUNTYFP	TRACTCE	GEOID	NAME	NAMLSAD	MTFCC	FUNCSTAT	ALAND	AWATER	INTPTLAT	INTPTLON	SCID
1	0	Polygon	45	041	000102	45041000102	1.02	Census Tract 1.02	G5020	S	12094023	220081	+34.1908107	-079.8738414	45041
2	1	Polygon	45	041	001505	45041001505	15.05	Census Tract 15.05	G5020	S	43397364	0	+34.0897395	-079.7823755	45041
3	2	Polygon	45	003	022100	45003022100	221	Census Tract 221	G5020	S	120040659	1729890	+33.3147374	-081.8384557	45003

5. Join tabular data with geographic layers

- a. Add in the **SouthCarolinatbl** CSV table that you prepared in Excel in section 2.
- b. Join the CSV table with the SCTracts layer by right-clicking on **SCTracts** in the Contents pane, selecting **Joins and Relates**, and clicking **Add Join**.
- c. In the Add Join window, enter the following:
 - i. Input Table: **SCTracts**
 - ii. Input Join Field: **SCID**
 - iii. Join Table: **SouthCarolinatbl.csv**
 - iv. Output Join Field: **SCGeoID**
 - v. Click **OK**

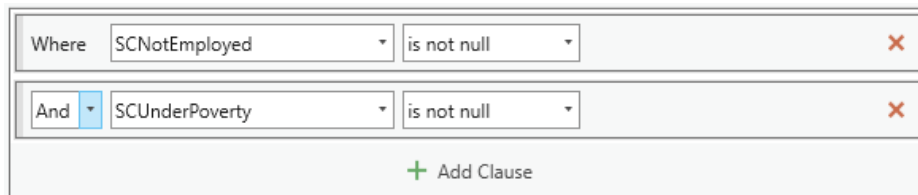
Q2. Open the attribute table of the SCTRacts layer again, how many fields/columns are there now after the join process?

Tip: You should have noticed that the last three columns in the SCTRacts layer's updated attribute table were brought over from the SouthCarolinatbl table as a part of the join process.

6. Analyze the statistical relationship between poverty and unemployment

- In case you have any features selected in your map, go to the **Map** tab in the main menu, and click **Clear**  to deselect them.
- To ensure that we are only running our analysis on census tracts with valid data, go to

the **Map** tab in the main menu, click **Select By Attributes** , and enter the conditions as shown in the screenshot below. Click **OK** when you are done.

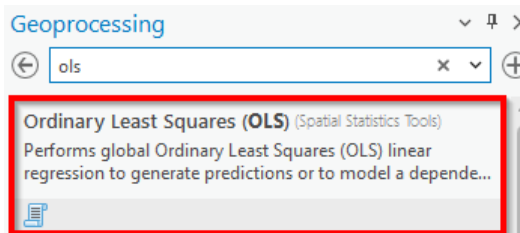


Where SCNotEmployed is not null

And SCUnderPoverty is not null

+ Add Clause

- With only those features that are reporting data selected, we can explore the relationship between the percentage of a census tract's population that is living below the poverty line (i.e. SCUnderPoverty) and the percentage of a census tract's population that is not employed (i.e. SCNotEmployed). Go to the **Analysis** tab in the main menu, click on **Tools**, and search for **OLS**. In the results, click on the **Ordinary Least Squares (OLS)** tool.

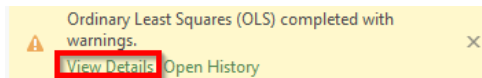


Q3. Write a hypothesis of how you think the percentage of the tract's population that is living below the poverty line (i.e. SCUnderPoverty) and the percentage of the tract's population that is not employed may be related. Are the two things positively correlated? (i.e. high number of not employed people is related to a high number of people living below the poverty line) Or are the two things negatively correlated? (i.e. high number of not employed people is related to a low number of people living below the poverty line)

- d. In the Ordinary Least Squares (OLS) window, enter the following:
 - i. Input Feature Class: **SCTracts**
 - ii. Unique ID Field: **tl_2018_45_tract.TID**
 - iii. Output Feature Class: **OLS**
 - iv. Dependent Variable: **SouthCarolinatbl.csv.SCUnderPoverty**
 - v. Explanatory Variable: **SCNotEmployed**
 - vi. Click **Run**

****Tip:** Since people are likely to fall below the poverty line due to not being able to find employment, rather than not being able to find employment due to falling below the poverty line, *BelowPov* was selected as the dependent variable and *NotEmploy* was selected as the explanatory variable (also known as the casual variable or the independent variable). In other words, we are hypothesizing that the poverty rate is dependent on the unemployment rate. ******

- e. When the Ordinary Least Squares (OLS) operation is complete, click **View Details** at the bottom of the pane.



- f. Look at the results in the details window (maximize the window if needed), the two numbers that we are particularly interested in are covered by box (1) and box (2) below:

Summary of OLS Results

Variable	Coefficient ^a	StdError	t-Statistic	Probability ^b	Robust_SE	Robust_t	Robust_Pr ^b
Intercept	4.458839	0.484798	9.197315	0.000000*	0.494075	9.024613	0.000000*
SOUTHCAROLINATBL.CSV.SCNOTEEMPLOYED	1	0.102545	21.098133	0.000000*	0.126420	17.113604	0.000000*

OLS Diagnostics

Input Features	SCTracts	Dependent Variable	SOUTHCAROLINATBL.CSV.SCUNDERPOVERTY
Number of Observations	1083	Akaike's Information Criterion (AICc) ^d	7662.637769
Multiple R-Squared ^d	0.291673	Adjusted R-Squared ^d	2
Joint F-Statistic ^e	445.131231	Prob(>F), (1,1081) degrees of freedom	0.000000*
Joint Wald Statistic ^e	292.875450	Prob(>chi-squared), (1) degrees of freedom	0.000000*
Koenker (BP) Statistic ^f	41.048440	Prob(>chi-squared), (1) degrees of freedom	0.000000*
Jarque-Bera Statistic ^g	372.962574	Prob(>chi-squared), (2) degrees of freedom	0.000000*

To understand the meaning of the two numbers, insert the value from box 1 and box 2 in the following sentences (continues on the next page).

Q4. For every 1 percentage point increase in a census tract's population that is not employed, the amount of people living below the poverty line in the census tract is expected to increase by (insert box 1 value here:) percentage point. (Round to two decimal places)

Q5. (Round box 2 value to two decimal places first and multiply by 100)
percent of the variation in various census tracts' poverty levels can be
explained by the unemployment levels in those tracts.

Q6. Create one map containing **two data frames**. One of the data frames
should show the variation in not employed population in various census tracts,
and the other one should show the variation in population living below the
poverty line in various census tracts. Please include relevant map elements.

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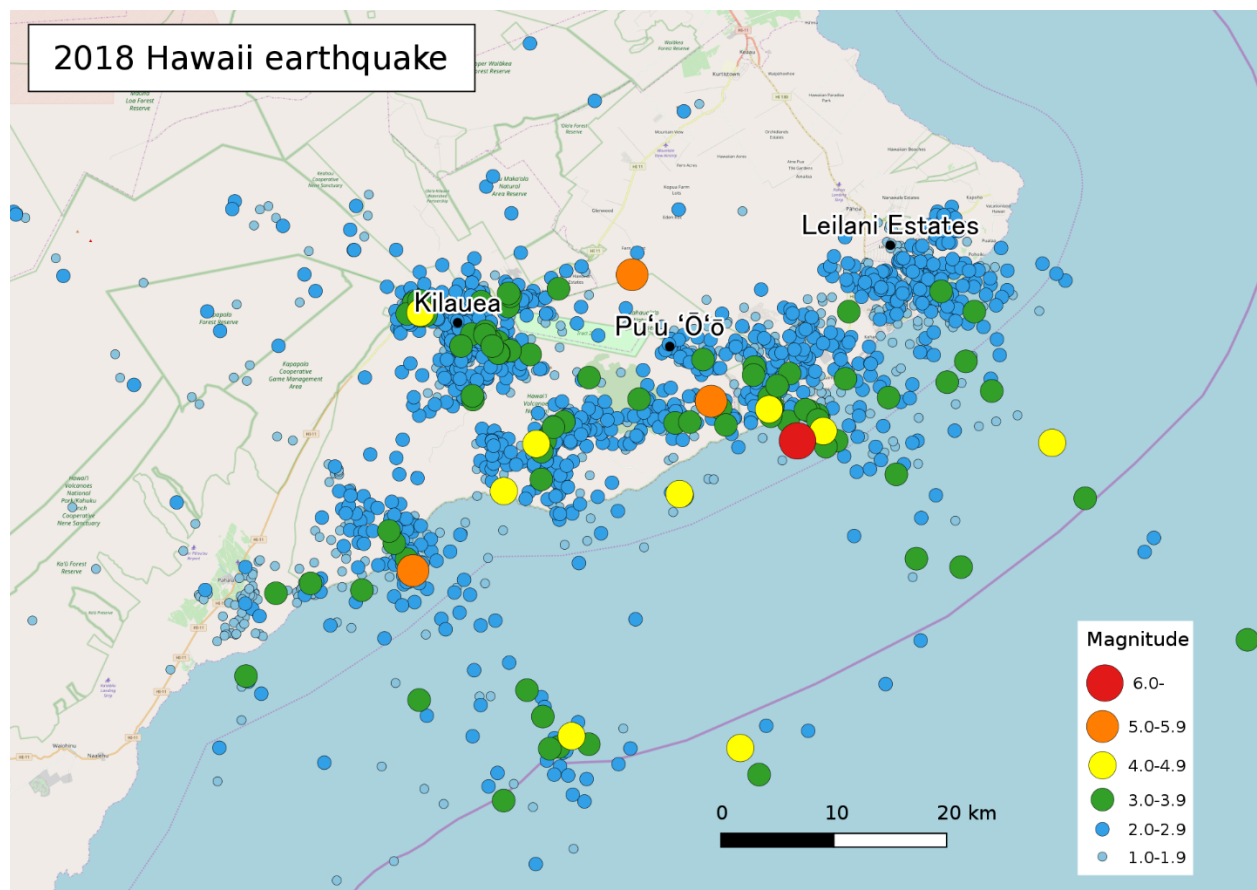
Chapter 6: Analyzing volcanic hazards near Kilauea Volcano, Hawai'i

In this exercise, you will:

- Georeference map images from different dates
- Create and edit polygons features
- Create and test task items
- Perform spatial analysis using the erase tool
- Perform attribute calculations using the summary statistics tool

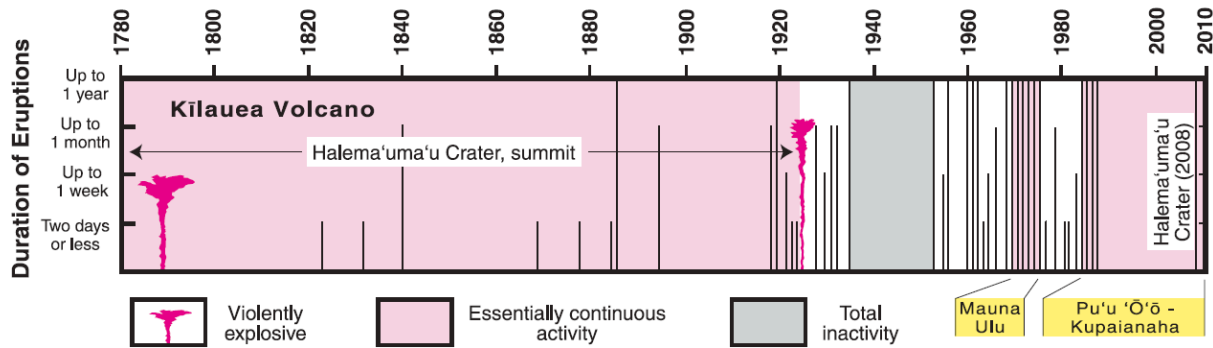
Introduction

May 3, 2018 marked the start of the lower Puna eruption on the East Rift Zone of the Kilauea Volcano on the island of Hawaii (Chiu and Bever, 2018). The eruption was accompanied by many earthquakes, active fissures and lava flow, dangerously high levels of sulfur dioxide (i.e. Vog), and the deposition of volcanic bombs (i.e. large ballistic blocks) and ashes in the region (BBC, 2018).



MAP OF 2018 EARTHQUAKES IN SELECTED PARTS OF HAWAII (SOURCE: PHOENIX7777 OWN WORK, CC BY-SA 4.0, VIA WIKIMEDIA COMMONS)

The Kilauea volcano is an example of a shield volcano, which is built up slowly over time with the accretion of fluid lava flow (Watson, 2011). It is not only regarded as one of the world's most active volcanoes, but it is also considered a relatively young volcano given that 90% of its surface is formed by lava flows of less than 1,100 years old (Oregon State University, 2019).



TIMELINE OF THE ERUPTION OF KILAUEA DURING THE LAST 200 YEARS (SOURCE: ROBERT I. TILLING, CHRISTINA HELIKER, AND DONALD A. SWANSON, PUBLIC DOMAIN, VIA WIKIMEDIA COMMONS)

In this exercise, you will georeference and examine two map images of the Kilauea Volcano area that were created roughly two weeks and fourteen weeks after the eruption on May 3, 2018. In particular, you will identify areas that were impacted by Kilauea's lava flow, in order to appreciate the scale of this eruption which destroyed 716 homes before abruptly settling down in August 2018 (Loomis, 2018a; Loomis, 2018b).

1. Georeference map images

- Open **ArcGIS Pro**, and start a new **project** and a new **map**.
- Add in the **5_15_18.jpg** image. Click **Yes** when asked about calculating statistics for the image. Right-click on the layer name in the Contents pane and select **Zoom to Layer** to see its approximate extent.

Q1. Zoom out and describe the approximate location of the 5_15_18 image in the world at the end of step 1b.



- Go to the **Map** tab, click **Locate**, and search for **Kapoho, HI**.
- Zoom out to approximately **1:100,000** map scale.
- With the **5_15_18** image selected in the contents pane, go to the **Imagery** tab, select



Georeference, and click **Fit to Display** (make sure that your ArcGIS Pro window is maximized if you do not see the Fit to Display icon).

- Find at least four point locations (e.g. obvious natural features, major intersections) that you can readily identify in the 5_15_18 image as well as the topographic basemap.

These points are called control points. They will enable us to carry out the process of **georeferencing** by aligning the image to the topographic basemap, which results in the image inheriting the spatial reference information (i.e. coordinate system, projection) of the topographic basemap.

- g. To specify each control point, make sure that the **5_15_18** image is still selected in the

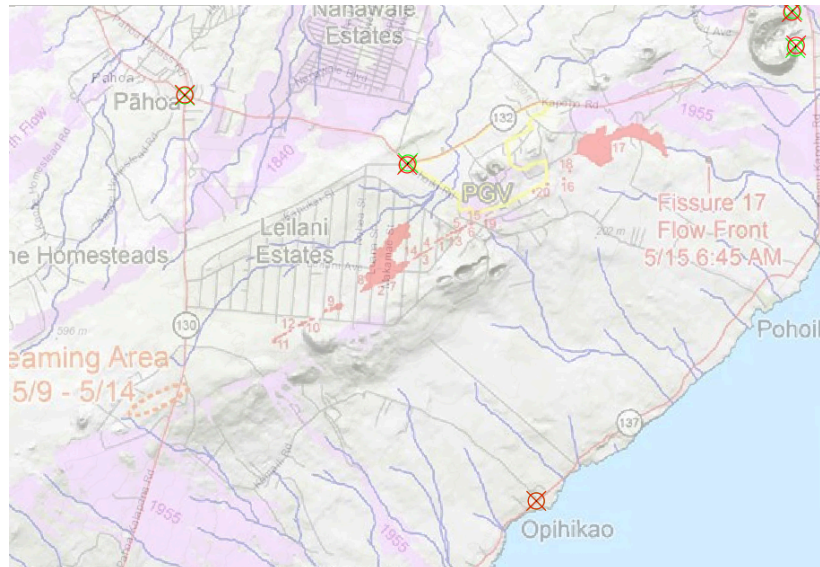
Contents pane. Click **Add Control Points**  (within the Georeference tab). **FIRST**, in the **5_15_18 image**, click on one of the readily identifiable points you noted in the previous step. **THEN**, temporarily **turn off (uncheck)** the 5_15_18 image layer in the Contents pane so you can see the basemap, and click on the point's corresponding location in the **topographic basemap**. **Turn on** the 5_15_18 image layer in the Contents pane, and **repeat** this process for at least 3 other control point pairs.

Tip: When georeferencing an image to a basemap, always put the control point on the image first, then the basemap.

Tip: To temporarily pan the map to a different area while you are placing your control points, press the C key on your keyboard while using the mouse to pan.

I have suggested 5 control points for you below (you can also feel free to use your own). Please refer to the image below for their locations.

- The north end of the cinder cone
- The south end of the cinder cone
- The intersection of Highway 132 and Pohoiki Rd
- The intersection of Highway 130 and Highway 132
- The intersection of Highway 137 and Kamaili Rd



- h. After you have placed **at least 4 control points**, you should see the 5_15_18 image aligns reasonably well with the topographic basemap. Click **Save** within the Georeference tab.

And select **Close Georeference** .

Q2. Zoom out and describe the approximate location of the 5_15_18 image in the world at the end of step 1h.

- i. Turn off the 5_15_18 image layer in the Contents pane to reduce clutter. **Modify and repeat** steps 1b thru 1i to add in the **8_14_18.jpg** image and georeferenced it.
- 2. Create a new feature class and perform basic edits**
- a. If the Catalog pane is not already opened, go to **View>Catalog Pane**.
 - b. Find the file geodatabase that was already created for you when you created this project, by expanding **Folders>(your project name)>(your project name).gdb**
 - c. Right-click on **(your project name).gdb** in the Catalog pane, and select **New>Feature Class**. The Create Feature Class pane should appear. Enter the following:
 - i. Name: **May15Lava**
 - ii. Feature Class Type: **Polygon**
 - iii. Click **Finish**

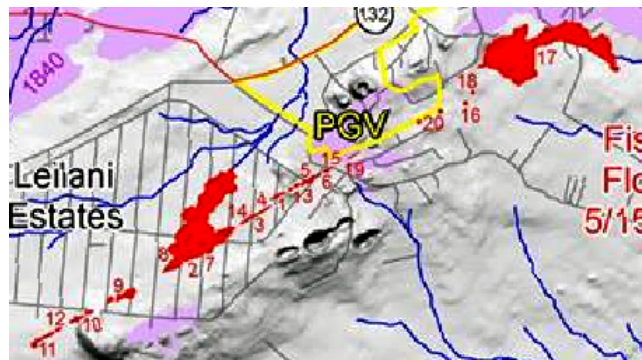
You have just created an empty polygon feature class that will be used to store the polygon trace that you will create in the next steps to document the lava extent around Kilauea on 5/15/18.

- d. Turn on only the **May15 Lava** layer and the **5_15_18** layer in the Contents pane.
- e. Select the **May15Lava** layer in the Contents pane, click the **Edit** tab, and select **Create**.
- f. In the Create Features pane, select **May15Lava**.
- g. Ensure that the **Polygon** tool is selected . **Trace** all of the bold red polygons (see image below) that represent active lava flow in the 5_15_18 image. (Only trace the polygons, ignore the points and the numbers)

Tip: There is no need to be painstakingly precise in your trace, and you may choose to ignore some of the smaller polygons

Tip: If you cannot see what you just finished tracing, double-check to make sure that the May15Lava feature class is turned on in the Contents pane.

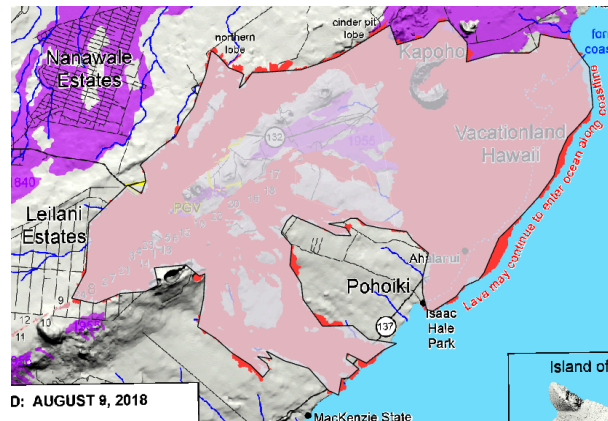
Tip: If you need to move to another part of the map after you have started tracing, hold down on the C key on your keyboard while using the mouse to pan.



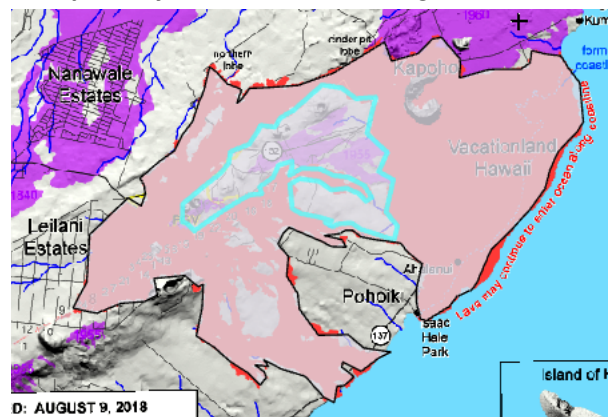
- h. When you have finished tracing the bold red polygons from the 5_15_18 image, be sure to go to the **Edit** tab, and hit **Save** to save your edits.
- i. **Modify and repeat** step 2c from this section to create a new polygon feature class in (your project name).gdb called **Aug14Lava**. **Modify and repeat** steps 2d thru 2h to trace the existing lava flow in the **8_14_18** image (**read all of the tips below before tracing the 8_14_18 image**). **Save** your edits.

Tip: Since the lava c polygons in the 8_14_18 images have “holes” (areas with no lava coverage) within them (see image below), you will need to:

1. Digitize the **outer boundary** of the polygon as one feature.

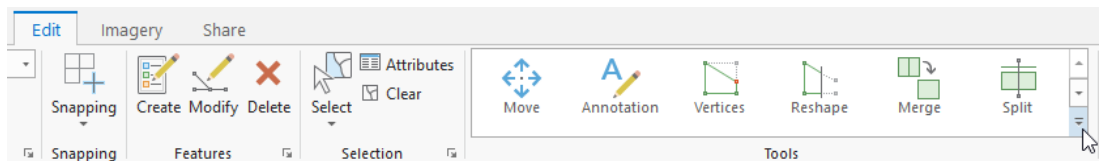


2. You will need to adjust the Aug14Lava to see the “hole” currently covered by the layer. With Aug14Lava selected in the Contents pane, go to the **Feature Layer** tab, and adjust **Transparency**  **Transparency**. Digitize one or more new polygons to cover the “hole”.

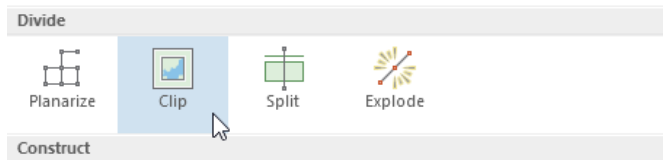


3. Select the “hole” polygon(s) that you just digitized by opening the Aug14Lava layer’s **attribute table** and **selecting** the record(s)/row(s) associated with the “hole” polygon (use the Ctrl key on your keyboard to select multiple rows at the same time), then go to the **Edit** tab, and **expand** all of the edit Tools options (see image 1 below). Select **Clip** (2). In the Clip pane, make sure that **Discard** is selected and **Clip all editable features** is checked, click **Clip** (3). Finally, verify the “hole” polygon(s) that you previously selected are still selected (or re-select them in the attribute table if needed), and hit the **delete** key on your keyboard to get rid of them. You should now see the lava polygon with a hole in it (4). **Save** your edits.

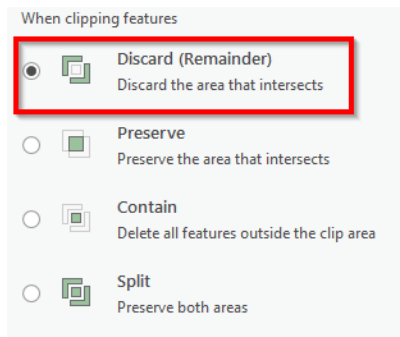
(1)



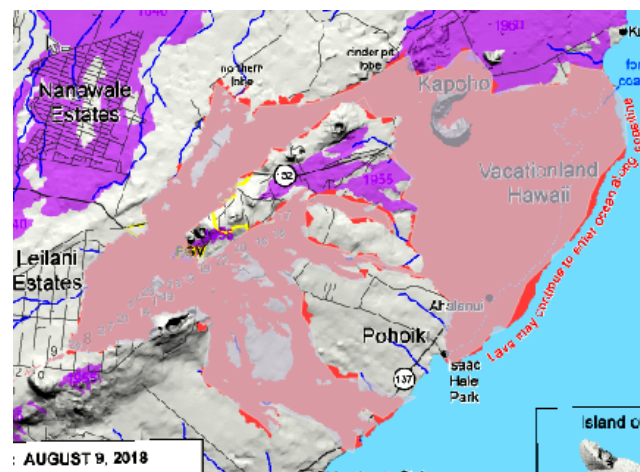
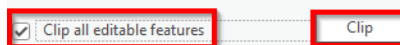
(2)



(3)



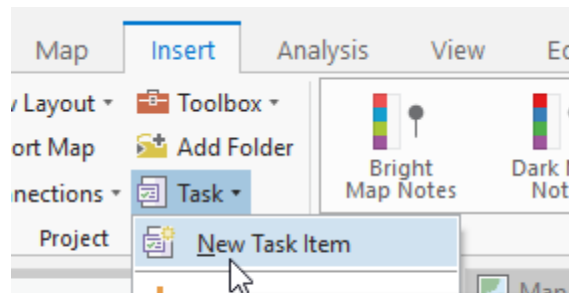
(4)



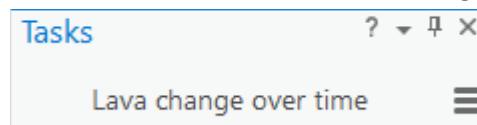
3. Creating a task

Now we will use a combination of geoprocessing tools (i.e. Erase, Summary Statistics) to calculate how the extent of lava flow changed between May 2018 and August 2018. Furthermore, we will document our workflow using **Task**, which is a collection of steps or a recipe for a common workflow.

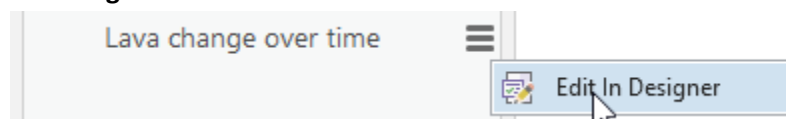
- a. Go to **Insert>New Map** to start a new map.
- b. To start building a Task, go to the **Insert** tab. Click on **Task**, and select **New Task Item**.



- c. In the Task Designer pane, enter the following:
 - i. Name: **Lava change over time**
 - ii. Author: **(your name)**
 - iii. Summary: **Follow these steps to show lava change around Kilauea.**
 - iv. Tags: **Editing, Kilauea**
 - v. **Close** the task designer pane (i.e. X in the upper right corner)
- d. You should now see the Lava change over time task in your Tasks window.



- e. Click the **options** button (3 horizontal bars) in the Tasks window, and select **Edit in Designer**.



- f. Within the Tasks window, click the **New Task** icon .
- g. In the Task Designer pane, enter the following:
 - i. Name: **Lava change**
 - ii. Description: **This task will calculate the area of lava change around Kilauea over time**
- h. In the Tasks window (not the Task Designer), click the **Record** icon  to record the commands/tools we will be using in the following steps to calculate the change in lava-covered areas. Specifically, these tools will be added to our task by just opening them in the following steps, but not actually running any of them yet.
- i. The first step is to add in the polygon feature showing lava flows from an earlier time point (i.e. earlier feature). Go to **Map>Add Data** and **close** the Add Data window.

- j. Note how the Add Data tool is automatically added to the Tasks window. Within the Task Designer (not the Tasks window) Enter the following and do not close the Tasks or Task Designer windows:
- Name: **Add earlier lava polygon**
 - Step Behavior: **Manual**

The screenshot shows the 'Task Designer' window with the following configuration:

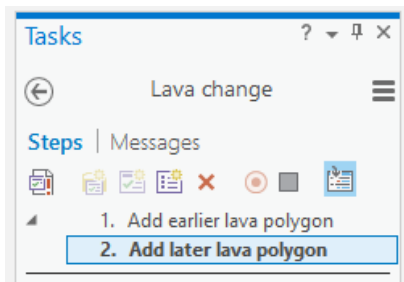
- Step:** Add earlier lava polygon
- General** | Actions | Views | Contents
- Name:** Add earlier lava polygon
- Tooltip:** (Empty text box)
- Instructions:** (Empty text box with formatting icons: B, I, U, and a link icon)
- Run/Proceed Instructions:** (Empty text box with formatting icons: B, I, U, and a link icon)
- 150 characters left**
- Step Behavior:**
 - ☒ **Manual** User runs and user proceeds
 - ☐ **Auto Run** Step runs and user proceed:
 - ☐ **Auto Proceed** User runs and step proc
 - ☐ **Automatic** Step runs and step proceer
 - ☐ **Hidden** User will not see the step

- k. The second step is to add in the polygon feature showing lava flows from a later time point (i.e. later feature). Go to **Map>Add Data** and **close** the Add Data window.

- I. Note how the Add Data tool is automatically pulled up again in the Tasks and Task Designer windows. Enter the following in the Task Designer, and do not close the Tasks or Task Designer windows:
 - i. Name: **Add later lava polygon**
 - ii. Step Behavior: **Manual**

The screenshot shows the 'Task Designer' window with the 'New Step' configuration. The 'Name' field is set to 'Add later lava polygon'. The 'Step Behavior' section has 'Manual' selected, with the description 'User runs and user proceeds'. Other options like 'Auto Run', 'Auto Proceed', 'Automatic', and 'Hidden' are unselected. The 'Instructions' and 'Run/Proceed Instructions' fields are empty.

- m. You should now see both steps that we added in the Tasks Window.



- n. The third step is to subtract the earlier lava polygon from the later lava polygon in order to see the new area covered by lava between the two time points. Go to **Analysis>Tools**, and search for the **Erase** tool. **Click** on the search result and **close** the Geoprocessing: Erase window (do not close the Tasks or Task Designer windows).

- o. Note how the Erase tool is automatically pulled up in the Task Designer and Tasks windows. Enter the following in the Task Designer and do not close the Tasks or Task Designer windows:
 - i. Name: **Difference in lava cover**
 - ii. Instructions: **Enter the later lava polygon layer as the Input Features, and the earlier lava polygon layer as the Erase Features. Name your output.**
 - iii. Step Behavior: **Manual**

Task Designer ? ▾ 🔍 ✕

Step: Erase

General | Actions | Views | Contents

Name
Difference in lava cover

Tooltip

Instructions **B** *I* U 🗑️

Enter the later lava polygon layer as the Input Features, and the earlier lava polygon layer as the Erase Features. Name your output.

Run/Proceed Instructions **B** *I* U 🗑️

150 characters left

Step Behavior

☒ **Manual** User runs and user proceeds 🗑️

☐ **Auto Run** Step runs and user proceeds!

☐ **Auto Proceed** User runs and step proceeds

☐ **Automatic** Step runs and step proceeds

☐ **Hidden** User will not see the step

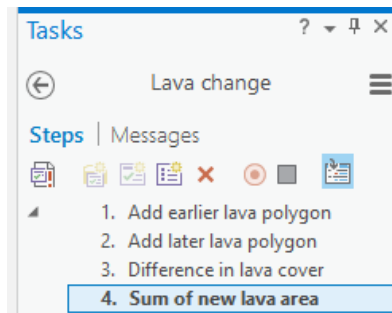
- p. The final step in this task is to sum the areas that are covered by the polygons generated by the previous step. In other words, we need to sum up the size of the new areas that are covered by lava between the earlier and the later dates. Go to **Analysis>Tools**, and search for the **Summary Statistics** tool. **Click** on the search result and **close** the Geoprocessing: Summary Statistics window (do not close the Tasks or Task Designer windows).

- q. Note how the Summary Statistics tool is automatically pulled up in the Task Designer and Tasks windows. Enter the following in the Task Designer:
- Name: **Sum of new lava area**
 - Instructions: **Enter the output from the previous step for the Input Table. Name the Output Table. Select Shape_Area for Field, and Sum for Statistics Type.**
 - Step Behavior: **Manual**

The screenshot shows the 'Task Designer' window with the following details:

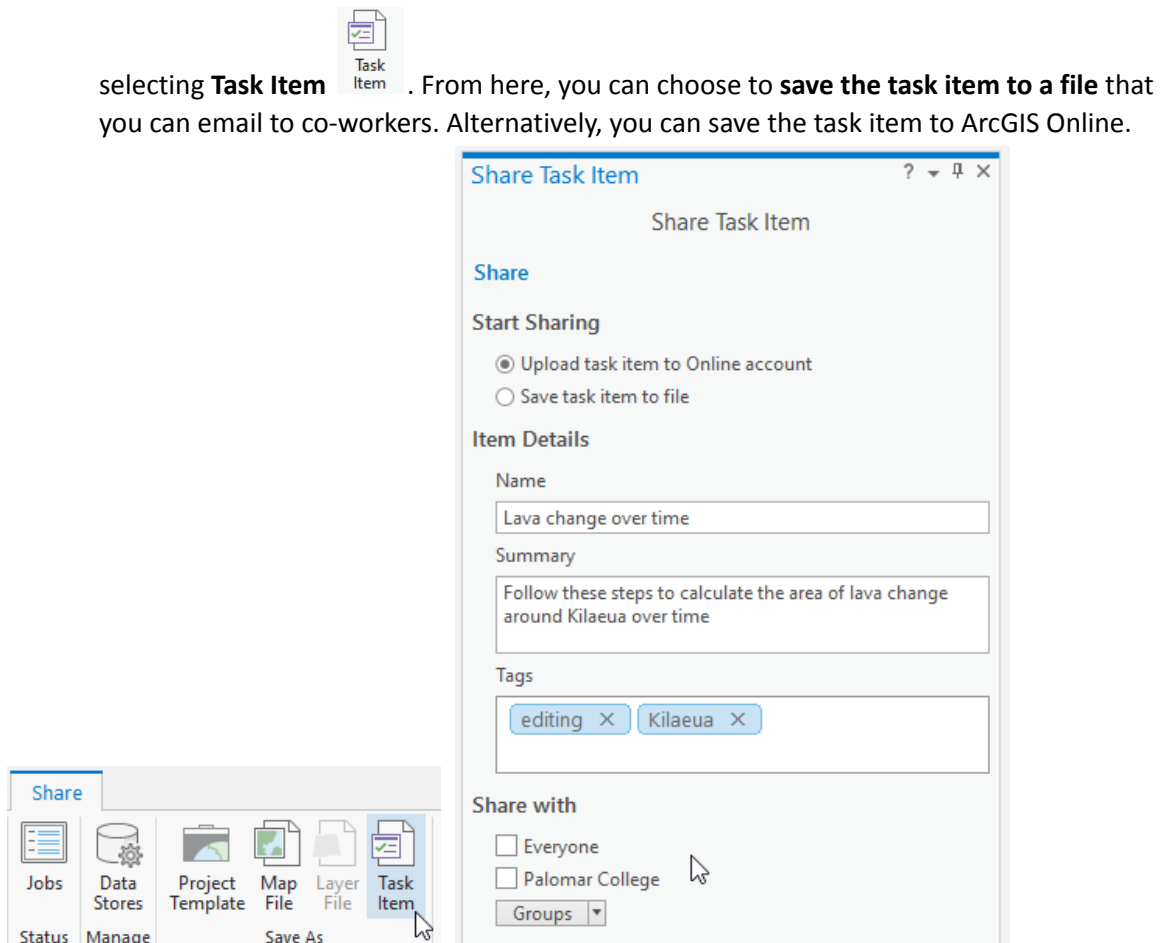
- Step:** Sum of new lava area
- General** | Actions | Views | Contents
- Name:** Sum of new lava area
- Tooltip:** (Empty text box)
- Instructions:** Enter the output from the previous step for the Input Table. Name the Output Table. Select Shape_Area for Field, and Sum for Statistics Type.
- Run/Proceed Instructions:** (Empty text box)
- 150 characters left**
- Step Behavior:**
 - ☒ **Manual** User runs and user proceeds
 - ☐ **Auto Run** Step runs and user proceeds
 - ☐ **Auto Proceed** User runs and step proceeds
 - ☐ **Automatic** Step runs and step proceeds
 - ☐ **Hidden** User will not see the step

- r. **Review** your Tasks window. It should consist of 4 steps similar to the following image:



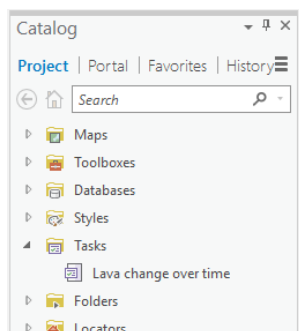
- s. Click the **Stop** icon  to stop recording your steps. Close the Task Designer pane (not the Tasks window).

- t. **(Optional)** You can also share your Tasks with co-workers by going to the **Share** tab, and

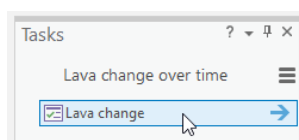


4. Test your Task Item

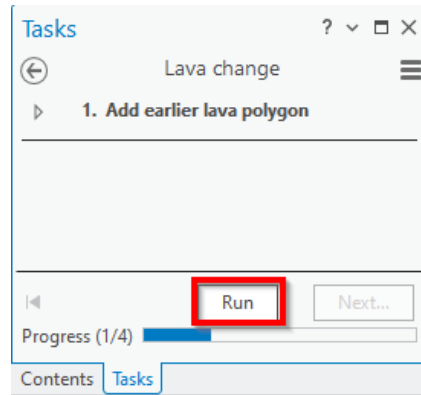
- a. You can access the Task you just created from the Tasks window, or if you closed the Tasks window, you can find it in the **Catalog pane** within the **Tasks** folder.



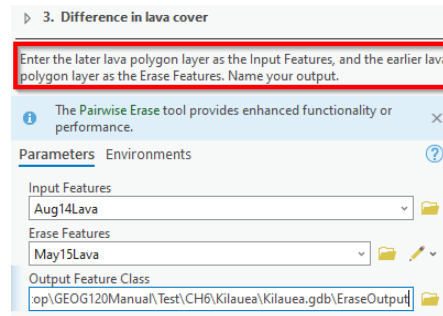
- b. Run the task by double-clicking the **Lava Change** task in the Tasks Window.



Click **Run** on the page for each step to run each of the four tools we previously added to the task. Remember to use your **May15Lava** layer as the earlier lava polygon layer for Step 1, and your **Aug14Lava** layer as the later lava polygon layer for Step 2. Click **Next Step** to advance between steps in the task.



- c. For Step 3 of your task, **follow the instructions** you previously typed in when creating the task. Click **Run**, then click **Next Step** to go onto Step 4.



- d. For Step 4 of your task, **follow the instructions** you previously typed in when creating the task. Click **Run**, then click **Finish. Close** the Tasks window.
- e. When you have finished going through all of the steps in the Task, go to the Contents pane. You should see:
- The May15Lava layer
 - The Aug14Lava layer
 - The Difference layer (i.e. Erase output) that you created in Step 3 of your task
 - A table containing the sum of the additional areas that were covered by lava between May 15th and August 14th.

Q3. Open the table containing the sum of the additional areas that were covered by lava, what is the value recorded for the Shape_Area field?

Q4. Turn off all layers in your map except the **May15Lava** layer. Go to the

Share tab, and click **Export Map to File**



. In the Export Map window, name the file **May15Lava**, and select **PDF** as the Save as type.

Q5. Turn off all layers in your map except the **Aug14Lava** layer. Go to the **Share**

tab, and click **Export Map to File**



. In the Export Map window, name the file **Aug14Lava**, and select **PDF** as the Save as type.

Q6. Turn off all layers in your map except the **Difference** feature class that you create in Step 3 of your task. Go to the **Share** tab, and click **Export Map to File**



. In the Export Map window, name the file **Difference**, and select **PDF** as the Save as type.

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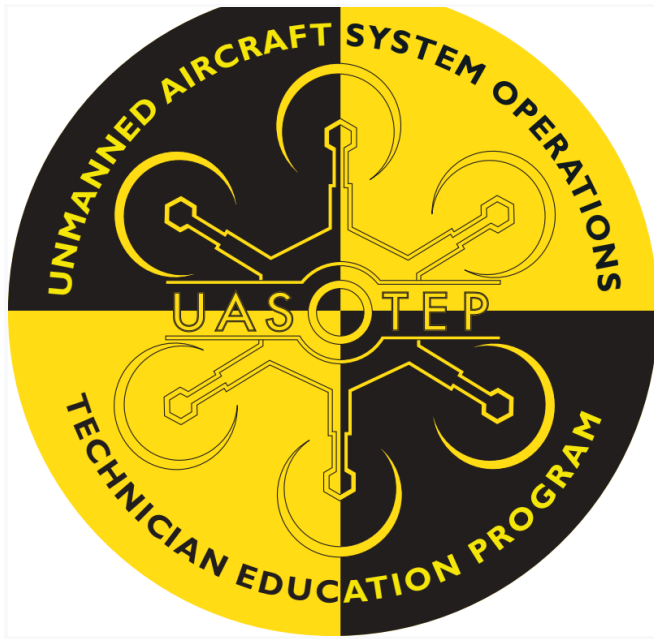
Chapter 7: Collecting field data with Field Maps for ArcGIS

In this exercise, you will:

- *Create a feature layer in ArcGIS Online*
- *Create, and share web maps in ArcGIS Online*
- *Enable attachments for feature layers*
- *Access a web map in Field Maps for ArcGIS*
- *Collect data using Field Maps for ArcGIS*
- *Customize pop-ups in web maps using Arcade expression*

Introduction

Gathering ground control points (GCPs) is an essential part of a drone data collection workflow. While a drone may have its own onboard global navigation satellite system (GNSS) receiver to help it navigate, the accuracy and capability of the onboard receiver are highly variable. As a result, for industrial applications requiring a high level of accuracy (e.g. land surveying, city planning, construction), the GNSS data from the drone should be supplemented with GCP data carefully gathered from the field.



AN EXAMPLE OF A GROUND CONTROL POINT TARGET WITH HIGH-CONTRAST COLORS

To enhance the locational accuracy of your drone data with GCPs, you will first need to create GCP targets that are large, clear, and have high-contrast colors, because you will eventually need to be able to identify these targets from your drone photos. Secondly, you will need to have a sufficient number of GCP targets distributed evenly across your study area. What is considered a sufficient number of targets greatly depends on the complexity, terrain, and size of your study area, but you should generally have at least 5 to 10 GCPs in a study area. Thirdly, you will need to use a high-accuracy GNSS unit, such as a real-time kinematic Global Positioning System (RTK GPS) receiver unit to record the location of each of the GCPs in your study area.

Lastly, you will match the recorded coordinates of each GCP target (provided by the RTK GPS) with its location as it appears in your drone photos.

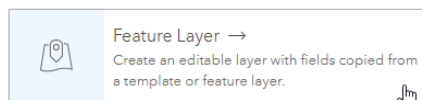


EXAMPLE OF A REAL-TIME KINEMATIC GLOBAL POSITIONING SYSTEM RECEIVER UNIT (SOURCE: ALEXANDER P KAPP / LEICA RTK WITH FINGER POST)

In this exercise, you will create a layer on ArcGIS Online (and use Field Maps for ArcGIS on your mobile device, if one is available) to record the coordinates of several targets that can be used as GCPs.

1. Create a feature layer

- Open an **internet browser** (preferably Chrome), go to www.arcgis.com and **Sign In**.
- Go to the **Content** tab Content near the top of the window, click **New item** + New item and select **Feature Layer**.



- In the Create a feature layer window, select the **Use a template** option to see some common types of features that people collect in the field. Click **Next**.



Use a template

Use the layers and fields from a template.

- d. Select **General** under Industry, and click **Ground Control**. Click **Next**.

The screenshot shows a web interface for selecting an industry. On the left, a list of industries is shown, with 'General' highlighted by a red box. On the right, four options are listed: 'Damage Assessment', 'Field Interviews', 'Field Notes', and 'Ground Control'. 'Ground Control' is selected with a radio button and highlighted by a red box. Below the options, there are 'Back', 'Cancel', and 'Next' buttons. The 'Next' button is highlighted with a red box.

- e. In the Create a feature layer window, click **Next**. Then on the next screen, under options, make sure that **Add GPS metadata fields** is turned on, and **Enable Z-values** is turned off. Click **Next**.

Options

The screenshot shows two toggle options. The first option, 'Add GPS metadata fields', is turned on and highlighted with a red box. The second option, 'Enable Z-values', is turned off and highlighted with a red box.

- f. Lastly, specify some more details for the feature layer you are creating as follows:
- Title: **DroneGCP_(enter the last 4 digits of your mobile number here)**
 - Click **Save**
- g. You should now see the Overview window for the newly created layer. To review what data fields/columns are created with the new layer, click **Data**.

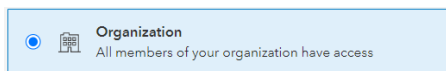
The screenshot shows a navigation bar with five tabs: 'Overview', 'Data', 'Visualization', 'Usage', and 'Settings'. The 'Data' tab is highlighted with a red box.

h. In the Data window, click **Fields**  on the right-hand side of the screen. You can see that there are a lot of fields/columns currently in the feature layer. We will delete some of the fields that we don't need. Check the box next to:

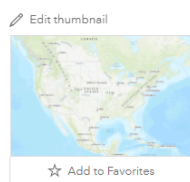
- i. **Speed (km/h)**
- ii. **Direction of travel (°)**
- iii. **Compass reading (°)**
- iv. **Position source type**
- v. **HDOP**
- vi. **VDOP**
- vii. **Average Horizontal Accuracy (m)**
- viii. **Average Vertical Accuracy (m)**
- ix. **Averaged Positions**
- x. **Standard Deviation (m)**

Click **Delete**  **Delete** in the upper right-hand side of the window and **confirm**. You cannot delete some of the fields needed by the system, such as GlobalID, CreationDate, Creator, EditDate, or Editor.

i. Go back to the **Overview** section, and click **Share**. Select **Organization**



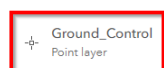
to ensure everyone in your ArcGIS organization can see your layer, and click **Save**. Then click the **layer's name** (i.e. Ground Control) under the Layers section, and make sure that the **Attachments** option is enabled. This will allow you to capture photos to accompany the spatial data (i.e. point locations) you will be gathering soon.



Map for ground control point collection for use in georeferencing aerial drone imagery.
Feature Layer (hosted) by [whcheung](#)
Created: Oct 13, 2022 Updated: Oct 13, 2022 View Count: 1

Description
[Add an in-depth description of the item.](#)

Layers



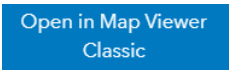
Description
Ground_Control

Credits (Attribution)
[Acknowledge this layers's source.](#)

URL 
<https://services1.arcgis.com/> 

Attachments
Enable Attachments 

2. Create, save, and share a web map

- a. Within the **Overview** tab, select **Open in Map Viewer Classic**  (or click the dropdown and select Open in Map Viewer Classic, if needed). You should now see a web map with your feature layer in the Contents pane (but the layer is currently empty). To access this map within the Field Maps App on your mobile device in the next section of this exercise, click the **Save** icon () and select **Save**.
- b. In the Save Map window, enter the following:
- Title: **GCP(last 4 digits of your mobile phone number)Map**
 - Tags: **GCPmaps**
 - Summary: **Collect GCPs for your drone project with this map**
 - Click **Save Map**

Save Map

Title:

Tags:

Summary:

Save in folder:

- c. Click the **Share** icon  and share the map with Palomar College (or your own organization) by clicking the checkbox next to **Palomar College**. This will grant everyone in the Palomar College ArcGIS Online Organization access to your web map. Click **Done**.

3. Test your map and layer

- a. It is a good practice to make sure that your map and layer are functioning properly prior to going to the field. To do this, click **Edit** in your web map.
- b. In the Add Features pane, click the **Ground_Control** icon  (your icon may have a different color), and then **click anywhere** in your map to manually place the point in the map.

Tip: When you are in the field, the location of the point will be provided by your mobile device's GPS or GNSS.

- c. Type in **Test1** for the Control Point Notes field in the pop-up window associated with your point. **Close** the pop-up window. There is no need to manually save the point edit, since ArcGIS Online automatically saves your edits.

- d. **Repeat** steps b and c to place **three more test points** in your map, so you should have a total of four test points. Remember to enter the Control Point Notes (i.e. Test2, Test3, Test4) for each point.

4. Customize pop-ups with Arcade expressions

- a. You can customize the information that is shown in the pop-up window for each of your points. To do so, click **Details** , then hover your mouse over the **name of your layer** in the Contents pane in the web map, and click the **More Options** icon.



- b. Select **Configure Pop-up**, and then select **Configure Attributes**.

- c. In the Configure Attributes window, you can choose which attributes you want to display by checking/unchecking their Display boxes. You can also choose which attributes you want to allow others to edit by checking/unchecking their Edit boxes. For this exercise, **uncheck all the Edit boxes except for Control**

Point Notes, and only leave the **Display** boxes for the following attributes checked:

- i. **Horizontal Accuracy (m)**
- ii. **Latitude**
- iii. **Longitude**
- iv. **PDOP**

☐ Display	☐ Edit	Field Name	Field Alias
☐	☒	{ControlPointNotes}	Control Point Notes
☐	☐	{ControlPointName}	Control Point Name
☐	☐	{esrignss_receiver}	Receiver Name
☒	☐	{esrignss_h_rms}	Horizontal Accuracy (m)
☐	☐	{esrignss_v_rms}	Vertical Accuracy (m)
☒	☐	{esrignss_latitude}	Latitude
☒	☐	{esrignss_longitude}	Longitude
☐	☐	{esrignss_altitude}	Altitude
☒	☐	{esrignss_pdop}	PDOP
☐	☐	{esrignss_fixtype}	Fix Type

- d. You can also change the names of the fields that appear in the pop-up window by clicking on a name under the Field Alias column, and changing it to a new name. For this exercise, change the Field Alias for **PDOP** to **Position Dilution of Precision**, select **1 decimal place** under Format, and uncheck its **Use 1000 Separator** check box. Click **OK**.

Configure Attributes

Check the fields you want to display and edit. Select a field to change its alias, order it, and format it.

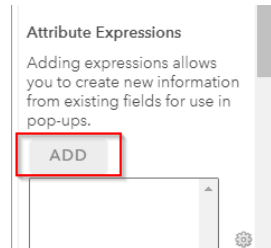
☐ Display	☐ Edit	Field Name	Field Alias
☒	☐	{esrignss_h_rms}	(m)
☐	☐	{esrignss_v_rms}	Vertical Accuracy (m)
☒	☐	{esrignss_latitude}	Latitude
☒	☐	{esrignss_longitude}	Longitude
☐	☐	{esrignss_altitude}	Altitude
☒	☐	{esrignss_pdop}	Position Dilution of Precision

Format

1 decimal place

☐ Use 1000 Separator

- e. Now we will use some values in our existing fields to create an **Arcade expression**, so that an informative custom message can be shown in the pop-up window when someone clicks on a point. Within the Configure Pop-up pane, click **ADD** in the Attribute Expressions section.



- f. When working with Arcade expression, keep in mind that custom text (i.e. those not coming from your layer's fields) must be enclosed in **single quote (')** marks. You can use the **plus (+)** sign to string together different elements.

We will write a simple expression that contains the location of each point along with its notes in the following steps:

- i. We will select the fields that we need for our expression. Click on the **arrow** to the right of the word **\$feature** to see all the fields in your layer.

\$feature



- ii. In the following order, single-click on
- **\$feature.ControlPointNotes**
 - **\$feature["esrignss_latitude"]**
 - **\$feature["esrignss_longitude"]**
- to add these field values to your expression box.

Custom [Edit](#)

Expression

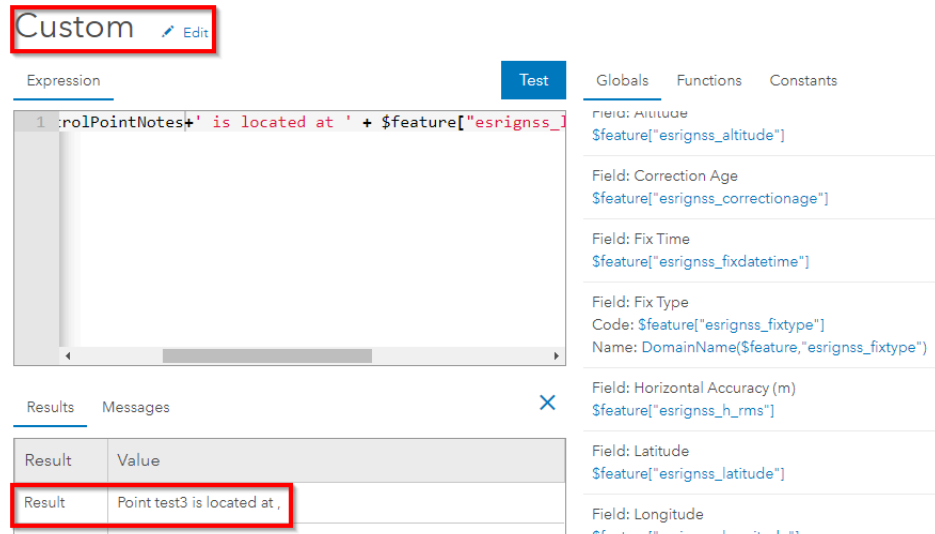
Test

\$feature.ControlPointNotes\$feature.ControlPointID\$feature["esrignss_latitude

- iii. Type in the **bolded and underlined elements** (see below) to add them to your expression within the expression box, so they will form a complete sentence with the fields you selected in the previous step. (Be very careful with spaces and putting the underlined elements below in the correct locations within the expression. Your expression will appear in a single line in the expression box.)

'Point ' + \$feature.ControlPointNotes + ' is located at ' +
\$feature["esrignss_latitude"] + ', ' + \$feature["esrignss_longitude"]

- iv. Click **Test** to preview the Result. Since we did not rename the Expression, it has the default name of Custom. And since the test points were entered manually, they lack the latitude and longitude information, which will be missing in the result message. Click **OK**.



- g. In the Configure Pop-up pane, click **Configure Attributes**. At the bottom of the Configure Attributes window, you should now see a new field with the Field Alias of Custom. Rename its Field Alias from Custom to **Description**.

Configure Attributes

Check the fields you want to display and edit. Select a field to change its alias, order it, format it.

☐ Display	☐ Edit	Field Name	Field Alias
☐	☐	{GlobalID}	GlobalID
☐	☐	{CreationDate}	CreationDate
☐	☐	{Creator}	Creator
☐	☐	{EditDate}	EditDate
☐	☐	{Editor}	Editor
☒	☐	{expression/expr1}	Custom

- h. Once you have renamed the Field Alias to Description, click the **up** arrow to move the Description field to the top of the list of attributes. Click **OK** to close the Configure Attribute window. Click **OK** to close the Configure Pop-up pane.

Configure Attributes

Check the fields you want to display and edit. Select a field to change its alias, order it, and format it.

☐ Display	☐ Edit	Field Name	Field Alias
☐	☐	{GlobalID}	GlobalID
☐	☐	{CreationDate}	CreationDate
☐	☐	{Creator}	Creator
☐	☐	{EditDate}	EditDate
☐	☐	{Editor}	Editor
☒	☐	{expression/expr1}	Description

- i. **Click** on any point in your map to see the new pop-up information. Note that your Latitude and Longitude may be blank in the pop-up window, because your test points were manually placed by you in the previous section, rather than collected with your mobile device's GPS/GNSS

Ground_Control: <i>Without GPS/GNSS Data</i>		Ground Control <i>With GPS/GNSS Data</i>	
Description	Point test2 is located at ,	Description	Point Test 258 is located at 33.8,-116.5
Horizontal Accuracy (m)		Horizontal Accuracy (m)	
Latitude		Latitude	33.80000000
Longitude		Longitude	-116.50000000
Position Dilution of Precision		Position Dilution of Precision	
Attachments:		Attachments:	
No attachments found		No attachments found	

- j. Click the **Save** icon  to save the latest changes to your web map.

Q1. Click the Share icon , write down the web address for your web map.
The instructor will use this link to check your work.

Link to this map

<https://arcg.is/> 

☒ Share current map extent

 Facebook  Twitter

5. Collect data using ArcGIS Field Maps

- a. On your mobile device, **download, install, and open ArcGIS Field**

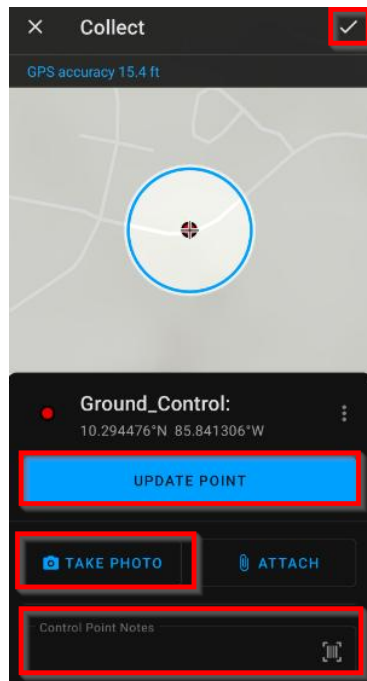
Maps . Let your instructor know if you need to borrow a compatible mobile device.

- b. Once the App is open on your mobile device, tap **Sign in with ArcGIS Online** and sign in. You should be able to see the web map that you created, saved, and shared in the previous sections (i.e. *GCP(last 4 digits of your mobile phone number)Map*). **Tap** the item to open it on your mobile device.

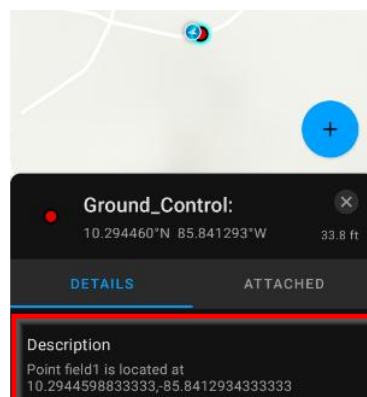
- c. You should now see the web map in the app. You will need to give **location permission** to the app. Click the **add data** icon  in the lower right corner of the map.

Tip: Do not worry if your app's and icon's color is different

- d. In the Collect window,
- Enter **Field1** in **Control Point Notes** for this first point you are collecting with your mobile device's GPS/GNSS in the field.
 - Tap **Take Photo** to include a photo with your point.
 - Tap **Update Point** to ensure you have the latest Latitude and Longitude data for your location
 - Tap the **checkmark** in the upper right corner to submit your location.



- e. **Clicking** on the point you just created in the map will show the information (i.e. Control Point Notes) that you entered within the custom message we configured using arcade expression.



Repeat steps c and d in this section to **collect four additional points** in the field. Remember to enter notes for each point (i.e. Field2, Field3, Field4, Field5) and take photos to go with them. When finished, you should have a total of five points that you collected in the field.

Tip: Preferably these points should be the locations of permanent features (e.g. lamp posts, fire hydrants) that you think will be readily identifiable from a drone flyover and can be used as ground control points

- f. Once you are finished with data collection, **go back to your computer and open your web map** using the web address you recorded for Question #1. You should be able to see each point that you collected in the field automatically added to the web map.

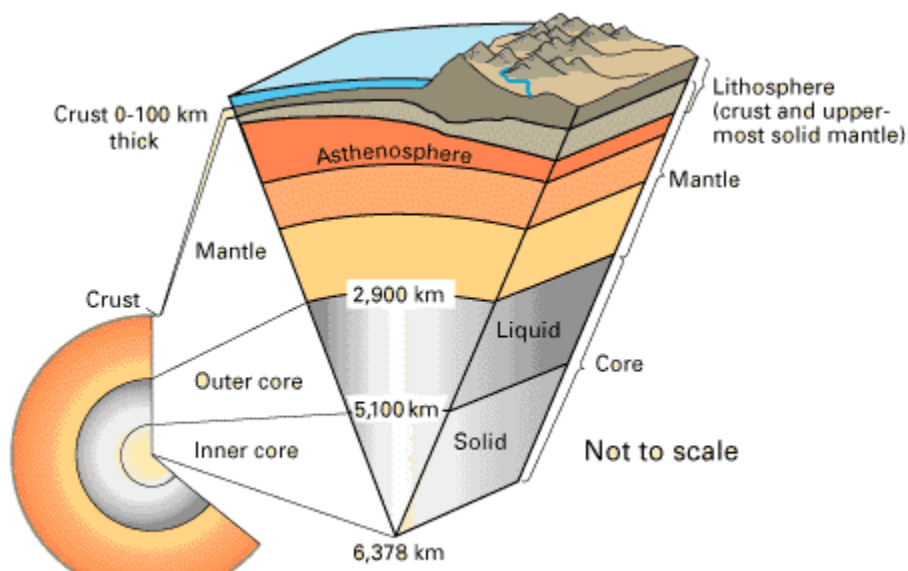
Chapter 8: Natural hazards on the edge

In this exercise, you will:

- Create a new project and map
- Add layers to a new map
- Change the symbology of features
- Repair layer's data source
- Perform definition queries
- Use the near tool to calculate the distance between features
- Explore summary statistics
- Use the Display XY function to map coordinates
- Select features by their locations

Introduction

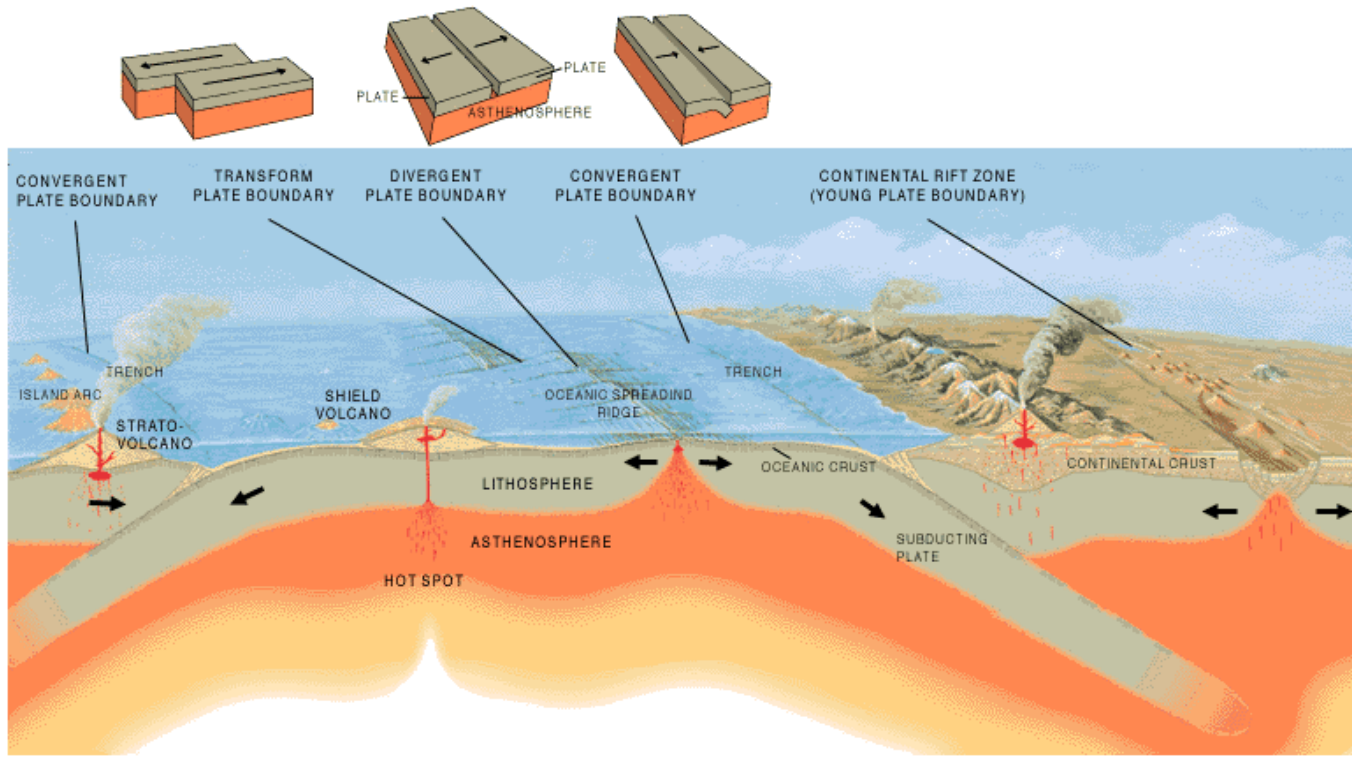
Some scientists have compared the earth's internal structure to an onion, because the earth consists of four main layers that wrap around each other, with the crust being the outermost layer and the inner core being the innermost layer (Williams, 2015).



THE EARTH CONSISTS OF FOUR CONCENTRIC LAYERS, WHICH ARE THE (1) CRUST, (2) MANTLE, (3) OUTER CORE, AND (4) INNER CORE. (SOURCE: USGS)

The crust and the uppermost zone of the mantle make up the solid lithosphere, which is broken into large continent-sized slabs known as tectonic plates (Mcknight and Hess, 2009). These solid plates move slowly over the warm and soft asthenosphere, which sits below the lithosphere (Mcknight and Hess, 2009). As a result of tectonic plate movement, some plates may collide with each other at convergent plate boundaries, other plates may slide past each other at

transform plate boundaries, and other plates (or sections of a plate) may move away from each other at divergent plate boundaries (Mcknight and Hess, 2009).

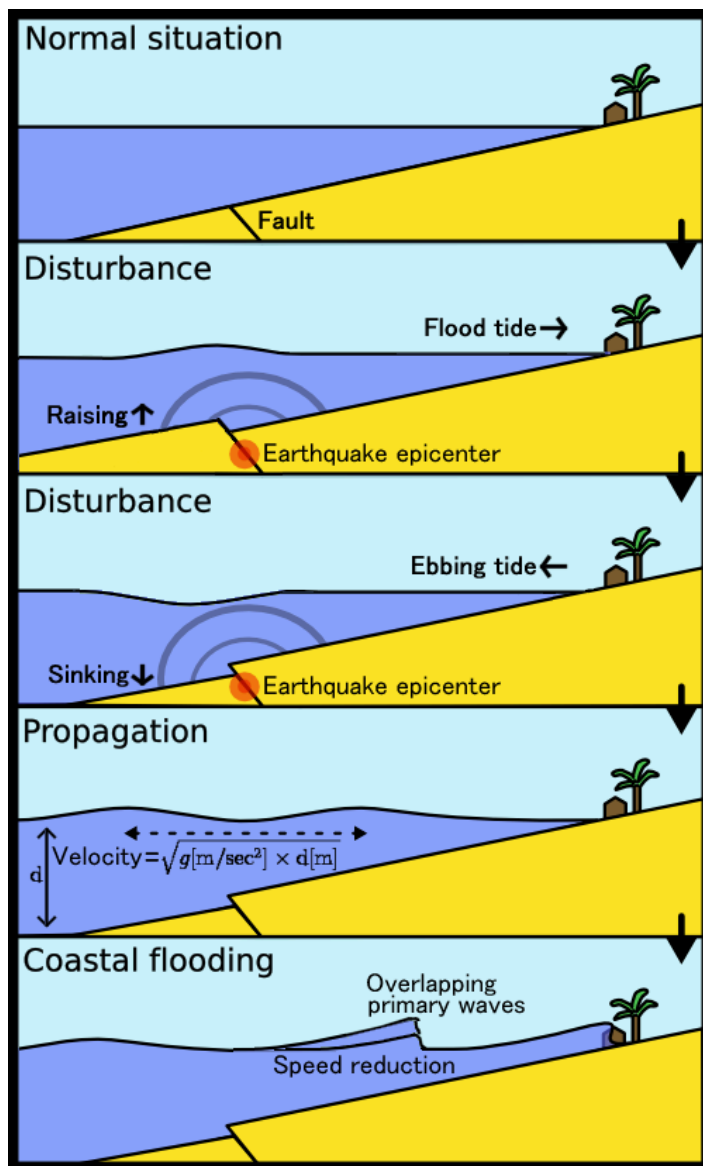


AN OVERVIEW OF DIFFERENT TYPES OF PLATE BOUNDARIES AND LANDSCAPE FEATURES ASSOCIATED WITH THEM. (SOURCE: USGS)

Many of the world's natural hazards occur as a result of the movement of tectonic plates, and tend to be most commonly found at the boundaries where plates may be colliding, sliding past, or pulling apart from each other. For example, earthquakes may occur at a convergent plate boundary, as it abruptly releases the energy that was accumulated at the area of stress between the plates. It is important to remember that earthquakes can occur at other types of plate boundaries (i.e. transform, divergent), and can even occur within a plate or continent in the case of sudden subsidence of the ground.

As a result of plate movements that may be associated with earthquakes, landslides, or volcanic eruptions, large waves of water may be set in motion. These large seismic sea waves are known as Tsunamis, and while they may be hardly noticeable in deep water, they can generate huge waves that may crash onto the shore (Mcknight and Hess, 2009). The largest recorded tsunami wave took place in the aftermath of a 7.8 magnitude earthquake in southeast Alaska on July 9, 1958, which created tsunami waves of up to 1,720 feet at Lituya Bay, Alaska that washed away vegetation, soil, and manmade structures in its path (Dutch, 1999).

To learn more about the movement of tectonic plates and the types of natural hazards caused by these movements, have a look at this [Esri Story Map on the Motion of Tectonic Plates](#).



AN ILLUSTRATION OF HOW TECTONIC PLATE MOVEMENTS SUCH AS EARTHQUAKES AT PLATE BOUNDARIES (FAULT) CAN LEAD TO COASTAL FLOODING AND TSUNAMI WAVES (SOURCE: CC BY-SA 3.0, [HTTPS://COMMONS.WIKIMEDIA.ORG/W/INDEX.PHP?CURID=34403](https://commons.wikimedia.org/w/index.php?curid=34403))

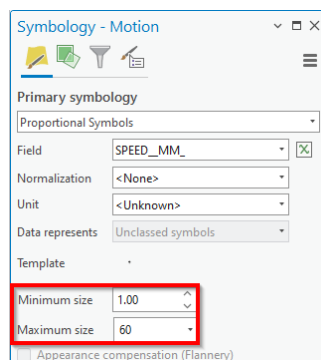
In this exercise, you will examine the locations of the world's major tectonic plates and plate boundaries. You will also explore the locations of historical tsunamis and earthquakes, and analyze the locations of those natural hazards with respect to the locations of plate boundaries.

1. **Start a project, add data, change symbology, and explore data**
 - a. Open **ArcGIS Pro**, and start a new **project** and a new **map**.

- b. Go to **Add Data**, and add the **Plate.shp** layer to your map. This layer shows the locations of the major tectonic plates in the world.
- c. Right-click the layer name (i.e. **Plate**) in the Contents pane, and select **Symbology**. Enter the following settings in the Symbology window in order to display each polygon with a unique PLATE_NAME using a unique color:
 - i. Primary symbology: **Unique Values**
 - ii. Field 1: **PLATE_NAME**
 - iii. Color scheme:  (or something similar)
 - iv. Click **Add all values** 
 - v. **Close** the Symbology window
- d. Go to **Add Data**, and add the **MotionLayer.lyr** layer file (NOT the Motion.shp) to your map. Drag the Motion layer above the Plate layer in the Contents pane to see its features (arrows) in the map. However, the features are not currently showing because the layer's data source is broken. You will fix this in the next step.
- e. To repair the layer's data source, click the **red exclamation mark**  next to the layer name in the Contents pane. Then, in the Change data source window, navigate to the location of the **Motion.shp** file, **select** it, and click **OK**.

Tip: Broken data source is quite common in GIS, and can be caused by a number of factors such as changes in file names, file locations, or files being saved or deleted improperly.
- f. The symbology of the Motion layer is configured such that the size of each arrow corresponds to the speed at which the plate is moving (e.g. bigger arrows represent faster movement), and the orientation of the arrow indicates the direction of the plate's movement. We will adjust the layer's symbology in order to increase the contrast between the size of the arrows.

Right-click the layer name (i.e. **Motion**) in the Contents pane, and select **Symbology**. In the Symbology window, enter **1** for the Minimum size, and enter **60** for the Maximum size.



Q1. Based on the relative size of the arrows indicated in the Motion layer, which polygon (i.e. plate) within the Plate layer that is **located almost entirely south of the equator** is moving the fastest? (Click on the polygon to get its name in the pop-up window)

Q2. Which direction is the plate identified in the previous question moving?

Q3. The margin between the South American plate and the Nazca plate is an example of which type of plate boundaries (select the correct answer):

- Divergent
- Convergent
- Transform

Q4. The margin between the African plate and the South American plate is an example of which type of plate boundaries (select the correct answer):

- Divergent
- Convergent
- Transform

2. Analyze data with definition query, summary statistics, and the Near tool

- Go to **Add Data**, and add the **PlateBoundary.shp** layer to your map. This layer shows the locations of the world's divergent, convergent, and transform plate boundaries.
- Modify and repeat** what you did in step 1c to symbolize each unique type of plate boundary (i.e. MARGIN_TYP) with a unique color.
- Go to **Add Data**, and add the **Tsunami.shp** layer to your map. This layer shows the locations of Tsunami events since the year 1900.
- Visually **examine** the relationship between the tsunami points (i.e. Tsunami) and the locations of the different types of plate boundaries (i.e. PlateBoundary). **Turn off** the Motion and Plate layer to help you better visualize the relationship.

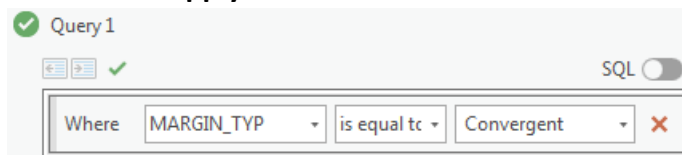
Q5. Most tsunami events occur near which type of plate boundaries (select the correct answer):

- Divergent
- Convergent
- Transform

- e. To verify your answer to the previous question, we will calculate the average distance between the tsunami points and each of the three types of plate boundaries in the following steps.

We will start with the convergent plate boundaries. Right-click on **PlateBoundary** in the Contents pane, and select **Properties**. Select **Definition Query** on the left. In the Definition Query section of the Layer Properties window, click **New definition query**, and enter the following:

- i. Select **MARGIN_TYP** for the first dropdown.
- ii. Select **is equal to** for the second dropdown.
- iii. Select **Convergent** for the third dropdown.
- iv. Click **Apply**. Click **Ok**.



Now you should only see the convergent plate boundaries being displayed.

- f. In order to calculate the average distance between each tsunami point and its nearest convergent plate boundary, go to the **Analysis** tab, and select **Tools**. In the Geoprocessing pane, search for **Near**. Select **Near (Analysis Tools)**.
- g. In the Near window, enter the following:
 - i. Input Features: **Tsunami**
 - ii. Near Features: **PlateBoundary**
 - iii. Click **Run**
- h. In the Contents pane, right-click on the **Tsunami** layer, and select **Attribute Table**.
- i. Find the NEAR_DIST field in the attribute table, this is the distance between a tsunami point and its nearest convergent plate boundary (calculated by the Near tool). Right-click on the field name **NEAR_DIST**, and select **Statistics**.

Q6. What is the mean distance between the tsunami points in the layer and their nearest convergent plate boundaries (in meters) (round to nearest whole number)?

- j. **Modify and repeat** steps e thru i above in order to find the mean (average) distance between each tsunami point and its nearest **divergent** plate boundary, as well as the average distance between each tsunami point and its nearest **transform** plate boundary. To complete this task, you need to **edit the definition query** that you built in step e to select the divergent plate margins (as opposed

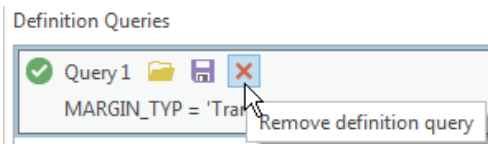
to the convergent ones), **run the Near tool**, and **calculate Statistics** (i.e. steps f thru i). Repeat this entire process for the transform plate boundaries as well.

Q7. What is the mean distance between the tsunami points in the layer and their nearest divergent plate boundaries (in meters) (round to nearest whole number)?

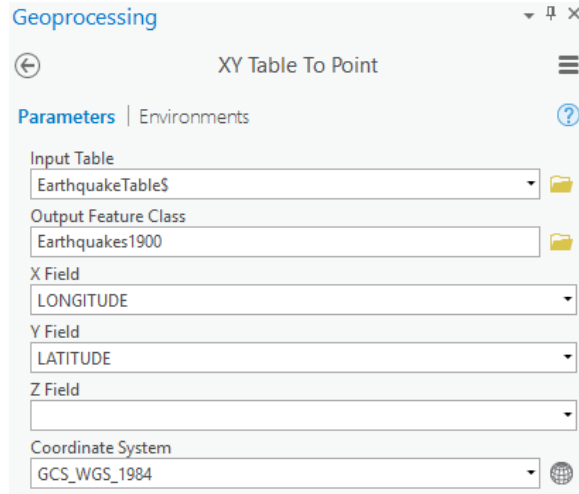
Q8. What is the mean distance between the tsunami points in the layer and their nearest transform plate boundaries (in meters) (round to nearest whole number)?

3. Map points using XY coordinates

- a. We will first clear our definition query, because we need to see all of the plate boundaries for the following steps. In the Contents pane, right-click **PlateBoundary** and select **Properties**. Select **Definition Query**. In the Definition Query section of the window, **hover** your mouse over any existing queries, and click the **red X** to delete them. **Confirm** your decision. Click **OK**.



- b. Go to **Add Data**, and add the **EarthquakeTable\$** table to your map (double click on the EarthquakeTable.xls to see the EarthquakeTable\$). This table recorded the earthquakes that occurred between 1900 and 2016.
- c. Right-click on the **EarthquakeTable\$** table in the Contents pane, and select **Display XY Data**.
- d. In the XY Table to Point window, enter the following:
 - i. Input Table: **EarthquakeTable\$**
 - ii. Output Feature Class: **Earthquakes1900**
 - iii. X Field: **LONGITUDE**
 - iv. Y Field: **LATITUDE**
 - v. Click **OK**



- e. In the Contents pane, **turn off** all layers except PlateBoundary, Earthquakes1900, and World Topographic Map.
- f. To explore the relationship between earthquake locations and plate boundary type, we will first run a definition query on PlateBoundary layer to only show convergent plate boundaries (as you did in the previous section).

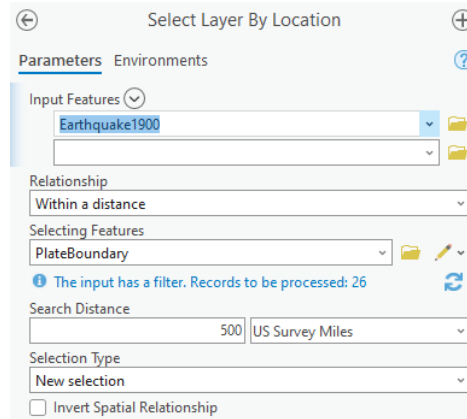
Right-click on **PlateBoundary** in the Contents pane, and select **Properties**. In the Definition Query section of the Layer Properties window, click **New definition query**, and enter the following:

- i. Select **MARGIN_TYP** for the first dropdown.
 - ii. Select **is equal to** for the second dropdown.
 - iii. Select **Convergent** for the third dropdown.
 - iv. Click **Apply**. Click **OK**.
- g. To see the number of earthquake points that are reasonably close (within 500

miles) to convergent plate boundaries, click on **Select By Location** within the **Map** tab. In the Select By Location window, enter the following:

- i. Input Feature Layer: **Earthquake1900**
- ii. Relationship: **Within a distance**
- iii. Selecting Features: **PlateBoundary**
- iv. Search Distance: **500 US Survey Miles**
- v. Click **Apply**





- h. Open the **Earthquakes1900** layer's attribute table to see how many earthquake points are selected (i.e. within 500 miles of a convergent plate boundary).

Q9. How many earthquake points are within 500 US Survey Miles of a convergent plate boundary?

- i. Within the Earthquakes1900 layer's attribute table, clear your current selection by clicking on the **clear selection** icon. Close the attribute table.

Field:	Selection:							
OBJECTID	Shape	OBJECTID	FLAG_TSUNA	YEAR	MONTH	DAY	hour	
1	Point	20076	Tsu	1900	1	10	0	

- j. **Modify and repeat** steps f thru i above to find the number of earthquake points that are reasonably close (within 500 US Survey Miles) to **divergent** plate boundaries and **transform** plate boundaries. To complete this task, you need to first **edit your definition query** in step f to show divergent plate margins, run the **Select Layer by Location** tool and examine your results (i.e. steps g thru i). Remember to **clear your selection** before repeating this process for transform plate boundaries as well.

Q10. How many earthquake points are within 500 US Survey Miles of a divergent plate boundary?

Q11. How many earthquake points are within 500 US Survey Miles of a transform plate boundary?

Works Cited

Dutch, Steven. "Giant Waves in Lituya Bay, Alaska." *Archive.Org*, Internet Archive, 29 Dec. 1999, web.archive.org/web/20131013223418/www4.uwsp.edu/geo/projects/geoweb/participants/Dutch/LituyaBay/Lituya0.HTM. Accessed 28 June 2019.

Mcknight, Tom L, and Darrel Hess. *Physical Geography*. Boston, Ma, Pearson Education, 2009.

Williams, Matt. "What Are the Earth's Layers?" *Phys.Org*, Universe Today, 7 Dec. 2015, phys.org/news/2015-12-earth-layers.html. Accessed 28 June 2019.

Chapter 9: Exploring the world's water resources

In this exercise, you will:

- *Create a new project and map*
- *Add layers to a new map*
- *Rename layers*
- *Change the symbology of features*
- *Import symbology from other layers*
- *Explore summary statistics*
- *Work with vector geoprocessing tools (i.e. Clip, Intersect)*
- *Share the map as a PDF*

Introduction

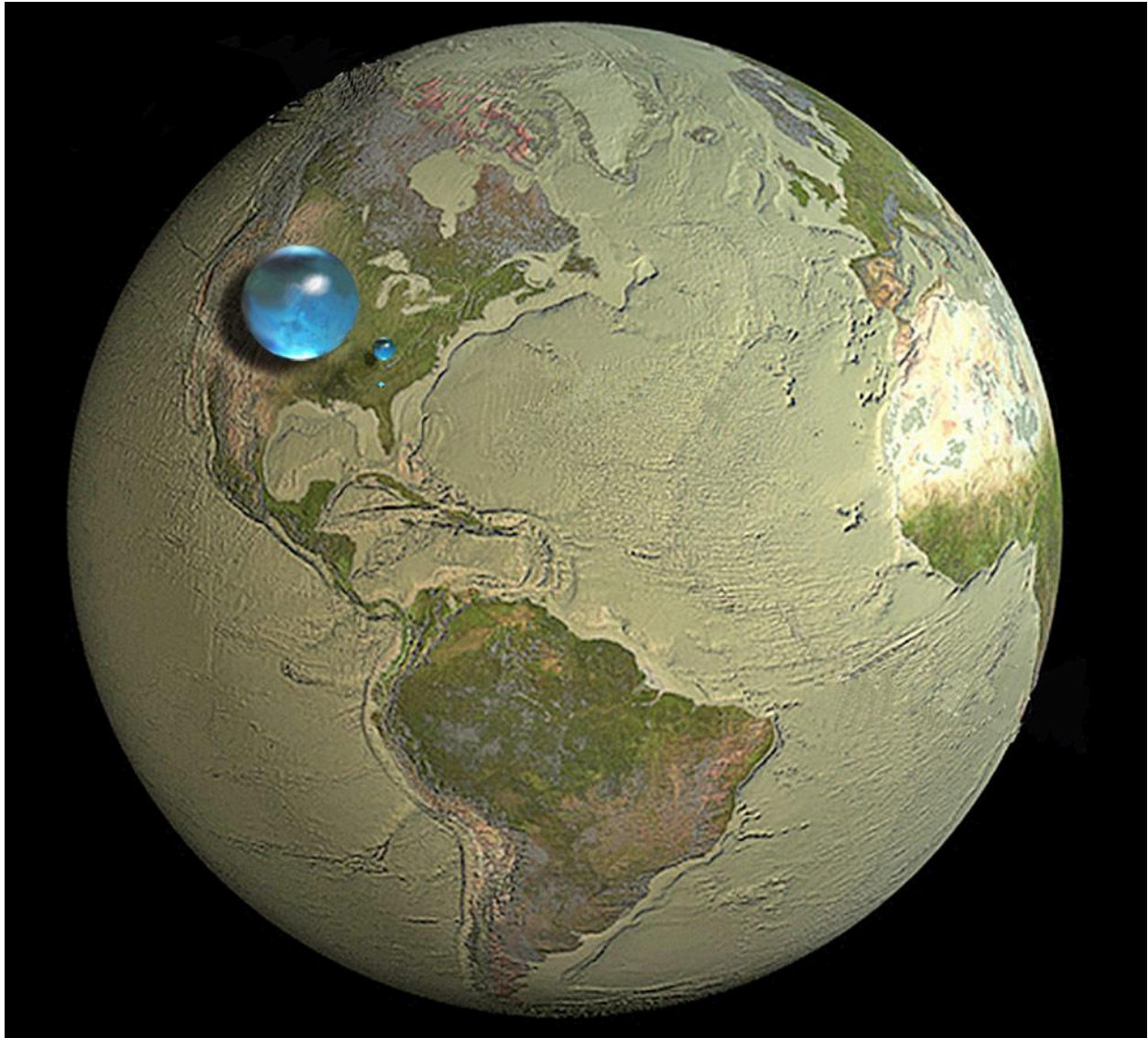
Water covers approximately 71 percent of the earth's surface (Perlman, 2005). Water can exist in three states or phases, which are:

(1) liquid (e.g. groundwater),

(2) solid (e.g. ice caps), and

(3) gases (e.g. water vapor in the atmosphere).

*In the following graphic, the biggest water sphere (about 332,500,000 cubic miles) represents the volume of **all water** found in oceans, ice caps, lakes, rivers, groundwater, atmospheric water, and water in living organisms (plants, people) relative to the volume of the earth. The second biggest water sphere (about 2,551,100 cubic miles) represents the world's **liquid freshwater** found in groundwater, lakes, swamp water, and rivers. The tiny water sphere (about 22,339 cubic miles) represents **surface freshwater** resources found in the world's lakes and rivers (Perlman, 2016).*



RELATIVE COMPARISON OF THE WORLD'S TOTAL WATER RESOURCES (LARGEST WATER SPHERE), THE WORLD'S LIQUID FRESHWATER RESOURCES (MEDIUM WATER SPHERE), AND THE WORLD'S SURFACE FRESHWATER RESOURCES (SMALLEST WATER SPHERE). (SOURCE: USGS)

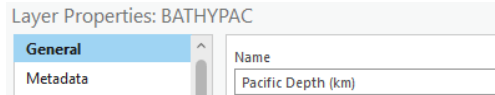
*We can find a majority of the world's water in **oceans**, which holds 96.5 percent of the earth's water. Meanwhile, the world's **ice caps, glaciers, and permanent snow** hold about 1.74 percent of the earth's water, and **groundwater** accounts for 1.69 percent of the earth's water (Perlman, 2005). Surface water sources such as **lakes, rivers, and the atmosphere** collectively constitute less than 1 percent of the earth's water. In fact, it is estimated that lakes, rivers, and the atmosphere only hold 0.013 percent, 0.0002 percent, and 0.001 percent of the earth's water, respectively. The following table shows the percentage of the world's water that you may find in different places, as well as the percentage of the world's freshwater found in these locations.*

<i>Water source</i>	<i>Percent of freshwater</i>	<i>Percent of total water</i>
<i>Oceans, Seas, & Bays</i>	--	96.54
<i>Ice caps, Glaciers, & Permanent Snow</i>	68.6	1.74
<i>Fresh Groundwater</i>	30.1	0.76
<i>Saline Groundwater</i>	--	0.93
<i>Soil Moisture</i>	0.05	0.001
<i>Ground Ice & Permafrost</i>	0.86	0.022
<i>Freshwater Lakes</i>	0.26	0.007
<i>Saline Lakes</i>	--	0.007
<i>Atmosphere</i>	0.04	0.001
<i>Swamp Water</i>	0.03	0.0008
<i>Rivers</i>	0.006	0.0002
<i>Biological Water</i>	0.003	0.0001
<i>Adapted from Igor Shiklomanov's chapter "World fresh water resources" in Peter H. Gleick (editor), 1993, Water in Crisis: A Guide to the World's Fresh Water Resources (Oxford University Press, New York).</i>		

In this exercise, you will explore three common sources of the world's water resources: (1) the world's oceans, (2) continental ice caps, and (3) the atmosphere. Specifically, you will approximate the volume of water that is stored in each of these three sources.

1. Start a project, rename layers, add data, and change and import symbology

- a. Open **ArcGIS Pro**, and start a new **project** and a new **map**.
- b. Go to **Add data**, and add the following layers to your map. These layers show the bathymetry of seafloors for the world's oceans.
 - i. **BATHYPAC**
 - ii. **BATHYIND**
 - iii. **BATHYATL**
 - iv. **BATHYARC**
 - v. **BATHYSOU**
- c. In the Contents pane, right-click **BATHYPAC**, and select **Properties**. In the General section of the Layer Properties window, change the **Name** from BATHYPAC to **Pacific Depth (km)**. Click **OK**.

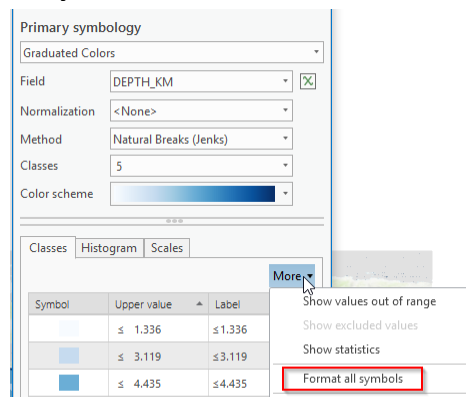


Repeat this procedure to change the names of the other layers:

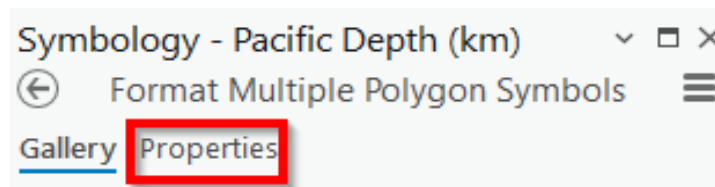
- i. **BATHYIND** → **Indian Depth (km)**
 - ii. **BATHYATL** → **Atlantic Depth (km)**
 - iii. **BATHYARC** → **Arctic Depth (km)**
 - iv. **BATHYSOU** → **Southern Depth (km)**
- d. Follow the directions below to symbolize each bathymetry layer using a blue color ramp that is more consistent with how we perceive ocean features.

Select **Pacific Depth (km)** in the Contents pane, click on the **Feature Layer** tab, and select **Symbology**. In the Symbology window, enter the following:

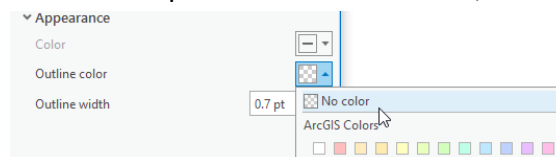
- i. Primary symbology: **Graduated Colors**
- ii. Field: **DEPTH_KM**
- iii. Method: **Natural Breaks (Jenks)**
- iv. Classes: **5**
- v. Color scheme: 
- vi. Click the **More** dropdown in the middle of the window, and select **Format all symbols**



- vii. In the Format Multiple Polygon Symbols window, select **Properties**

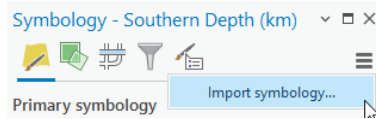


- viii. Click the dropdown for Outline color, and select **No color**.

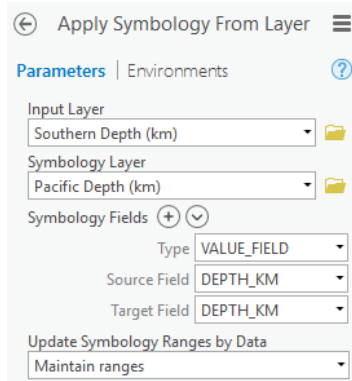


- ix. Hit **Apply** to save your changes. **Close** the Symbology window
- e. To apply the symbology from the Pacific Depth (km) layer to the other layers in your map, right-click on the other layer's name (e.g. **Southern Depth (km)**) in the

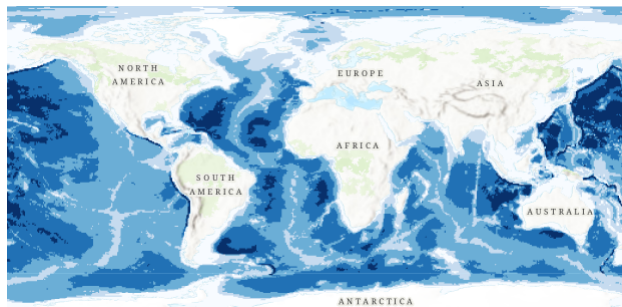
Contents pane. Select **Symbology**. Click the **hamburger button** , and select **Import symbology...**



- f. In the Apply Symbology From Layer window, ensure that **Southern Depth (km)** is the Input Layer, select **Pacific Depth (km)** for the Symbology Layer, and click **Run**.



- g. **Modify and repeat** the previous step to change the symbology for the remaining layers except for the topographic and hillshade layers (i.e. the remaining bathymetry layers), so they match the symbology of the Pacific Depth (km) layer.



2. Explore your data and summary statistics

- a. In the Contents pane, right-click on the **Pacific Depth (km)** layer, and select **Attribute Table**.
- b. Within the attribute table, right-click the column name of **DEPTH_KM**, and select **Statistics**. This field/column stores the depth information for various locations in the Pacific Ocean in kilometers.

Record the Mean DEPTH_KM value for the Pacific Depth (km) layer in **Table 1** (round to 1 decimal place).

- c. **Modify and repeat** steps a and b in this section for the other ocean depth layers.

Record the Mean DEPTH_KM value for the other ocean depth layers in **Table 1** (round to 1 decimal place). Submit a copy/photo of the completed **Table 1** as a part of your assignment.

Q1. TABLE 1	Mean Depth (km)	Surface Area (km²)	Volume (km³)
Pacific Ocean		172000000	
Southern Ocean			
Indian Ocean			
Atlantic Ocean			
Arctic Ocean			

- d. Go to **Add data**, and add the **5BASINS** layer to your map. This layer shows the rough extents of the 5 major ocean basins.
- e. Open the 5BASINS layer's **attribute table**, and find each ocean's surface area value under **AREA_KM2** column.

Copy each ocean's surface area value over to Table 1 (be careful to copy over the correct number of zeros). The surface area value for Pacific Ocean has been entered for you in Table 1 as an example.

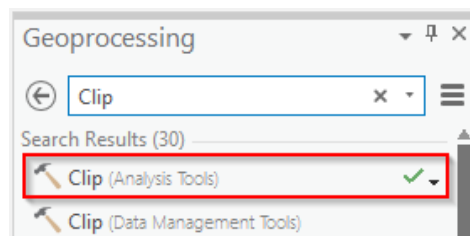
Multiply each ocean's mean depth value with its corresponding surface area value to approximate each ocean's volume, and carefully record it under the **Volume (km³)** column in Table 1.

Q2. Sum the volume values across the 5 oceans recorded in Table 1, this is the approximate volume of water that is stored in the world's 5 major oceans. What is the sum (in km³)?

3. Analyze water vapor data with the Clip tool

- a. Go to **Add data**, and add the following layers to your map. (Hold down the Ctrl key on your keyboard while selecting multiple items to add them to your map simultaneously.)
 - i. **H2OVAPOR** (layer showing the volume of water vapor across the world)
 - ii. **ICE_ANT** (layer showing the location of the Antarctic continent and ice sheet)
 - iii. **ICE_GRN** (layer showing the location of Greenland and its ice sheet)
- b. In order to compare the volume of water vapor over Antarctica, Greenland, and the world's oceans, we will first use the Clip tool to extract the sections of the H2OVAPOR layer that coincide with the ICE_ANT layer.

Go to the **Analysis** tab, and select **Tools**. In the Geoprocessing pane, type in **Clip**. Select **Clip (Analysis Tools)**.



- c. In the Clip window, enter the following:
 - i. Input Features or Dataset: **H2OVAPOR** (i.e. this is the cookie dough, we only want a part of this layer)
 - ii. Clip Features: **ICE_ANT** (i.e. this is the cookie cutter, this defines the specific part of the input layer that we want)
 - iii. Output Features or Dataset: **vaporclipant** (i.e. this is the cookie, the parts of the input layer that is inside the clip feature)
 - iv. Click **Run**
- d. **Turn off** all the layers in the Contents pane except the vaporclipant layer to see your output. Open the vaporclipant layer's **Attribute Table**, right-click the column name **VAP_KM3**, and select **Statistics**. This field/column stores the volume of water vapor recorded in specific parts of the world.

Q3. What is the mean water vapor volume over Antarctica (in km³)? (Round to three decimal places)

- e. **Modify and repeat** steps b thru d in this section with the **ICE_GRN** and the **5BASINS** layers as the Clip Features to find the mean water vapor volume over Greenland and the world's oceans, respectively. Name the output for the Greenland water vapor layer **vaporclipgrn**, and the output for the oceans water vapor layer **vaporclipoceans**.

Q4. What is the mean water vapor volume over Greenland (in km³)? (Round to three decimal places)

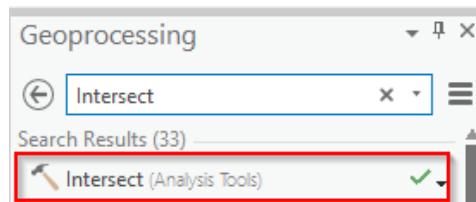
Q5. What is the mean water vapor volume over the world's oceans (in km³)? (Round to three decimal places)

Review the attribute table for the oceans water vapor layer (i.e. vaporclipoceans), note that while you can find the average water vapor volume over all of the world's oceans, there is no way for us to find out the volume of water vapor over a specific ocean (e.g. Pacific Ocean) or compare the water vapor volume over different oceans. We will do this in the next section.

4. Analyze water vapor data with the Intersect tool

- a. To compare the water vapor volume over specific oceans, we need to get an output that tells us which part of the H2OVAPOR layer coincided with which specific ocean feature within the 5BASIN layer.

Go to the **Analysis** tab, and select **Tools**. In the Geoprocessing pane, type in **Intersect**. Select **Intersect (Analysis Tools)**.



- b. In the Intersect window, enter the following:
 - i. Input Features: select **H2OVAPOR**, then select **5BASINS** (leave the Ranks empty)
 - ii. Output Feature Class: **vaporinteroceans**
 - iii. Click **Run**
- c. **Compare** the attribute tables for the **vaporclipoceans** and the **vaporinteroceans** layer.

Q6. List some of the field/column names that are present in the vaporinteroceans layer, but are missing in the vaporclipoceans layer.

- d. To calculate the water vapor volume over a specific ocean basin, we will first select the rows in the vaporinteroceans layer that are associated with a specific ocean and then calculate its statistics. To reduce clutter, **turn off** all layers in the Contents pane except the vaporinteroceans layer.

Go to the **Map** tab, and click **Select By Attribute**. In the Select Layer By Attribute window, enter the following:

- i. Input Rows: **vaporinteroceans**
- ii. Selection type: **New selection**
- iii. Expression:
 1. First dropdown: **OCEAN_NAME**
 2. Second dropdown: **is equal to**
 3. Third dropdown: **Pacific Ocean**
- iv. Click **Apply**

You should now see the features in the vaporinteroceans layer that coincide with the Pacific Ocean are selected in your map.

- e. Open the **vaporinteroceans** layer's Attribute Table, right-click the column name of **VAP_KM3**, and select **Statistics**. To get statistics that are calculated based solely on the selected features, make sure that the **Filter by Selection** icon in the Chart window has

been selected.  Selection  Extent

Q7. What is the mean water vapor volume over the Pacific Ocean (in km³)?
(Round to three decimal places) (Be sure to get this value in the chart window
or under the selection column, not the dataset column, in the Chart Properties
pane)

- f. **Repeat and modify** the expression in step d to find out the mean water vapor volume over the Arctic Ocean, the Atlantic Ocean, the Indian Ocean, and the Southern Ocean.

Q8. What is the mean water vapor volume over the Arctic Ocean (in km³)?
(Round to three decimal places) (Be sure to get this value in the chart window
or under the selection column, not the dataset column, in the Chart Properties
pane)

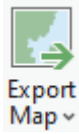
Q9. What is the mean water vapor volume over the Atlantic Ocean (in km³)?
(Round to three decimal places) (Be sure to get this value in the chart window
or under the selection column, not the dataset column, in the Chart Properties
pane)

Q10. What is the mean water vapor volume over the Indian Ocean (in km³)? (Round to three decimal places) (Be sure to get this value in the chart window or under the selection column, not the dataset column, in the Chart Properties pane)

Q11. What is the mean water vapor volume over the Southern Ocean (in km³)? (Round to three decimal places) (Be sure to get this value in the chart window or under the selection column, not the dataset column, in the Chart Properties pane)

5. Share your map as a PDF document

- i. Go to the **Contents** pane, and **turn off** all layers except Pacific Depth (km), Indian Depth (km), Atlantic Depth (km), Arctic Depth (km), Southern Depth (km), and World Topographic Map. Mark sure that your layers are properly symbolized as instructed in section 1.
- ii. To share your map as a PDF document, go to the **Share** tab, and click **Export**



- iii. In the Export Map window, be sure to select **PDF** as the File Type, specify the **Name** and **location** for your file, and hit **Export**.

Q11. Submit a copy of your exported PDF map for this assignment.

Works Cited

Perlman, Howard. "How Much Water Is There on Earth." *The USGS Water Science School*, US Geological Survey, 2005, water.usgs.gov/edu/earthhowmuch.html.

Perlman, Howard. "Water Science Photo Gallery." *The USGS Water Science School*, US Geological Survey, 2016, water.usgs.gov/edu/gallery/global-water-volume.html. Accessed 27 June 2019.

Chapter 10: Visualizing global climate change under different emission scenarios

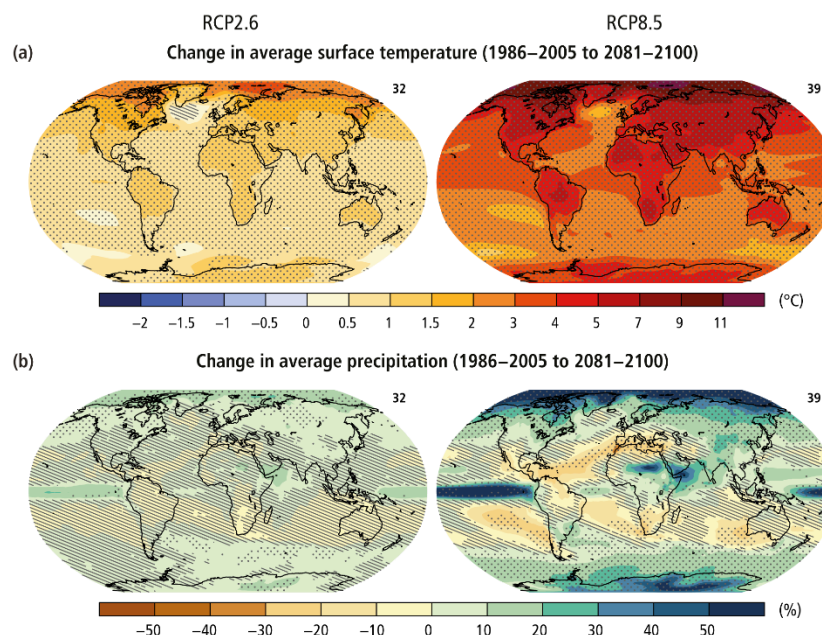
In this exercise, you will:

- Create a new project and map
- Add layers to a new map
- Change the symbology of features
- View basic metadata information
- Edit metadata information using the Contents pane and the Catalog pane
- Change metadata stylesheet
- Verify metadata compliance with established standards

Introduction

The Intergovernmental Panel on Climate Change (IPCC) is an intergovernmental organization made up of member states from the United Nations and the World Meteorological Organization (IPCC, 2019). The thousands of staff members and scientists who work on behalf of the IPCC assess and review scientific findings, in order to provide governments with updated information to develop policies to mitigate the risks associated with climate change (IPCC, 2019).

The IPCC's assessments of the future impacts of climate change are based on four representative concentration pathways (RCPs) (i.e. scenarios), which take into account a variety of natural (e.g. natural system feedback) and man-made (e.g. greenhouse gas emission) contributing factors (Alexander et al., 2013).



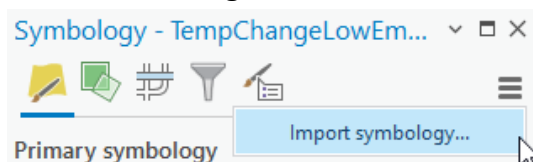
Forecasted long-term changes to average surface temperature (a) and average precipitation (b) in 2081-2100 relative to conditions in 1986-2005 under two scenarios (RCP 2.6, RCP 8.5).

In this exercise, we will look at climate change's projected impact on global temperatures under two of the four RCPs adopted by the IPCC. Specifically, in the "best case scenario" (i.e. RCP 2.6), it assumes that global carbon dioxide emissions will peak soon and decline, the global population growth is minimal, and there will be a high level of global economic growth (Furphy, 2013). In contrast, the "worst case scenario" (i.e. RCP 8.5) assumes that emissions and population will continue to increase rapidly, while economic growth will be much slower, especially among lower-income populations (Furphy, 2013).

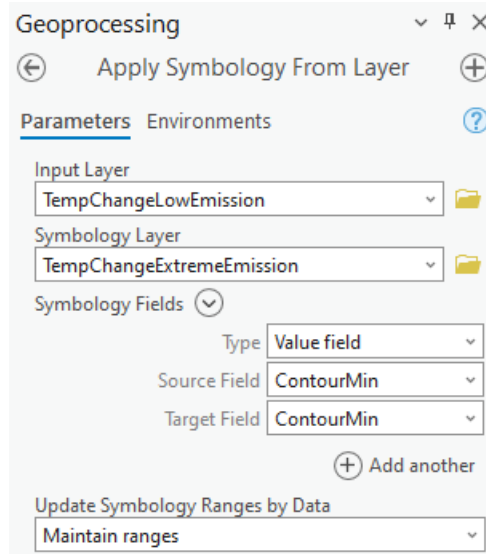
1. Start a project, add data, and change and import symbology

- a. Open **ArcGIS Pro**, and start a new **project** and a new **map**.
- b. Go to **Add data** (found under the Map tab), and add the **TempChangeExtremeEmission** and **TempChangeLowEmission** layers to your map. These layers show two scenarios of how the global temperatures (in Celsius) may be warmer (positive values) or cooler (negative values) by the year 2040-2059 in comparison to temperatures recorded in earlier periods (1986-2005). The scenarios (i.e. Extreme Emission, Low Emission) were compiled by the Intergovernmental Panel on Climate Change.
- c. Select **TempChangeExtremeEmission** in the Contents pane, click on the **Feature Layer** tab, and select **Symbology**. In the Symbology window, enter the following:
 - i. Primary symbology: **Graduated Colors**
 - ii. Field: **ContourMin**
 - iii. Method: **Natural Breaks (Jenks)**
 - iv. Classes: **5**
 - v. Color scheme: 
 - vi. **Close** the Symbology window.
- d. We will apply the symbology from the TempChangeExtremeEmission layer to the TempChangeLowEmission layer so that the same colors represent the same levels of temperature changes across both layers. Right-click on the **TempChangeLowEmission** layer's name in the Content pane. Select **Symbology**.

Click the **hamburger button** , and select **Import symbology...**

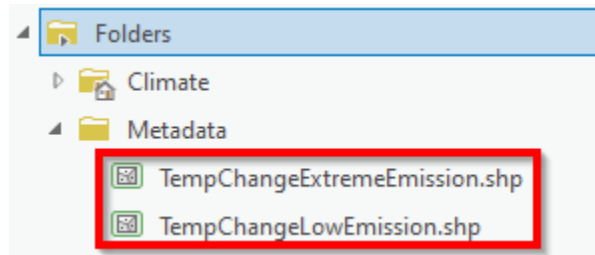


- e. In the Apply Symbology From Layer window, select **TempChangeLowEmission** as the Input Layer, **TempChangeExtremeEmission** for the Symbology Layer, and click **Run**.



2. View basic metadata in the Catalog pane pop-up

- Open the Catalog pane by going to **View>Catalog Pane**.
- In the Catalog pane, right-click on **Folders** and select **Add Folder Connection** to make a connection to the **Folder** (not individual files) where you stored your data (i.e. the Metadata folder) for this exercise.
- Expand** the Folders icon in the Catalog Pane to see your new folder connection. Find the **TempChangeExtremeEmission** and **TempChangeLowEmission** layers within the folder, and **hover** your mouse pointer over each of them. You should see a pop-up with basic information for each layer.

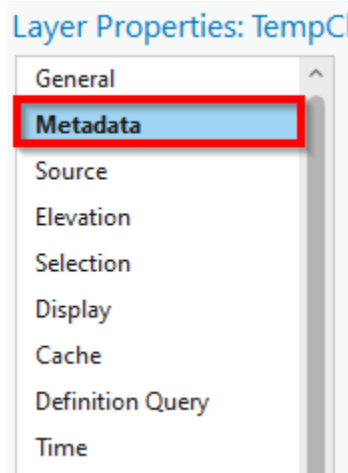


Q1. What is the File Type (i.e. Type) for the TempChangeExtremeEmission layer?

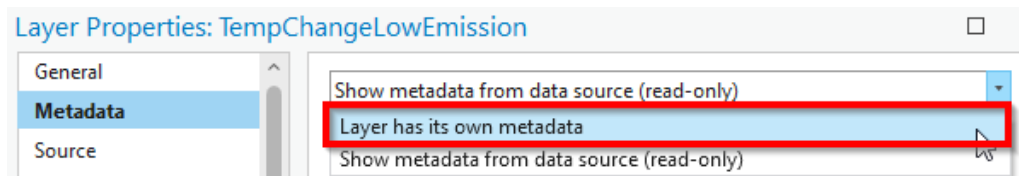
Q2. What is the File Type (i.e. Type) for the TempChangeLowEmission layer?

3. View and edit metadata from the Contents pane

- a. Go back to the Contents pane (not the Catalog pane), right-click the **TempChangeLowEmission** layer, and select **Properties**.
- b. Click **Metadata** found on the left side of the Layer Properties window.



- c. Within the Metadata section, on the right side of the window, select **Layer has its own metadata**.



- d. **Fill out** the Metadata section by copying and pasting from the contents below.
(Source: <https://www.arcgis.com/home/item.html?id=b92294d54afc48499e1a8db479023e72>)

- i. Title

Feature layer showing projected temperature and precipitation anomaly for RCP 2.6 from 2040 to 2059 using a CMIP5 multimodel ensemble of ten climate models.

- ii. Tags

[climate](#), [climate change](#), [temperature](#), [Precipitation](#), [IPCC](#), [model](#), [RCP](#), [CMIP](#)

- iii. Summary

RCPs (Representative Concentration Pathways) represent the ratio of energy absorbed by the Earth's atmosphere to the energy reflected back into space in watts per square meter. Climate scientists call this radiative forcing. As levels of greenhouse gasses increase or decrease in the atmosphere, radiative forcing values will correspondingly increase or

decrease. As radiative forcing values increase, the atmosphere becomes warmer because more energy (heat) is retained in the atmosphere.

In 2014, the IPCC (Intergovernmental Panel on Climate Change) adopted four standard RCPs: 2.6, 4.5, 6.0, and 8.5 W/m². These scenarios give us a range from the best case (2.6) to the worst (8.5) for greenhouse gasses being emitted into the atmosphere by human and natural sources.

RCP 2.6 is a "peak and decline" scenario where radiative forcing levels not only peak prior to, but decline to 2.6 W/m² by 2100 (Van Vuuren, D.P., et al., 2011). Scenario 2.6 is no longer considered a realistic possibility.

Global climate projections are produced by general circulation models (GCMs). This particular layer was derived from the mean results of 10 of the best GCMs, called a multi-model ensemble (MME). Esri partnered with the Research Applications Laboratory (RAL) from the National Center for Atmospheric Research (NCAR) to select the model members.

iv. Description

This layer was generated from data associated with the Learn ArcGIS Lesson "Explore Future Climate Projections."

A projected climate anomaly is the difference between the current climate, or "baseline" conditions, and a future climate scenario, or "projection". This layer contains contoured features representing annual precipitation (mm) and temperature (°C) anomalies. These are computed for 2040 to 2059 using a scenario called "RCP 2.6". In this case, the baseline conditions are from 1986 to 2005.

v. Credits

NCAR, Esri

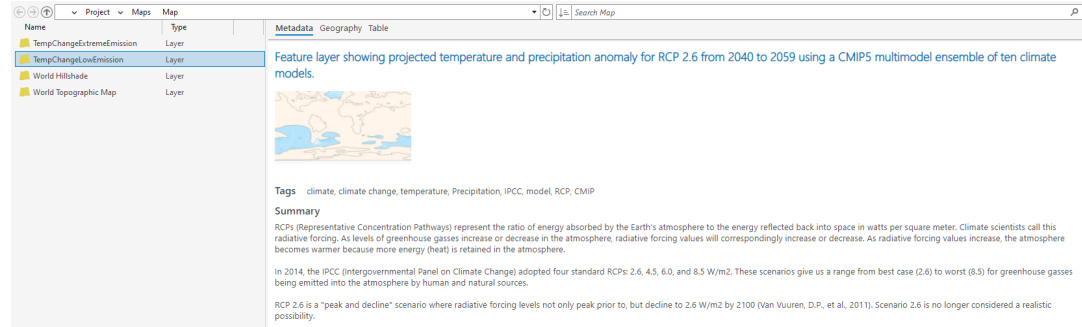
vi. Use Limitations

This work is licensed under the Esri Master License Agreement.

e. Click **Generate Thumbnail**

f. Click **OK**

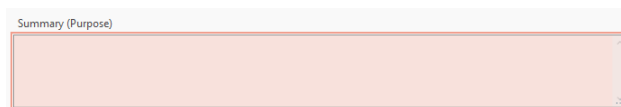
g. To confirm that the information we entered for the layer is saved, right-click on the **TempChangeLowEmission** layer's name in the Contents pane, and select **View Metadata**. You should be able to see the information that we entered in the previous section. ***Tip: Do not worry if your thumbnail is different***



4. View and edit metadata from the Catalog pane

- We can also view and edit metadata from the Catalog pane. Open the Catalog pane if needed by going to **View>Catalog pane**.
- Navigate to the **TemperatureChangeExtremeEmission** layer within the Catalog pane. It should be under the Folders icon.
- Right-click on the **TemperatureChangeExtremeEmission** layer's name in the Catalog pane and select **Edit Metadata**.
- A new window showing the metadata form for the layer appears. Depending on the metadata style you have chosen (we will learn how to change between different metadata styles in the next section), it will have its own set of required and optional fields.

In your window, the **fields/boxes that are in red** are those that must be completed to comply with the metadata style that you are currently using in ArcGIS.



The **red exclamation signs** at the top of the window provide an overview of the missing requirements for the specific metadata style.



- Given that the metadata form window is saying the only required sections with missing information are (1) purpose/summary and (2) tags (assuming that you are currently using the default metadata style of Item Description), fill in **only the summary and tags** sections by copying and pasting from the appropriate sections below. (Source:

<https://www.arcgis.com/home/item.html?id=8bdb6bfd60d346cfb272b6892ad45252>)

- Tags

climate, climate change, Precipitation, Temperature, CMIP, RCP, model, IPCC

ii. Summary (Purpose)

RCPs (Representative Concentration Pathways) represent the ratio of energy absorbed by the Earth's atmosphere to the energy reflected back into space in watts per square meter. Climate scientists call this radiative forcing. As levels of greenhouse gasses increase or decrease in the atmosphere, radiative forcing values will correspondingly increase or decrease. As radiative forcing values increase, the atmosphere becomes warmer because more energy (heat) is retained in the atmosphere.

In 2014, the IPCC (Intergovernmental Panel on Climate Change) adopted four standard RCPs: 2.6, 4.5, 6.0, and 8.5 W/m². These scenarios give us a range from the best case (2.6) to the worst (8.5) for greenhouse gasses being emitted into the atmosphere by human and natural sources.

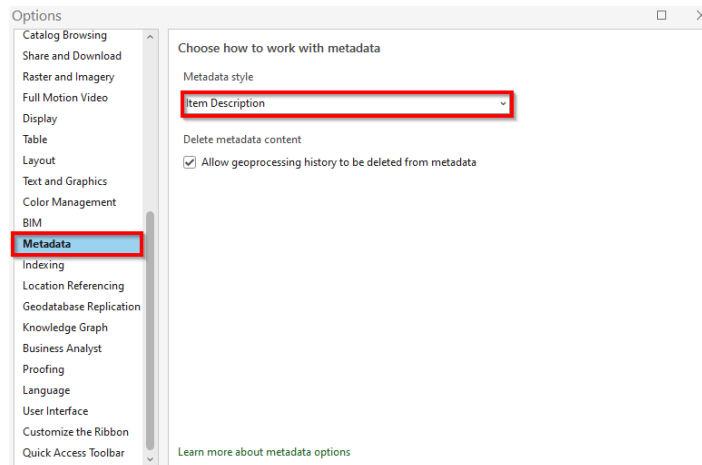
RCP 8.5 is a scenario where radiative forcing levels are modeled with different assumptions than 2.6, 4.5, and 6.0. In 8.5 the assumptions are very high levels of radiative forcing due in part to high levels of population growth and continued lower incomes in developing countries will reach 8.50 W/m² by 2100 (Van Vuuren, D.P., et al., 2011). At the time, RCP 8.5 was the most extreme model scenario, but it is now considered very likely.

Global climate projections are produced by general circulation models (GCMs). This particular layer was derived from the mean results of 10 of the best GCMs, called a multi-model ensemble (MME). Esri partnered with the Research Applications Laboratory (RAL) from the National Center for Atmospheric Research (NCAR) to select the model members.

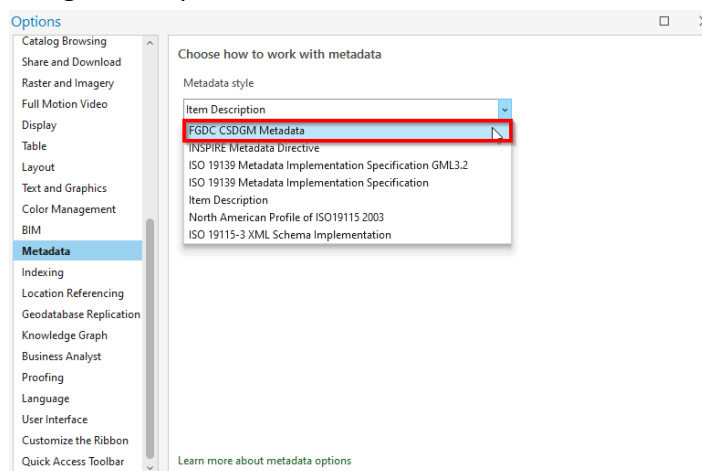
- f. Hit the **Save** icon  under the Metadata tab to save your changes. And save your project by going to **Project>Save**.
- g. **Close** the metadata form window.

5. Choose a different metadata stylesheet and verify metadata standard compliance

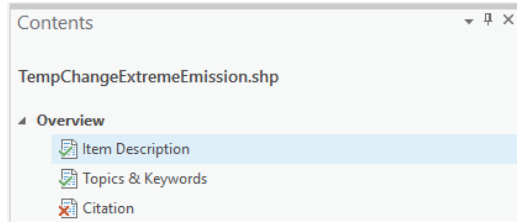
- a. To choose a different metadata stylesheet, click on **Project>Options**.
- b. Click **Metadata** on the left side of the Options window. On the right-hand side of the Metadata section window, you should now see a dropdown for different Metadata stylesheets. You can see that the metadata style we have been working with up to this point is the **Item Description** style, which is an Esri-specific metadata style.



Using the dropdown, select the **FGDC CSDGM Metadata stylesheet**. Click **OK**.



- c. Click the **back arrow**  in the upper left to return to your map.
- d. To verify that the information we entered in the previous section of this exercise is sufficient to be in compliance with the FGDC CSDGM Metadata standard, navigate to the **TemperatureChangeExtremeEmission** layer within the **Catalog pane**.
- e. Right-click on the **TemperatureChangeExtremeEmission** layer's name in the **Catalog pane** and select **Edit Metadata**.
- f. The metadata form window for the file appears. Switch to the Contents pane by going to **View>Contents**, and click on one of the many sections (e.g. Citation) of the FGDC CSDGM metadata standard that has a **red X mark**  **Citation** in the **Contents pane** (not Catalog pane). Then, study the red exclamation sign at the top of the metadata form window to see what requirements are we missing for that section.
- g. **Look** at other sections of the FGDC CSDGM metadata standard in the **Contents pane**.



Q3. List the names of sections in which we have missing requirements according to the FGDC CSDGM metadata standard.

Close the metadata form window.

- h. **Modify and repeat** steps a thru e for the **TemperatureChangeExtremeEmission** layer, except select **North American Profile of ISO 19115 2003** for the metadata stylesheet in step b to see if our metadata is compliant with the ISO metadata standard.

Q4. Look at the various sections of the ISO metadata standard in the **Contents pane**, and list the names of the sections in which we have missing requirements.

Close the metadata form window.

- i. **Save** your project and **close** ArcGIS Pro.

Works Cited

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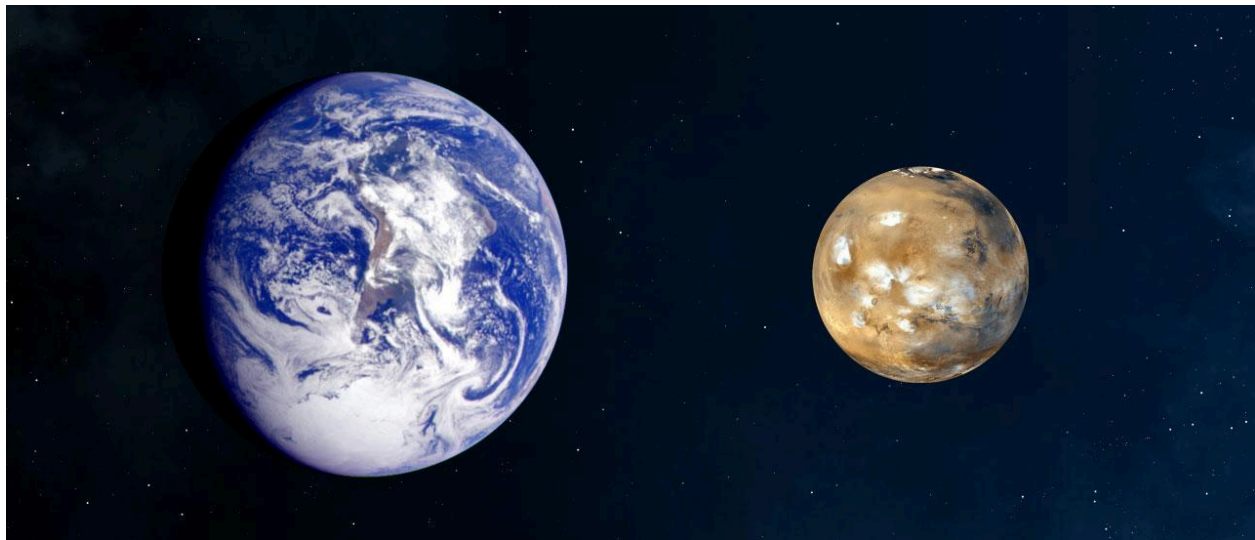
Chapter 11: Plan your Mars landing

In this exercise, you will:

- *Reclassify raster layers*
- *Extract cells from a raster layer based on their attributes*
- *Perform basic terrain analysis (i.e. slope)*
- *Perform raster calculations*
- *Convert raster layers to the vector format*
- *Add and calculate new fields*
- *Perform zonal statistical analysis*

Introduction

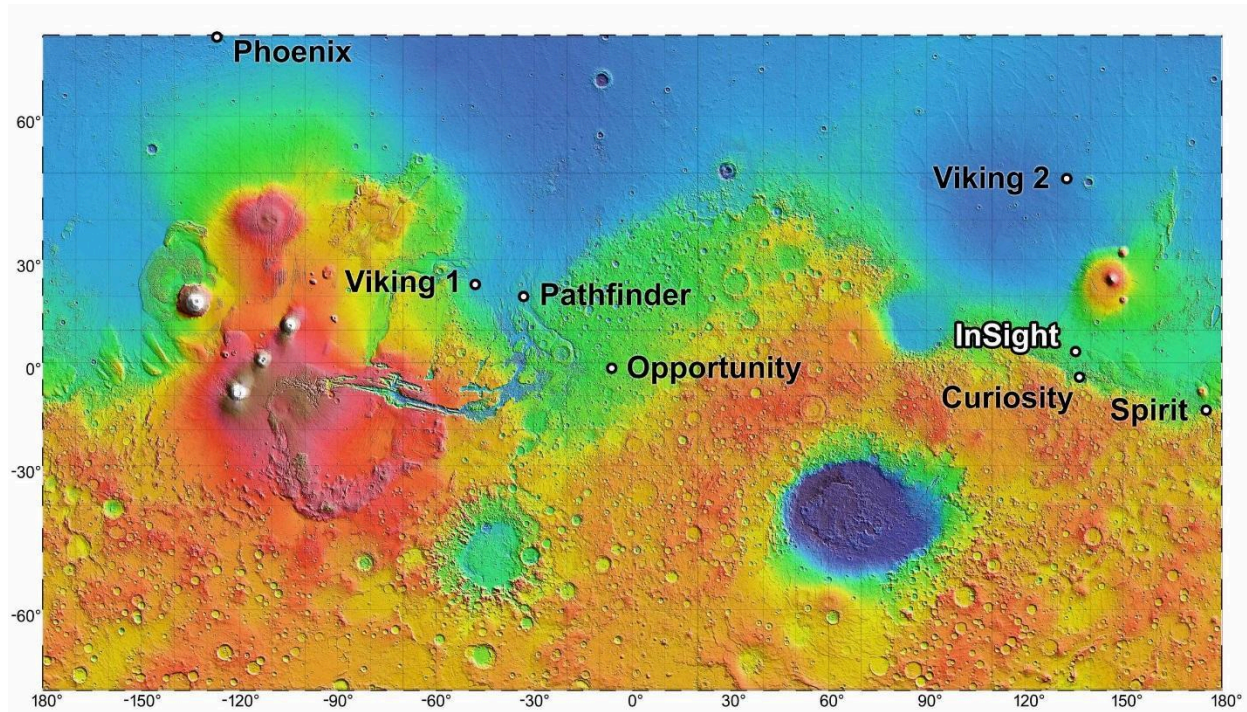
Known as the Red Planet due to the oxidation of iron minerals (i.e. rust) in its soils, the diameter of the planet Mars (i.e. 6,790 kilometers) is only slightly more than half the size of the Earth's diameter (i.e. 12,750 kilometers) (Kiefer et al., 1997; Erickson et al., 2019). The former Soviet Union and the United States have launched probes to Mars since 1960, and the first successful flyby of Mars was performed by the Mariner 4 mission in 1965 (Erickson et al., 2019). While the Soviet probes, Mars 2 and Mars 3, were the first to contact the Martian surface in 1971, these missions failed seconds after landing (Williams, 2019a; Williams, 2019b). NASA's Viking 1 and Viking 2 missions provided us with the first color panoramic images of the planet. Both spacecraft landed on Mars' surface in 1976 and remained operational for six and three years, respectively (NASA/JPL).



A COMPARISON OF THE SIZE OF EARTH WITH THE SIZE OF MARS (SOURCE: NASA)

NASA's Viking program was subsequently followed by other successful missions to Mars in the 1980s (e.g. Soviet's Phobos 1, 2), 1990s (e.g. NASA's Mars Global Surveyor, Mars Pathfinder &

Sojourner), 2000s (e.g. NASA's *Phoenix*, European Space Agency's *Mars Express*), and up to the present day (e.g. NASA's *Curiosity*, *InSight*) (The Planetary Society).



SELECTED SOVIET AND NASA LANDING SITES ON MARS (SOURCE: NASA)

As various countries and space agencies continue to be fascinated by the Red Planet, the research objective of Mars Exploration has also evolved from a “skin-deep” exploration that seeks to photograph and study the planet’s surface features, to one that will explore the planet’s geological processes and deep interior (Byrd, 2018). Ultimately, scientists hope to learn more about Mars’ history of life and habitability by understanding the forces that shaped the planet’s evolution over time (NASA, 2014).

A crucial obstacle for many expeditions is the ability to land rovers with scientific instruments on Mars. In this exercise, you will find a suitable landing site for future rovers based on elevation and terrain data, as well as dust level information that has been gathered in past missions. Specifically, in this simplified exercise to determine a suitable landing site, you will look for areas that fulfill the following criteria:

- *Low dust level: Since dust storm is a common phenomenon on Mars, areas with relatively low dust level is preferred, because high dust level may prevent rovers from accessing the sunlight needed to charge their solar panels (Witze, 2018).*
- *Low slope value: When looking at many of the past landing sites, smooth and flat surfaces were intentionally chosen, because rocky terrain may not only damage the spacecraft, but also cause it to tip over (Koren, 2017).*

- *Low elevation: Relative to Earth, Mars's atmosphere is about 100 times thinner (Sharp, 2017). Consequently, landing spots at low elevations are preferred, because Mar's atmosphere is thickest at these locations, and provides sufficient atmospheric friction for the spacecraft's landing gear to slow its landing (JPL, 2018).*

1. Identify areas of low elevation

- Open **ArcGIS Pro**, and start a new **project** and a new **map**.
- Add in the **Elevation** layer. This will take several minutes to open, as it contains the elevation information for the entire planet of Mars. Explore the layer's properties.

Q1. What is the Projected Coordinate System used by this layer? ***Tip: Check the Layer Properties, the Source section, and the Spatial Reference subsection***

- Go to **Analysis>Tools**. Look for **Reclassify (Spatial Analyst Tools)**. Enter the following:
 - Input raster: **Elevation**
 - Reclass field: **Value**
 - (Click **Classify**), select **Equal Interval** for Method, enter **10** for the Number of classes, and click **OK**.

In the table below, fill out the Start and End values for each New class. You will be submitting a copy or photo of this table for the assignment.

Reclassification of the ELEVATION layer		
Start	End	New
		1
		2
		3
		4
		5
		6

		7
		8
		9
		10

- vii. Output raster: **ReclElev** (**Tip: Replace the characters after the last backslash (\) with ReclElev. Do not delete everything and just type in ReclElev**)
- iv. Click **Run** (This process will take several minutes)
- d. We want to land at relatively low elevations since Mars' atmosphere is thickest at these locations, which enables the spacecraft to make a slow and stable descent. In this step, we will single out the low-elevation areas that we previously reclassified to the new class value of 1.

Go to **Analysis>Tools**. Look for **Extract by Attributes (Spatial Analyst Tools)**.

Enter the following:

- i. Input raster: **ReclElev**
- ii. Use the dropdown menus to enter **Value** for the first dropdown, **is equal to** for the second dropdown, and **1** for the third dropdown.

Where Value is equal to 1

- iii. Output raster: **ClipElev**
- iv. Click **Run** (This process will take several minutes)

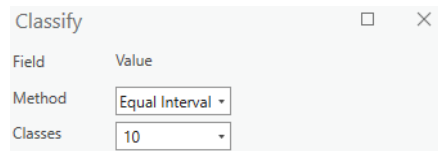


- e. **Turn off** the ReclElev layer in the Contents pane, so you can see the ClipElev output clearly.

2. Identify relatively flat areas

- a. Go to **Analysis>Tools**. Look for **Slope (Spatial Analyst Tools)**. Enter the following:
 - i. Input raster: **Elevation**
 - ii. Output raster: **MarsSlope**
 - iii. Click **Run** (This process will take several minutes)
- b. Go to **Analysis>Tools**. Look for **Reclassify (Spatial Analyst Tools)**. Enter the following:
 - i. Input raster: **MarsSlope**

- ii. Reclass field: **Value**
- iii. (Click **Classify**), select **Equal Interval** for Method, enter **10** for the Number of classes, and click **OK**.



Classify

Field: Value

Method: Equal Interval

Classes: 10

In the table below, fill out the Start and End values for each New class.

You will be submitting a copy or photo of this table for the assignment.

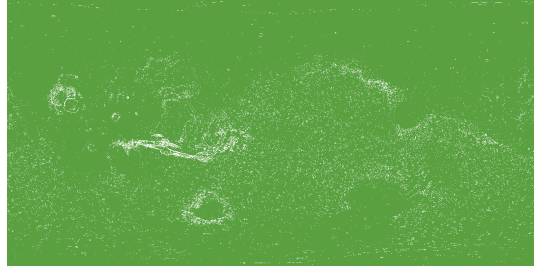
Reclassification of the SLOPE layer		
Start	End	New
		1
		2
		3
		4
		5
		6
		7
		8
		9
		10

- iv. Output raster: **ReclSlope**
- v. Click **Run** (This process will take several minutes)
- c. We want to land on relatively flat areas much like previous landings, especially taking care to avoid overly steep areas (i.e. areas with high slope values), in order to avoid tipping over or damaging the spacecraft.

Go to **Analysis>Tools**. Look for **Extract by Attributes (Spatial Analyst Tools)**.

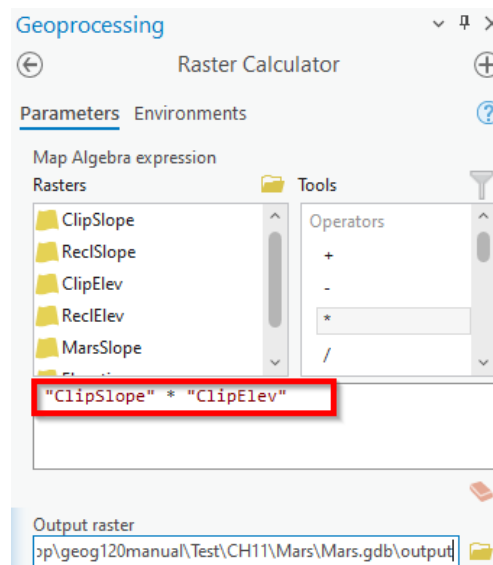
Enter the following:

- i. Input raster: **ReclSlope**
- ii. Use the dropdown menus to enter **Value** for the first dropdown, **is equal to** for the second dropdown, and **1** for the third dropdown.
- iii. Output raster: **ClipSlope**
- iv. Click **Run** (This process will take several minutes)



3. Find areas at low elevations that are also flat

- a. **Turn off** all layers in the Contents pane except **ClipElev** and **ClipSlope**. All the cells that are left in the ClipElev layer have a class value of 1 (i.e. low elevation), and all other areas have a value of no data (hidden). Similarly, all the cells that are left in the ClipSlope layer have a class value of 1 (i.e. relatively flat), and all other areas have no data values (hidden).
- b. Go to **Analysis>Tools**. Look for **Raster Calculator (Spatial Analyst Tools)**. Enter the following equation by double-clicking the appropriate items from the **Rasters** and **Tools** sections:



- i. Output raster: **Output**
- ii. Click **Run** (This process will take several minutes)

4. Convert raster output into vector features

- a. **Turn off** all layers in the Contents pane except Output. Only those cells which have a value of 1 in the ClipSlope layer (relatively flat) AND a value of 1 in the ClipElev layer (low elevation) should be found in the Output layer, because:
 - *1 in ClipSlope (flat) * 1 in ClipElev (low elevation) = 1 in output*
 - *nodata in ClipSlope (not flat) * 1 in ClipElev (low elevation) = nodata in output*
 - *1 in ClipSlope (flat) * nodata in ClipElev (not low elevation) = nodata in output*
- b. We will need to convert the raster output into a polygon layer in order to calculate the area of suitable (low elevation, flat) landing sites. To do this, go to **Analysis>Tools**. Look for **Raster to Polygon (Conversion Tools)**, and enter the following:
 - i. Input raster: **Output**
 - ii. Field: **Value**
 - iii. Output polygon features: **OutputPoly**
 - iv. **(Keep Simplify polygons checked)**
 - v. Click **Run**

5. Add field and calculate area

- a. To calculate the area of the suitable sites in our OutputPoly layer, we will first need to add a new field in the layer's attribute table. Go to **Analysis>Tools**, and look for **Add Field (Data Management Tools)**. Enter the following:
 - i. Input Table: **OutputPoly**
 - ii. Field Name: **NewArea**
 - iii. Field Type: **Double (64-bit floating point)**
 - iv. Click **Run**
- b. To perform our area calculation and store the result in the NewArea field, go to **Analysis>Tools**, and look for **Calculate Geometry Attributes (Data Management Tools)**. Enter the following:
 - i. Input Features: **OutputPoly**
 - ii. Geometry Attributes: Field (Existing or New)-**NewArea**, Property-**Area**
 - iii. Area Unit: **Square US Survey Miles**
 - iv. Coordinate System: **Current Map [Map]**
 - v. Click **Run**

- c. Open the attribute table of the **OutputPoly** layer, and **sort** the NewArea field in a Descending order.

Q2. What is the size of the largest site which is relatively flat and at a low elevation (i.e. largest value in the NewArea column)?

Submit a screenshot of your map with the largest site selected.

- d. **Clear** any selected feature(s) before moving on to the next step.

6. Compare the dust level at our potential landing sites

- a. Add in **MarsDust** layer.
- b. Go to **Analysis>Tools**. Look for **Zonal Statistics (Spatial Analyst Tools)**. Enter the following parameters to calculate the average dust level at each of the zone/site polygons that are in the OutputPoly layer:
 - i. Input raster or feature zone data: **OutputPoly**
 - ii. Zone field: **Id**
 - iii. Input value raster: **MarsDust**
 - iv. Output raster: **DustStat**
 - v. Statistics type: **Mean**
 - vi. Click **Run**
- c. **Turn off** all layers in the Contents pane except DustStat.
- d. We want to avoid areas with high levels of dust, as Martian dust storms can prevent the sun's radiation from reaching our spacecraft's solar panels and recharging its battery. **Symbolize** the new DustStat layer (under the **Raster Layer** tab) to increase the contrast between the levels of dust at various landing sites/zones if necessary.
- e. Using the **Explore** tool under the **Map** tab, click on the approximate location of the polygon that was in your screenshot for your answer to Question 2.

Q3. What is the average/mean level of dust [i.e. Stretch.Pixel Value] at that site?

- f. Refer to the **DustStat** (not MarsDust) layer's legend in the Contents pane.

Q4. Of the average dust levels among our suitable sites, what is the minimum dust level value?

Q5. What is the maximum dust level value?

Q6. Is the value that you reported for Question 3 closer to the minimum or maximum dust level value?

Be sure to submit all the tables for this lab exercise with your answers.

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